

MAT 247 Algebra II

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Chapter 1

Week 1

1.1 Diagonalization

Note:-

Related homework problems: see 1 for a diagonalization computation, 2 for characteristic-polynomial coefficients, 3 for square roots of diagonalizable matrices, and 4 for nonzero eigenvalues of AB and BA .

Note:-

Related tutorial problems: see 9.1.1 for characteristic polynomials under similarity, 9.1.2 for the distinct-eigenvalues diagonalization criterion, 9.1.3 for a full diagonalization computation, and 9.1.4 for the cyclic shift matrix.

Note:-

Related additional exercises: see 1 for a past midterm diagonalization computation, 2 for diagonalization true/false checks, and 11 for kernels of powers of a diagonalizable operator.

Definition 1.1.1: Eigenvalues and Eigenvectors

Let $A \in F^{n \times n}$. A scalar $\lambda \in F$ is called an eigenvalue of A if there exists a nonzero vector $v \in F^n$ such that

$$Av = \lambda v.$$

Such a vector v is called an eigenvector corresponding to λ .

If $T : V \rightarrow V$ is a linear operator and β is a basis of V , then

$$Tv = \lambda v \iff [T]_{\beta}[v]_{\beta} = \lambda[v]_{\beta}.$$

Hence T and $[T]_{\beta}$ have the same eigenvalues, and eigenvectors correspond via the coordinate map.

Definition 1.1.2: Eigenspace

Let $T : V \rightarrow V$ be linear and let $\lambda \in F$. The eigenspace of λ is

$$E_\lambda(T) = \ker(\lambda I - T) = \{v \in V : Tv = \lambda v\}.$$

Definition 1.1.3: Characteristic Polynomial

Let $A \in F^{n \times n}$. The characteristic polynomial of A is defined by

$$p_A(t) = \det(A - tI_n).$$

Note:-

Some authors define the characteristic polynomial as

$$p_A(t) = \det(A - tI_n) = (-1)^n \det(tI_n - A).$$

Both definitions yield the same eigenvalues.

Proposition 1.1.1

If $A, B \in F^{n \times n}$ are similar matrices, then A and B have the same characteristic polynomial.

Proof: Since A and B are similar, there exists an invertible matrix $P \in \mathbb{F}^{n \times n}$ such that

$$B = P^{-1}AP.$$

Then for any indeterminate t ,

$$B - tI = P^{-1}AP - tI = P^{-1}AP - tP^{-1}IP = P^{-1}(A - tI)P,$$

where we used $P^{-1}IP = I$. Taking determinants and using multiplicativity of determinant,

$$\det(B - tI) = \det(P^{-1}(A - tI)P) = \det(P^{-1}) \det(A - tI) \det(P) = \det(A - tI),$$

since $\det(P^{-1}) \det(P) = 1$. Therefore,

$$p_B(t) = \det(B - tI) = \det(A - tI) = p_A(t),$$

so A and B have the same characteristic polynomial. ☺

Proposition 1.1.2

Let $A \in F^{n \times n}$ and $\lambda \in F$. Then λ is an eigenvalue of A if and only if

$$p_A(\lambda) = 0.$$

Proof: $\Rightarrow$ Suppose λ is an eigenvalue. Then there exists $v \neq 0$ such that

$$Av = \lambda v \iff (\lambda I - A)v = 0.$$

Thus $\lambda I - A$ is not invertible, so

$$\det(\lambda I - A) = 0,$$

hence $p_A(\lambda) = 0$.

$\Leftarrow$ If $p_A(\lambda) = 0$, then $\det(\lambda I - A) = 0$, so $\lambda I - A$ is not invertible. Hence there exists $v \neq 0$ such that

$$(\lambda I - A)v = 0 \iff Av = \lambda v,$$

so λ is an eigenvalue. 😊

Theorem 1.1.1

Let $A \in \mathbb{F}^{n \times n}$. Then $p_A(x)$ is a monic polynomial of degree n .

A polynomial

$$f(x) = \sum_{i=0}^n a_i x^i$$

is called monic if $a_n = 1$.

Theorem 1.1.2

Let $B \in \mathbb{F}^{n \times n}$. Then

$$\det(B) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n b_{i, \sigma(i)}.$$

Here,

$$S_n = \{ \sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\} \mid \sigma \text{ bijective} \}$$

is the set of all permutations of $\{1, \dots, n\}$, and

$$\text{sgn}(\sigma) \in \{1, -1\}.$$

Note:-

The determinant formula above is known as the Leibniz formula.

Degree of the Characteristic Polynomial

We compute

$$p_A(x) = \det(xI_n - A).$$

The identity permutation contributes the term

$$\prod_{i=1}^n (x - a_{ii}) = x^n + r(x), \quad \deg r \leq n - 1.$$

All other permutations contribute terms of degree at most $n - 1$.

Hence,

$$p_A(x) = x^n + q(x), \quad \deg q \leq n - 1,$$

so $p_A(x)$ is monic of degree n .

Roots and Eigenvalues

Note:-

A polynomial of degree n has at most n roots (counted with multiplicity).

Example:

$$f(x) = (x - 1)^2(x + 3)$$

has two distinct roots, with multiplicities.

Corollary 1.1.1

If $A \in \mathbb{F}^{n \times n}$, then A has at most n eigenvalues (counted with algebraic multiplicity).

For a linear map $T : V \rightarrow V$, define

$$p_T(x) := p_{[T]_\beta}(x),$$

which is independent of the choice of basis β .

Definition 1.1.4: Splitting of a Polynomial over a Field

A polynomial $f(x)$ splits over a field $\mathbb{F}$ if there exist $c, a_1, \dots, a_n \in \mathbb{F}$ such that

$$f(x) = c \prod_{i=1}^n (x - a_i).$$

The numbers $a_1, \dots, a_n$ are the roots of f (with multiplicity).

Theorem 1.1.3 Fundamental Theorem of Algebra

Every polynomial splits over $\mathbb{C}$.

Lemma 1.1.1

If $A \in \mathbb{F}^{n \times n}$ is diagonalizable, then its characteristic polynomial $p_A(x)$ splits over $\mathbb{F}$.

Proof: Since A is diagonalizable, it is similar to a diagonal matrix

$$D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Hence,

$$p_A(x) = p_D(x) = \det(xI - D) = \prod_{i=1}^n (x - \lambda_i),$$

which splits over $\mathbb{F}$.



Corollary 1.1.2

If $A \in \mathbb{F}^{n \times n}$, then A has at most n eigenvalues (counted with algebraic multiplicity).

For a linear map $T : V \rightarrow V$, define

$$p_T(x) := p_{[T]_\beta}(x)$$

for any basis β of V .

Chapter 2

Week 2

2.1 Diagonalization (continued)

Definition 2.1.1: Algebraic and Geometric Multiplicity

Let V be a vector space with $\dim(V) < \infty$, and let $T : V \rightarrow V$ be a linear operator. Let λ be an eigenvalue of T .

1. The *algebraic multiplicity* of λ is the multiplicity of λ as a root of the characteristic polynomial $p_T(x)$. Equivalently, it is the largest integer m such that

$$(x - \lambda)^m \mid p_T(x).$$

2. The *geometric multiplicity* of λ is

$$\dim E(\lambda, T),$$

where

$$E(\lambda, T) = \ker(\lambda I - T)$$

is the eigenspace corresponding to λ .

Theorem 2.1.1 Eigenbasis Characterization of Diagonalizability

Let $T : V \rightarrow V$ be a linear operator on a finite-dimensional vector space V . If there exists a basis of V consisting entirely of eigenvectors of T , then T is diagonalizable.

Proof: Let

$$B = (v_1, \dots, v_n)$$

be a basis of V consisting of eigenvectors of T . Then for each i there exists λ_i such that

$$Tv_i = \lambda_i v_i.$$

With respect to the basis B , the matrix of T is

$$[T]_B = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

which is diagonal. Hence T is diagonalizable. 😊

Proposition 2.1.1

Let λ be an eigenvalue of T . If m is the algebraic multiplicity of λ and d is the geometric multiplicity of λ , then

$$d \leq m.$$

Proof: Let $\{v_1, \dots, v_d\}$ be a basis of $E(\lambda, T)$, and extend it to an ordered basis

$$\beta = \{v_1, \dots, v_d, v_{d+1}, \dots, v_n\}$$

of V , where $n = \dim(V)$.

With respect to β , the matrix of T has block form

$$[T]_\beta = \begin{pmatrix} \lambda I_d & * \\ 0 & X \end{pmatrix},$$

where X is an $(n - d) \times (n - d)$ matrix.

Then

$$p_T(x) = \det(xI - [T]_\beta) = \det \begin{pmatrix} (x - \lambda)I_d & * \\ 0 & xI_{n-d} - X \end{pmatrix} = (x - \lambda)^d \det(xI_{n-d} - X).$$

Thus $(x - \lambda)^d$ divides $p_T(x)$, so $d \leq m$. 😊

Theorem 2.1.2 Diagonalizability via Multiplicities

Let V be a vector space with $\dim(V) = n$, and let $T : V \rightarrow V$ be a linear operator. Assume that the characteristic polynomial $p_T(x)$ splits over the field. Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T .

The following are equivalent:

1. T is diagonalizable.
2. $\sum_{i=1}^k \dim E(\lambda_i, T) = n$.
3. The algebraic multiplicity of λ_i equals the geometric multiplicity of λ_i for all i .

Proof: (1) $\Rightarrow$ (2): If T is diagonalizable, then V has a basis consisting of eigenvectors. Each eigenvector belongs to exactly one eigenspace $E(\lambda_i, T)$, so the sum of their dimensions is n .

(2) $\Rightarrow$ (3): Let m_i be the algebraic multiplicity and $d_i = \dim E(\lambda_i, T)$. Since

$$\sum_{i=1}^k d_i = n \quad \text{and} \quad \sum_{i=1}^k m_i = n,$$

and we already know $d_i \leq m_i$ for all i , it follows that $d_i = m_i$ for all i .

(3) $\Rightarrow$ (1): Choose a basis β_i for each eigenspace $E(\lambda_i, T)$. Since eigenspaces corresponding to distinct eigenvalues intersect trivially, the union

$$\beta = \beta_1 \cup \cdots \cup \beta_k$$

is linearly independent. Moreover,

$$|\beta| = \sum_{i=1}^k d_i = \sum_{i=1}^k m_i = n,$$

so β is a basis of V consisting of eigenvectors. Hence T is diagonalizable. ☺

Lemma 2.1.1

If $A \in \mathbb{F}^{n \times n}$ is diagonalizable, then its characteristic polynomial $p_A(x)$ splits over $\mathbb{F}$.

Proof: If A is diagonalizable, then A is similar to a diagonal matrix

$$D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Since similar matrices have the same characteristic polynomial,

$$p_A(x) = p_D(x) = \prod_{i=1}^n (x - \lambda_i),$$

which splits over $\mathbb{F}$. ☺

Lemma 2.1.2

Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of a linear operator $T : V \rightarrow V$. Let $v_i \neq 0$ be an eigenvector corresponding to λ_i . Then the set $\{v_1, \dots, v_k\}$ is linearly independent.

Proof: We prove the result by induction on k .

Base case ($k = 1$): The set $\{v_1\}$ is linearly independent since $v_1 \neq 0$.

Inductive step: Assume the statement holds for $k - 1 \geq 1$. Suppose

$$a_1 v_1 + \cdots + a_k v_k = 0.$$

We must show $a_1 = \cdots = a_k = 0$.

Apply $T - \lambda_k I$ to both sides:

$$0 = (T - \lambda_k I) \left(\sum_{i=1}^k a_i v_i \right) = \sum_{i=1}^k a_i (T - \lambda_k I) v_i.$$

Since $T(v_i) = \lambda_i v_i$, we have

$$(T - \lambda_k I) v_i = (\lambda_i - \lambda_k) v_i,$$

and the $i = k$ term vanishes. Thus,

$$0 = \sum_{i=1}^{k-1} a_i (\lambda_i - \lambda_k) v_i.$$

By the induction hypothesis, $\{v_1, \dots, v_{k-1}\}$ is linearly independent. Since $\lambda_i \neq \lambda_k$, it follows that

$$a_i = 0 \quad \forall 1 \leq i \leq k-1.$$

Returning to the original equation gives

$$a_k v_k = 0.$$

Since $v_k \neq 0$, we conclude $a_k = 0$. ☺

Theorem 2.1.3 Vandermonde Determinant

Let $a_1, \dots, a_n \in \mathbb{F}$. Define the Vandermonde matrix

$$M = \begin{pmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \cdots & a_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^{n-1} \end{pmatrix}.$$

Then

$$\det(M) = \prod_{1 \leq i < j \leq n} (a_j - a_i).$$

Corollary 2.1.1

If $a_1, \dots, a_n$ are distinct, then $\det(M) \neq 0$.

Proof: Assume

$$\sum_{i=1}^k a_i v_i = 0,$$

where v_i is an eigenvector of T with eigenvalue λ_i . Apply T^j to both sides:

$$0 = \sum_{i=1}^k a_i T^j(v_i).$$

Since $T(v_i) = \lambda_i v_i$, it follows by induction that

$$T^j(v_i) = \lambda_i^j v_i \quad \forall j \geq 0.$$

Hence,

$$\sum_{i=1}^k a_i \lambda_i^j v_i = 0 \quad \forall j \geq 0.$$

Now assume $V = \mathbb{F}^n$ and $v_1, \dots, v_k \in \mathbb{F}^n$. For $0 \leq j \leq k-1$, we obtain

$$\begin{pmatrix} 1 & \cdots & 1 \\ \lambda_1 & \cdots & \lambda_k \\ \vdots & & \vdots \\ \lambda_1^{k-1} & \cdots & \lambda_k^{k-1} \end{pmatrix} \begin{pmatrix} a_1 v_1 \\ \vdots \\ a_k v_k \end{pmatrix} = 0.$$

Let this matrix be M . Since the λ_i are distinct,

$$\det(M) = \prod_{1 \leq i < j \leq k} (\lambda_j - \lambda_i) \neq 0.$$

Thus M is invertible. Multiplying both sides by M^{-1} yields

$$a_i v_i = 0 \quad \forall 1 \leq i \leq k.$$

Because $v_i \neq 0$, we conclude $a_i = 0$ for all i . ⊖

Theorem 2.1.4

Let $T : V \rightarrow V$ be a linear operator on an n -dimensional vector space V . Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T . For each i , let

$$d_i = \dim E(\lambda_i, T) \quad \text{and} \quad m_i = \text{algebraic multiplicity of } \lambda_i.$$

Then the following are equivalent:

1. T is diagonalizable.
2. $\sum_{i=1}^k d_i = n$.
3. $d_i = m_i$ for all $i = 1, \dots, k$.

(1) $\Rightarrow$ (3)

Proof: Assume that T is diagonalizable. Then V has a basis B consisting of eigenvectors of T .

For each i , define

$$B_i = B \cap E(\lambda_i, T) = \{v \in B \mid v \text{ is an eigenvector with eigenvalue } \lambda_i\}.$$

Then

$$B = \bigcup_{i=1}^k B_i \quad (\text{disjoint union}).$$

Since $B_i \subset E(\lambda_i, T)$,

$$|B_i| \leq \dim E(\lambda_i, T) = d_i.$$

Therefore,

$$n = \dim V = |B| = \sum_{i=1}^k |B_i| \leq \sum_{i=1}^k d_i \leq \sum_{i=1}^k m_i = n,$$

where we used the fact that $\sum_{i=1}^k m_i = n$.

Thus equality holds throughout, and we conclude

$$d_i = m_i \quad \forall i.$$

😊

(3) $\Rightarrow$ (2)

Proof: Assume $d_i = m_i$ for all i . Then

$$\sum_{i=1}^k d_i = \sum_{i=1}^k m_i = n.$$

😊

(2) $\Rightarrow$ (1)

Proof: Assume

$$\sum_{i=1}^k d_i = n.$$

For each i , let B_i be a basis of the eigenspace $E(\lambda_i, T)$. Define

$$B = \bigcup_{i=1}^k B_i.$$

Then

$$|B| = \sum_{i=1}^k |B_i| = \sum_{i=1}^k d_i = n.$$

Claim: B is linearly independent.

Suppose

$$\sum_{v \in B} a_v v = 0.$$

Rewrite this sum as

$$\sum_{i=1}^k \sum_{v \in B_i} a_v v = 0.$$

Each inner sum lies in the eigenspace $E(\lambda_i, T)$.

By the lemma that eigenvectors corresponding to distinct eigenvalues are linearly independent, the vectors

$$\left\{ \sum_{v \in B_i} a_v v \right\}_{i=1}^k$$

are linearly independent. Hence,

$$\sum_{v \in B_i} a_v v = 0 \quad \forall i.$$

Since each B_i is a basis, it follows that $a_v = 0$ for all $v \in B$.

Thus B is a linearly independent set of n vectors, hence a basis of V . Therefore V has a basis of eigenvectors, and T is diagonalizable. ☺

2.2 Direct Sums

Note:-

Related homework problems: see 6 for invariant subspaces of diagonalizable maps, 10 for the stabilized kernel-image direct sum, and 20 for projections and direct-sum decompositions.

Note:-

Related tutorial problems: see 9.2.1 for equivalent criteria for direct sums, 9.2.1 for characteristic polynomials over invariant direct sums, and 9.2.2 for the decomposition of an idempotent map into kernel plus image.

Definition 2.2.1: Sum of Subspaces

Let V be a vector space and let $V_1, V_2 \leq V$ be subspaces. Define

$$V_1 + V_2 = \{ v + w \mid v \in V_1, w \in V_2 \}.$$

Theorem 2.2.1

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

In particular,

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 \iff V_1 \cap V_2 = \{0\}.$$

Definition 2.2.2: Direct Sum (Two Subspaces)

If $V_1 \cap V_2 = \{0\}$, then $V_1 + V_2$ is called a *direct sum*, denoted

$$V_1 \oplus V_2.$$

Equivalently, every $v \in V_1 \oplus V_2$ can be written uniquely as $v = v_1 + v_2$ with $v_i \in V_i$.

Definition 2.2.3: Direct Sum (Multiple Subspaces)

Let $V_1, \dots, V_r \leq V$ be subspaces. Define

$$V_1 + \dots + V_r = \{v_1 + \dots + v_r \mid v_i \in V_i\}.$$

This sum is called *direct*, denoted

$$V_1 \oplus \dots \oplus V_r,$$

if every $v \in V_1 + \dots + V_r$ has a unique representation $v = v_1 + \dots + v_r$ with $v_i \in V_i$.

Theorem 2.2.2

The following are equivalent:

1. $V_1 \oplus \dots \oplus V_r$ is a direct sum.
2. If $v_1 + \dots + v_r = 0$ with $v_i \in V_i$, then $v_i = 0$ for all i .
3. For each i ,

$$V_i \cap \left(\sum_{j \neq i} V_j \right) = \{0\}.$$

4. If B_i is a basis of V_i , then

$$\bigcup_{i=1}^r B_i$$

is a basis of $V_1 \oplus \dots \oplus V_r$.

- 5.

$$\dim(V_1 + \dots + V_r) = \sum_{i=1}^r \dim V_i.$$

Proposition 2.2.1

Let $T : V \rightarrow V$ be a linear operator with distinct eigenvalues $\lambda_1, \dots, \lambda_k$. Then T is diagonalizable if and only if

$$V = E(\lambda_1, T) \oplus \dots \oplus E(\lambda_k, T).$$

Proof: ($\Leftarrow$) Let B_i be a basis of $E(\lambda_i, T)$. By part (4) of the previous theorem,

$$B = \bigcup_{i=1}^k B_i$$

is a basis of V . Since every vector in B is an eigenvector, T is diagonalizable.

($\Rightarrow$) Assume T is diagonalizable and let B be a basis of V consisting of eigenvectors. Define

$$B_i = B \cap E(\lambda_i, T).$$

Then

$$V = \text{span}(B) = \sum_{i=1}^k \text{span}(B_i) = \sum_{i=1}^k E(\lambda_i, T).$$

Thus

$$V = E(\lambda_1, T) + \cdots + E(\lambda_k, T).$$

Moreover,

$$\sum_{i=1}^k \dim E(\lambda_i, T) = \sum_{i=1}^k d_i = n = \dim V,$$

so the sum is direct by the previous theorem. 

Corollary 2.2.1 Multiplicities and diagonalizability

By the theorem above, it's clear that for a linear operator $T : V \rightarrow V$,

$$\begin{aligned} \sum_i m_g(\lambda_i) &= \sum_i m_a(\lambda_i) = n \\ \Leftrightarrow T &\text{ is diagonalizable} \\ \Leftrightarrow V &= E(\lambda_1) \oplus \cdots \oplus E(\lambda_k) \end{aligned}$$

2.3 Invariant Subspaces

Note:-

Related homework problems: see 5 for showing that an eigenvector sum lying in an invariant subspace forces each summand into the subspace, 6 for decomposing invariant subspaces into eigenspace pieces, 9 for invariant eigenspaces in simultaneous diagonalization, and 23 for invariant subspaces of normal operators.

Note:-

Related tutorial problems: see 9.3.3 for invariant subspaces when there are no real eigenvalues, and 9.8.1 for a projection-operator characterization of invariant subspaces in inner product spaces.

Definition 2.3.1: Invariant Subspace

Let $T : V \rightarrow V$ be a linear operator and let $W \leq V$. We say W is *T-invariant* if

$$T(W) \subseteq W.$$

From this definition, we can intuitively see that an invariant subspace is a **self-contained space** where applying the linear transformation T doesn't bring you out of W .

It's useful in the fact that it allows us to divide V into different W , and understand T on W rather than understanding T on all of V . Which is the linear algebra way to **divide and conquer**.

Theorem 2.3.1

$\ker(T)$ and $\text{im}(T)$ are T -invariant subspaces.

Proof: Let $v \in \ker(T)$. Then $T(v) = 0$, so

$$T(T(v)) = T(0) = 0,$$

hence $T(v) \in \ker(T)$.

Let $v \in \text{im}(T)$. Then $v = T(u)$ for some u , and

$$T(v) = T(T(u)) \in \text{im}(T).$$



Theorem 2.3.2 Eigenvectors and eigenspaces are invariant subspaces

Let v be an eigenvector, then

$$T(v) = \lambda v,$$

then $\text{span}\{v\}$ is a 1-d invariant subspace. Eigenspaces $E(\lambda)$ are also invariant subspaces. Direct sums of eigenspaces are invariant subspaces.

Proof: 1. **$\text{span}\{v\}$ is T -invariant.** Let $w \in \text{span}\{v\}$. Then $w = cv$ for some scalar c . Using linearity and $Tv = \lambda v$,

$$Tw = T(cv) = cTv = c\lambda v = \lambda(cv) = \lambda w \in \text{span}\{v\}.$$

Hence $T(\text{span}\{v\}) \subseteq \text{span}\{v\}$.

2. **$E(\lambda)$ is T -invariant.** Recall

$$E(\lambda) = \{x \in V : Tx = \lambda x\}.$$

Let $x \in E(\lambda)$. Then $Tx = \lambda x$. Apply T to both sides:

$$T(Tx) = T(\lambda x) = \lambda Tx.$$

Thus $Tx \in E(\lambda)$, and so $T(E(\lambda)) \subseteq E(\lambda)$.

3. **A direct sum of eigenspaces is T -invariant.** Let $\lambda_1, \dots, \lambda_k$ be eigenvalues and set

$$W = E(\lambda_1) \oplus \dots \oplus E(\lambda_k).$$

Take $w \in W$. Then there exist (unique) $w_i \in E(\lambda_i)$ such that

$$w = w_1 + \dots + w_k.$$

By linearity,

$$Tw = T(w_1 + \dots + w_k) = Tw_1 + \dots + Tw_k.$$

Since each $E(\lambda_i)$ is T -invariant, $Tw_i \in E(\lambda_i)$ for all i . Hence $Tw \in E(\lambda_1) + \dots + E(\lambda_k) = W$, so W is T -invariant.



Note:-

Eigenvalues are really about finding invariant subspaces of minimal size

Theorem 2.3.3

If $V_1, \dots, V_r$ are T -invariant subspaces, then so are

$$V_1 \cap \dots \cap V_r \quad \text{and} \quad V_1 + \dots + V_r.$$

Proof: Assume each V_i is T -invariant, i.e. $T(V_i) \subseteq V_i$.

(1) **Intersection.** Let $v \in V_1 \cap \dots \cap V_r$. Then $v \in V_i$ for every i . By T -invariance of V_i , we have $Tv \in V_i$ for every i . Hence $Tv \in V_1 \cap \dots \cap V_r$. Therefore,

$$T(V_1 \cap \dots \cap V_r) \subseteq V_1 \cap \dots \cap V_r.$$

(2) **Sum.** Let $v \in V_1 + \dots + V_r$. Then there exist $v_i \in V_i$ such that

$$v = v_1 + \dots + v_r.$$

Applying T and using linearity,

$$Tv = T(v_1 + \dots + v_r) = Tv_1 + \dots + Tv_r.$$

Since each V_i is T -invariant, $Tv_i \in V_i$ for all i . Thus $Tv \in V_1 + \dots + V_r$. Therefore,

$$T(V_1 + \dots + V_r) \subseteq V_1 + \dots + V_r.$$

This proves both subspaces are T -invariant. ☺

Lemma 2.3.1

Let $S, T : V \rightarrow V$ be linear operators such that $ST = TS$. Then $\ker(S)$ is T -invariant.

Proof: Let $v \in \ker(S)$. Then $Sv = 0$. Using $ST = TS$, we compute

$$S(Tv) = T(Sv) = T(0) = 0.$$

Hence $Tv \in \ker(S)$, so $\ker(S)$ is T -invariant. ☺

Corollary 2.3.1

For any polynomial f , the subspace $\ker(f(T))$ is T -invariant.

Proof: Let $S = f(T)$. Since S is a polynomial in T , we have $ST = TS$. By the previous lemma, $\ker(S) = \ker(f(T))$ is T -invariant. ☺

Lemma 2.3.2

Let $T : V \rightarrow V$ be a linear operator and let $W \leq V$ be a T -invariant subspace. Then the characteristic polynomial of $T|_W$ divides the characteristic polynomial of T .

Proof: Let $\dim W = k$ and choose a basis

$$B_0 = (v_1, \dots, v_k)$$

of W . Extend B_0 to a basis of V ,

$$B = (v_1, \dots, v_k, v_{k+1}, \dots, v_n).$$

Since W is T -invariant, we have

$$Tv_i \in W = \text{span}(v_1, \dots, v_k) \quad \text{for all } 1 \leq i \leq k.$$

Hence, with respect to the basis B , the matrix of T has the block form

$$[T]_B = \begin{pmatrix} X & Y \\ 0 & Z \end{pmatrix},$$

where

$$X = [T|_W]_{B_0}.$$

Now compute the characteristic polynomial:

$$\begin{aligned} p_T(x) &= \det(xI - [T]_B) \\ &= \det \begin{pmatrix} xI_k - X & -Y \\ 0 & xI_{n-k} - Z \end{pmatrix}. \end{aligned}$$

Since this matrix is block upper triangular, its determinant is the product of the determinants of the diagonal blocks:

$$p_T(x) = \det(xI_k - X) \det(xI_{n-k} - Z).$$

But

$$\det(xI_k - X) = p_{T|_W}(x),$$

the characteristic polynomial of the restriction $T|_W$. Therefore,

$$p_{T|_W}(x) \mid p_T(x).$$

☺

Lemma 2.3.3

Let $T : V \rightarrow V$ and suppose that

$$V = V_1 \oplus \dots \oplus V_r$$

for T -invariant subspaces $V_1, \dots, V_r$. Then

$$p_T(x) = \prod_{i=1}^r p_{T|_{V_i}}(x).$$

Proof: Since each V_i is T -invariant, the restriction

$$T_i := T|_{V_i} : V_i \rightarrow V_i$$

is a linear operator.

Choose an ordered basis B_i for each V_i . Then the union

$$B = B_1 \cup \cdots \cup B_r$$

is an ordered basis of V , since the sum is direct.

With respect to this basis, the matrix of T has block diagonal form

$$[T]_B = \begin{pmatrix} [T_1]_{B_1} & 0 & \cdots & 0 \\ 0 & [T_2]_{B_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & [T_r]_{B_r} \end{pmatrix}.$$

Therefore,

$$p_T(x) = \det(xI - [T]_B) = \prod_{i=1}^r \det(xI - [T_i]_{B_i}) = \prod_{i=1}^r p_{T|_{V_i}}(x),$$

where we used the fact that the determinant of a block diagonal matrix is the product of the determinants of its diagonal blocks.

This completes the proof. ☺

2.4 Polynomials of Operators

Note:-

Related homework problems: see [7](#) for writing A^{-1} as a polynomial in A , [8](#) for Bezout identities applied to operators, and [15](#) for powers of a Jordan block.

Note:-

Related tutorial problems: see [9.4.1](#) for a Cayley-Hamilton application to powers of a matrix, [9.3.1](#) for minimal polynomials of diagonalizable maps, and [9.4.3](#) for an image-kernel identity when the minimal polynomial has coprime factors.

Note:-

Related additional exercises: see [3](#) for a Cayley-Hamilton reduction of polynomials in a matrix, [9](#) for minimal polynomials on invariant subspaces and coprime polynomials, and [18](#) for nilpotence detected through a polynomial action.

Definition 2.4.1: Polynomial of a Linear Operator

Let $T : V \rightarrow V$ be a linear operator and let

$$f(x) = \sum_{i=0}^d a_i x^i$$

be a polynomial. We define

$$f(T) = \sum_{i=0}^d a_i T^i,$$

where $T^i = \underbrace{T \circ \cdots \circ T}_{i \text{ times}}$ and $T^0 = I$.

Definition 2.4.2: Polynomial of a Matrix

Let $T : V \rightarrow V$ be a linear operator and let

$$f(x) = \sum_{i=0}^d a_i x^i$$

be a polynomial. We define

$$f(A) = \sum_{i=0}^d a_i A^i,$$

Lemma 2.4.1

Let f, g be polynomials. Then

$$(fg)(T) = f(T) g(T).$$

Proof: Write

$$f(x) = \sum_{i=0}^d a_i x^i \quad \text{and} \quad g(x) = \sum_{j=0}^m b_j x^j.$$

Then the product polynomial satisfies

$$(fg)(x) = \sum_{k=0}^{d+m} \left(\sum_{i+j=k} a_i b_j \right) x^k.$$

Evaluating at the operator T gives

$$(fg)(T) = \sum_{k=0}^{d+m} \left(\sum_{i+j=k} a_i b_j \right) T^k.$$

On the other hand, by linearity of operator addition and scalar multiplication, we have

$$f(T)g(T) = \left(\sum_{i=0}^d a_i T^i \right) \left(\sum_{j=0}^m b_j T^j \right) = \sum_{i=0}^d \sum_{j=0}^m a_i b_j T^i T^j.$$

Since composition of linear operators is associative, $T^i T^j = T^{i+j}$, and thus

$$f(T)g(T) = \sum_{i=0}^d \sum_{j=0}^m a_i b_j T^{i+j} = \sum_{k=0}^{d+m} \left(\sum_{i+j=k} a_i b_j \right) T^k.$$

Comparing with the expression for $(fg)(T)$, we conclude that

$$(fg)(T) = f(T)g(T).$$

☺

Corollary 2.4.1

For any polynomials f, g ,

$$f(T)g(T) = g(T)f(T).$$

Lemma 2.4.2

Let $v \neq 0$ be an eigenvector of T with eigenvalue λ . Then

$$f(T)v = f(\lambda)v.$$

Proof: We compute

$$f(T)v = \left(\sum_{i=0}^d a_i T^i \right) v = \sum_{i=0}^d a_i T^i v.$$

Since $Tv = \lambda v$, we have $T^i v = \lambda^i v$, hence

$$f(T)v = \sum_{i=0}^d a_i \lambda^i v = \left(\sum_{i=0}^d a_i \lambda^i \right) v = f(\lambda)v.$$

☺

2.5 Block Matrices

Definition 2.5.1: Block Matrix

Let $A \in \mathbb{F}^{n \times n}$. A *block matrix* is a partition of A into submatrices

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1k} \\ A_{21} & A_{22} & \cdots & A_{2k} \\ \vdots & \vdots & & \vdots \\ A_{\ell 1} & A_{\ell 2} & \cdots & A_{\ell k} \end{pmatrix},$$

where each A_{ij} is a matrix.

Definition 2.5.2: Compatible Block Sizes

Assume

$$A_{ij} \in \mathbb{F}^{h_i \times r_j}, \quad B_{jk} \in \mathbb{F}^{r_j \times s_k}.$$

Then the block products $A_{ij}B_{jk}$ are well-defined.

Lemma 2.5.1 Block Matrix Multiplication

Let

$$A = \begin{pmatrix} A_{11} & \cdots & A_{1k} \\ \vdots & & \vdots \\ A_{\ell 1} & \cdots & A_{\ell k} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & \cdots & B_{1m} \\ \vdots & & \vdots \\ B_{k1} & \cdots & B_{km} \end{pmatrix}.$$

Then the product $C = AB$ is the block matrix

$$C = \begin{pmatrix} C_{11} & \cdots & C_{1m} \\ \vdots & & \vdots \\ C_{\ell 1} & \cdots & C_{\ell m} \end{pmatrix},$$

where

$$C_{ij} = \sum_{t=1}^k A_{it} B_{tj}.$$

Proof: This follows directly from the definition of matrix multiplication: the (i, j) -block of AB is obtained by multiplying the i -th block row of A with the j -th block column of B . $\odot$

Definition 2.5.3: Square Block Matrices

If each A_{ii} is a square matrix and all blocks are compatible, then A is called a *square block matrix*. In this case, all blocks C_{ij} in the product are also well-defined.

$$A = \begin{array}{|c|c|} \hline A_{11} & A_{12} \\ \hline A_{21} & A_{22} \\ \hline \end{array} \quad B = \begin{array}{|c|c|} \hline B_{11} & B_{12} \\ \hline B_{21} & B_{22} \\ \hline \end{array} \quad AB = \begin{array}{|c|c|} \hline C_{11} & C_{12} \\ \hline C_{21} & C_{22} \\ \hline \end{array}$$

Lemma 2.5.2 Block Matrix Multiplication (Entrywise Proof)

Let

$$A = \begin{pmatrix} A_{11} & \cdots & A_{1k} \\ \vdots & & \vdots \\ A_{\ell 1} & \cdots & A_{\ell k} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & \cdots & B_{1m} \\ \vdots & & \vdots \\ B_{k1} & \cdots & B_{km} \end{pmatrix},$$

with compatible block sizes. Then the product $C = AB$ is the block matrix

$$C = \begin{pmatrix} C_{11} & \cdots & C_{1m} \\ \vdots & & \vdots \\ C_{\ell 1} & \cdots & C_{\ell m} \end{pmatrix}, \quad C_{ij} = \sum_{t=1}^k A_{it} B_{tj}.$$

Proof: Let (p, q) be an entry inside the block C_{ij} . By definition of block multiplication,

$$(C_{ij})_{pq} = \sum_{r=1}^{h_i} \sum_{t=1}^k (A_{it})_{pr} (B_{tj})_{rq}.$$

Rearranging the sums gives

$$(C_{ij})_{pq} = \sum_{t=1}^k \left(\sum_{r=1}^{h_i} (A_{it})_{pr} (B_{tj})_{rq} \right) = \sum_{t=1}^k (A_{it} B_{tj})_{pq}.$$

Since this holds for all (p, q) , we conclude

$$C_{ij} = \sum_{t=1}^k A_{it} B_{tj}.$$

☺

2.6 Determinant of Block Upper Triangular Matrices

Lemma 2.6.1 Determinant of a Block Upper Triangular Matrix

Let

$$A = \begin{pmatrix} A_1 & * & \cdots & * \\ 0 & A_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_k \end{pmatrix},$$

where each A_i is an $n_i \times n_i$ square matrix. Then

$$\det(A) = \prod_{i=1}^k \det(A_i).$$

Proof: We prove the statement by induction on k .

Base case ($k = 1$): Then $A = A_1$, and the result is trivial.

Induction hypothesis: Assume the result holds for block upper triangular matrices with $k - 1$ diagonal blocks.

Inductive step: Write

$$A = \begin{pmatrix} A_1 & B \\ 0 & A' \end{pmatrix},$$

where

$$A' = \begin{pmatrix} A_2 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_k \end{pmatrix}.$$

Then A' is block upper triangular with $k - 1$ diagonal blocks.

By properties of determinants for block triangular matrices,

$$\det(A) = \det(A_1) \det(A').$$

By the induction hypothesis,

$$\det(A') = \prod_{i=2}^k \det(A_i).$$

Therefore,

$$\det(A) = \det(A_1) \prod_{i=2}^k \det(A_i) = \prod_{i=1}^k \det(A_i).$$



Chapter 3

Week 3

3.1 Cyclic Subspaces

Note:-

Related homework problem: see 12 for constructing a cyclic vector across a direct sum with pairwise coprime minimal polynomials.

Note:-

Related tutorial problem: see 9.3.2 for cyclic vectors of diagonalizable maps, including the Vandermonde argument when the eigenvalues are distinct.

Note:-

Related additional exercise: see 17 for cyclic vectors after decomposing into invariant direct summands.

An important example of invariant subspaces are *cyclic subspaces*, defined as follows.

Definition 3.1.1: Cyclic Subspace

Let $T : V \rightarrow V$ and let $v \in V \setminus \{0\}$. The T -cyclic subspace generated by v is

$$\text{span}\{v, Tv, T^2v, T^3v, \dots\} = \text{span}\{T^i v : i \geq 0\}.$$

A vector $v \in V$ is called *cyclic for T* if the T -cyclic subspace generated by v equals V .

Proposition 3.1.1

The T -cyclic subspace generated by v is T -invariant. Moreover, it is the smallest T -invariant subspace containing v .

Proof: Let

$$W = \text{span}\{T^i v : i \geq 0\}.$$

First, we show that W is T -invariant. Let $x \in W$. Then

$$x = \sum_{i \in I} \alpha_i T^i v$$

for some finite set $I \subset \mathbb{N}$. Applying T , we obtain

$$Tx = \sum_{i \in I} \alpha_i T^{i+1} v \in W,$$

as required.

Now let U be any T -invariant subspace containing v . Then $T^i v \in U$ for all $i \geq 0$ (by induction, using T -invariance), so

$$W = \text{span}\{T^i v : i \geq 0\} \subset U.$$

Hence W is the smallest T -invariant subspace containing v . ⊙

Next, we show that the T -cyclic space generated by $v \neq 0$ has a basis of a specific natural form.

Lemma 3.1.1

Let W be the T -cyclic subspace generated by v , and let $d := \dim(W)$. Then

$$\{v, Tv, T^2v, \dots, T^{d-1}v\}$$

is a basis for W .

Proof: Let k be maximal such that

$$S := \{v, Tv, T^2v, \dots, T^{k-1}v\}$$

is linearly independent. We will show that $k = d$, which implies the lemma.

Let $U = \text{span}(S)$. Clearly $U \subset W$. It suffices to show that $T^\ell v \in U$ for all $\ell \geq k$.

By maximality of k , there exist scalars $\alpha_0, \dots, \alpha_k$, not all zero, such that

$$\sum_{i=0}^k \alpha_i T^i v = 0,$$

with $\alpha_k \neq 0$. Thus,

$$T^k v \in \text{span}\{v, Tv, \dots, T^{k-1}v\} = U.$$

Now let $\ell > k$, and suppose inductively that $T^{\ell-1}v \in U$. Then $T^{\ell-1}v$ is a linear combination of $v, Tv, \dots, T^{k-1}v$. Applying T , we see that $T^\ell v$ is a linear combination of $Tv, T^2v, \dots, T^k v$, all of which lie in U . Hence $T^\ell v \in U$.

This proves that $U = W$, and therefore

$$k = \dim(U) = \dim(W) = d,$$

and

$$\{v, Tv, T^2v, \dots, T^{d-1}v\}$$

is a basis for W . ☺

An equivalent formulation of Lemma 3.13 is as follows: Let $v \in V \setminus \{0\}$ and let k be maximal such that

$$\{v, Tv, \dots, T^{k-1}v\}$$

is linearly independent. Let W be the cyclic subspace generated by v . Then $\dim(W) = k$ and this set is a basis for W .

Next, we consider the characteristic polynomial of a linear map admitting a cyclic vector.

Lemma 3.1.2

Let $T : V \rightarrow V$, $\dim(V) = n$. Suppose that v is a cyclic vector for T . Let $a_0, \dots, a_{n-1}$ be the unique scalars satisfying

$$T^n v + \sum_{i=0}^{n-1} a_i T^i v = 0.$$

Then

$$p_T(x) = x^n + \sum_{i=0}^{n-1} a_i x^i.$$

Proof: Since v is cyclic, the set

$$B = (v, Tv, T^2v, \dots, T^{n-1}v)$$

is a basis for V .

Denote the basis elements by

$$v_i := T^i v, \quad i = 0, \dots, n-1.$$

Then

$$Tv_0 = v_1, \quad Tv_1 = v_2, \quad \dots, \quad Tv_{n-2} = v_{n-1},$$

and

$$Tv_{n-1} = T^n v = - \sum_{i=0}^{n-1} a_i v_i.$$

Thus, with respect to the ordered basis B , the matrix of T is

$$[T]_B = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}.$$

We now compute the characteristic polynomial:

$$p_T(x) = \det(xI - [T]_B) = \det \begin{pmatrix} x & 0 & \cdots & 0 & a_0 \\ -1 & x & \cdots & 0 & a_1 \\ 0 & -1 & \ddots & \vdots & a_2 \\ \vdots & \vdots & \ddots & x & a_{n-2} \\ 0 & 0 & \cdots & -1 & x + a_{n-1} \end{pmatrix}.$$

We prove by induction on n that this determinant equals

$$x^n + \sum_{i=0}^{n-1} a_i x^i.$$

The base case $n = 1$ is immediate. For the induction step, expand along the first row to obtain

$$p_T(x) = x \det \begin{pmatrix} x & \cdots & 0 & a_1 \\ -1 & \ddots & \vdots & a_2 \\ \vdots & \ddots & x & a_{n-2} \\ 0 & \cdots & -1 & x + a_{n-1} \end{pmatrix} + (-1)^{n+1} a_0 \det \begin{pmatrix} -1 & x & \cdots & 0 \\ 0 & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & x \\ 0 & \cdots & 0 & -1 \end{pmatrix}.$$

By the induction hypothesis, the first determinant equals

$$x^{n-1} + \sum_{i=0}^{n-2} a_{i+1} x^i,$$

and the second determinant equals $(-1)^{n-1}$. Therefore,

$$p_T(x) = x \left(x^{n-1} + \sum_{i=0}^{n-2} a_{i+1} x^i \right) + (-1)^{n+1} (-1)^{n-1} a_0 = x^n + \sum_{i=0}^{n-1} a_i x^i,$$

as required. ☺

Corollary 3.1.1

For every monic polynomial $f(x)$ of degree n , there exists an $n \times n$ matrix A such that

$$p_A(x) = f(x).$$

Proof: Write

$$f(x) = x^n + \sum_{i=0}^{n-1} a_i x^i,$$

and define

$$A := \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}.$$

☺

Note:-

The matrix above is called the *companion matrix* of the polynomial $f(x)$.

Finally, we arrive at the following central theorem. Recall that for a polynomial

$$f(x) = \sum_{i=0}^n a_i x^i$$

and a linear map T , we define

$$f(T) := \sum_{i=0}^n a_i T^i.$$

Theorem 3.1.1

(*Cayley–Hamilton Theorem*) For every linear map $T : V \rightarrow V$,

$$p_T(T) = 0.$$

Similarly, for every matrix A , it holds that

$$p_A(A) = 0.$$

Proof: We need to show that $p_T(T)v = 0$ for all $v \in V$. If $v = 0$, this is clear. Let now $v \in V \setminus \{0\}$.

Let W be the T -cyclic subspace generated by v , and let $d := \dim(W)$. Let $a_0, \dots, a_{d-1}$ be the unique scalars satisfying

$$T^d v + \sum_{i=0}^{d-1} a_i T^i v = 0.$$

By Lemma 3.14,

$$p_{T|_W}(x) = x^d + \sum_{i=0}^{d-1} a_i x^i.$$

Now plug T into $p_{T|_W}(x)$ and apply it to v :

$$p_{T|_W}(T)v = \left(T^d + \sum_{i=0}^{d-1} a_i T^i \right) v = T^d v + \sum_{i=0}^{d-1} a_i T^i v = 0.$$

On the other hand, by Lemma 3.9, $p_{T|_W}(x)$ divides $p_T(x)$. Thus there exists a polynomial $q(x)$ such that

$$p_T(x) = q(x) p_{T|_W}(x).$$

Therefore,

$$p_T(T) v = q(T) p_{T|_W}(T) v = q(T) 0 = 0,$$

as required. ☺

Proposition 3.1.2

Let $T : V \rightarrow V$, $\dim(V) = n$, and suppose that T admits a cyclic vector v . Then

$$p_T = q_T.$$

Proof: Recall that v being cyclic for T means that

$$\{v, Tv, T^2v, \dots, T^{n-1}v\}$$

is a basis for V .

To prove the proposition, it is enough to show that $\deg(q_T) = n$. Indeed, we also know that $\deg(p_T) = n$, that q_T divides p_T , and that both p_T, q_T are monic. This implies that $p_T = q_T$.

Let us show that for every nonzero polynomial f with $\deg(f) \leq n - 1$, we have $f(T) \neq 0$. Write

$$f(x) = \sum_{i=0}^{n-1} a_i x^i$$

(some leading coefficients may be zero). Suppose by contradiction that $f(T) = 0$. Then

$$0 = f(T) v = \sum_{i=0}^{n-1} a_i T^i v.$$

Since $f \neq 0$, not all a_i are zero, which implies that

$$\{v, Tv, T^2v, \dots, T^{n-1}v\}$$

is linearly dependent — a contradiction. ☺

Note:-

The converse of the above proposition is also true: if $p_T = q_T$, then T admits a cyclic vector.

3.2 The Primary Decomposition Theorem

Note:-

Related homework problems: see 8 for the Bezout argument behind coprime factors, and 10 for the kernel-image splitting that underlies nilpotent/invertible decompositions.

Note:-

Related tutorial problems: see 9.4.2 for the nilpotent-invertible decomposition from primary decomposition, and 9.4.3 for the identity $\text{Im}(g(T)) = \ker(f(T))$ when $q_T = fg$ with $\gcd(f, g) = 1$.

By the Cayley–Hamilton theorem, $p_T(T) = 0$ for every linear map $T : V \rightarrow V$. In particular, there exists a nonzero polynomial f such that $f(T) = 0$. It is therefore natural to consider the polynomial of smallest possible degree with this property.

Definition 3.2.1: Minimal Polynomial

The *minimal polynomial* of a linear map $T : V \rightarrow V$ is the unique monic nonzero polynomial $q_T(x)$ of minimal degree such that

$$q_T(T) = 0.$$

Note:-

We can similarly define the minimal polynomial of a matrix $A \in \mathbb{F}^{n \times n}$ as the unique monic nonzero polynomial q_A of minimal degree such that $q_A(A) = 0$. For every linear map $T : V \rightarrow V$ and ordered basis B of V , the operators T and $[T]_B$ have the same minimal polynomial. In particular, similar matrices have the same minimal polynomial.

To justify uniqueness, we prove the following result.

Proposition 3.2.1

Let f be a polynomial such that $f(T) = 0$. Then q_T divides f .

Proof. By polynomial division with remainder, there exist polynomials g, h such that

$$f = gq_T + h,$$

where $\deg(h) < \deg(q_T)$. Evaluating at T , we obtain

$$h(T) = f(T) - g(T)q_T(T) = 0.$$

If $h \neq 0$, this contradicts the minimality of q_T . Hence $h = 0$, so $f = gq_T$, and q_T divides f . $\odot$

Corollary 3.2.1

The minimal polynomial q_T divides the characteristic polynomial p_T .

Corollary 3.2.2

If f is a monic polynomial of degree $\deg(q_T)$ such that $f(T) = 0$, then $f = q_T$.

Proof. By the previous proposition, q_T divides f . Since both polynomials are monic and have the same degree, they must be equal. $\odot$

Thus, the minimal polynomial is uniquely determined.

Next, we identify the roots of the minimal polynomial.

Proposition 3.2.2

$q_T(\lambda) = 0$ if and only if λ is an eigenvalue of T .

Proof. If λ is a root of q_T , then since q_T divides p_T , it is also a root of p_T , hence an eigenvalue of T .

Conversely, if λ is an eigenvalue of T and $v \neq 0$ is an eigenvector, then

$$0 = q_T(T)v = q_T(\lambda)v.$$

Since $v \neq 0$, we must have $q_T(\lambda) = 0$. ☺

We now show that if T admits a cyclic vector, then the minimal and characteristic polynomials coincide.

Proposition 3.2.3

Let $T : V \rightarrow V$ with $\dim(V) = n$, and suppose that T admits a cyclic vector v . Then

$$p_T = q_T.$$

Proof. Since v is cyclic, the set

$$\{v, Tv, T^2v, \dots, T^{n-1}v\}$$

is a basis for V . It suffices to show that $\deg(q_T) = n$.

Let f be a nonzero polynomial with $\deg(f) \leq n - 1$. Write

$$f(x) = \sum_{i=0}^{n-1} a_i x^i.$$

Suppose for contradiction that $f(T) = 0$. Then

$$0 = f(T)v = \sum_{i=0}^{n-1} a_i T^i v.$$

Since not all a_i are zero, this yields a nontrivial linear dependence among $\{v, Tv, \dots, T^{n-1}v\}$, contradicting cyclicity. Thus $f(T) \neq 0$, so $\deg(q_T) = n$. As p_T is monic of degree n and q_T divides p_T , we conclude $p_T = q_T$. ☺

Note:-

The converse is also true: if $p_T = q_T$, then T admits a cyclic vector.

We now prove the following fundamental theorem.

Theorem 3.2.1

(*Primary Decomposition Theorem*) Let $T : V \rightarrow V$ and let q_T be the minimal polynomial

of T . Suppose that

$$q_T = \prod_{i=1}^r f_i,$$

where $f_1, \dots, f_r$ are pairwise relatively prime (and monic). Define

$$V_i := \ker(f_i(T)).$$

Then

$$V = V_1 \oplus \dots \oplus V_r,$$

and

$$q_{T|_{V_i}} = f_i.$$

In words, factoring the minimal polynomial into relatively prime factors corresponds to decomposing V as a direct sum of T -invariant subspaces, where each V_i is the kernel of $f_i(T)$. Note also that each V_i is T -invariant (by Corollary 3.8).

Before proving the theorem, we establish the following lemma.

Lemma 3.2.1

Let f_1, f_2 be relatively prime polynomials. Let $T : V \rightarrow V$. Then

$$\ker(f_1(T) f_2(T)) = \ker(f_1(T)) \oplus \ker(f_2(T)).$$

Proof: For convenience, put

$$W_1 := \ker(f_1(T)), \quad W_2 := \ker(f_2(T)), \quad W := \ker(f_1(T) f_2(T)).$$

First, observe that $W_1 + W_2 \subset W$. Indeed, let $v \in W_1$. Then $f_1(T)v = 0$, hence

$$f_1(T)f_2(T)v = f_2(T)0 = 0,$$

so $v \in W$. Similarly, $W_2 \subset W$, hence $W_1 + W_2 \subset W$.

For the reverse inclusion, since f_1, f_2 are relatively prime, there exist polynomials a_1, a_2 such that

$$a_1 f_1 + a_2 f_2 = 1. \tag{18}$$

Let $v \in W$, and define

$$v_1 := a_2(T) f_2(T) v, \quad v_2 := a_1(T) f_1(T) v.$$

Then

$$f_1(T)v_1 = a_2(T) f_1(T) f_2(T) v = a_2(T) 0 = 0,$$

and

$$f_2(T)v_2 = a_1(T) f_1(T) f_2(T) v = 0.$$

Thus $v_i \in W_i$ for $i = 1, 2$.

Moreover,

$$v_1 + v_2 = (a_1(T)f_1(T) + a_2(T)f_2(T))v = Iv = v,$$

where we used (18). Hence $v \in W_1 + W_2$, so $W \subset W_1 + W_2$.

We conclude that $W = W_1 + W_2$. It remains to show that the sum is direct. If $v \in W_1 \cap W_2$, then $f_i(T)v = 0$ for $i = 1, 2$. Using (18),

$$v = Iv = a_1(T)f_1(T)v + a_2(T)f_2(T)v = 0,$$

so $W_1 \cap W_2 = \{0\}$. ⊕

We now generalize Lemma 4.10 to an arbitrary number of polynomials.

Lemma 3.2.2

Let $f_1, \dots, f_r$ be pairwise relatively prime polynomials and put $f = \prod_{i=1}^r f_i$. Let $T : V \rightarrow V$. Then

$$\ker(f(T)) = \ker(f_1(T)) \oplus \dots \oplus \ker(f_r(T)).$$

Proof: We proceed by induction on r .

Base case. If $r = 2$, the statement is exactly Lemma 3.2.1, so there is nothing to prove.

Induction step. Assume $r \geq 3$ and that the statement holds for any product of $r - 1$ pairwise relatively prime polynomials.

Let

$$f = \prod_{i=1}^r f_i,$$

where $f_1, \dots, f_r$ are pairwise relatively prime. Define

$$W := \ker(f(T)), \quad W_i := \ker(f_i(T)) \quad (1 \leq i \leq r).$$

Now set

$$g := f_1, \quad h := \frac{f}{f_1} = \prod_{i=2}^r f_i.$$

Since the polynomials $f_1, \dots, f_r$ are pairwise relatively prime, it follows that g and h are relatively prime and that

$$f = gh.$$

By Lemma 3.2.1 applied to g and h , we obtain

$$W = \ker(g(T)) \oplus \ker(h(T)).$$

That is,

$$W = W_1 \oplus U, \tag{19}$$

where we define

$$U := \ker(h(T)).$$

Note that U is T -invariant, so the restriction

$$S := T|_U : U \rightarrow U$$

is a well-defined linear operator.

Since

$$h = \prod_{i=2}^r f_i$$

is a product of $r-1$ pairwise relatively prime polynomials, we may apply the induction hypothesis to the operator S . This yields

$$\ker(h(S)) = \ker(f_2(S)) \oplus \cdots \oplus \ker(f_r(S)).$$

Observe that for any polynomial p , the operator $p(S)$ is simply the restriction of $p(T)$ to U . Consequently,

$$\ker(p(S)) = \ker(p(T)) \cap U.$$

Applying this to the previous equation gives

$$\ker(h(T)) \cap U = \bigoplus_{i=2}^r (\ker(f_i(T)) \cap U).$$

Since $U = \ker(h(T))$, the left-hand side equals U . We now show that for each $i \geq 2$,

$$\ker(f_i(T)) \cap U = \ker(f_i(T)).$$

Indeed, because f_i divides h , we may write $h = f_i q_i$ for some polynomial q_i . If $v \in \ker(f_i(T))$, then

$$f_i(T)v = 0 \implies h(T)v = q_i(T)f_i(T)v = 0,$$

so $v \in \ker(h(T)) = U$. This proves the claim.

Therefore,

$$U = W_2 \oplus \cdots \oplus W_r.$$

Finally, combining this decomposition with (19), we obtain

$$W = W_1 \oplus \cdots \oplus W_r,$$

which completes the induction step.

Hence,

$$\ker(f(T)) = \ker(f_1(T)) \oplus \cdots \oplus \ker(f_r(T))$$

for all $r \geq 2$, as required. ☺

We also need the following simple lemma.

Lemma 3.2.3

Let $T : V \rightarrow V$ and let $V_1, \dots, V_r$ be subspaces of V such that

$$V = V_1 + \dots + V_r.$$

Let $f_1, \dots, f_r$ be polynomials such that $f_i(T)|_{V_i} = 0$. Then, for $f = \prod_{i=1}^r f_i$, we have $f(T) = 0$.

Proof: We need to show that $f(T)v = 0$ for all $v \in V$. Since $V = V_1 + \dots + V_r$, there exist $v_i \in V_i$ such that $v = v_1 + \dots + v_r$. It suffices to show that $f(T)v_i = 0$.

Indeed,

$$f(T)v_i = \left(\prod_{j \neq i} f_j(T) \right) f_i(T)v_i = \left(\prod_{j \neq i} f_j(T) \right) 0 = 0,$$

using that $f_i(T)$ is zero on V_i . ☺

We are now ready to prove the Primary Decomposition Theorem.

Proof of Primary Decomposition Theorem: By Lemma (3.2.2) 3.2.2, we have

$$V = \ker(q_T(T)) = V_1 \oplus \dots \oplus V_r. \quad (20)$$

It remains to show that the minimal polynomial of $T|_{V_i}$ is f_i . By definition, $f_i(T)|_{V_i} = 0$, since $V_i = \ker(f_i(T))$.

Suppose, by contradiction, that there exists a polynomial g with $\deg(g) < \deg(f_i)$ and $g(T)|_{V_i} = 0$. Define

$$h := g \cdot \prod_{j \neq i} f_j.$$

By Lemma (3.2.3) 3.2.3, $h(T) = 0$. However,

$$\deg(q_T) = \sum_{i=1}^r \deg(f_i), \quad \deg(h) = \deg(g) + \sum_{j \neq i} \deg(f_j),$$

so $\deg(h) < \deg(q_T)$, contradicting the minimality of q_T . ☺

The Primary Decomposition Theorem has the following important corollary, which we will use later on.

Corollary 3.2.3

Let V be a finite-dimensional vector space over $\mathbb{F}$ and let $T : V \rightarrow V$. Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of T , and let t_i be the multiplicity of λ_i as a root of q_T . Set

$$V_i := \ker((T - \lambda_i I)^{t_i}), \quad i = 1, \dots, r.$$

Then:

1. $V = U \oplus V_1 \oplus \cdots \oplus V_r$, where U is a subspace such that $T|_U$ has no eigenvalues (over $\mathbb{F}$).
2. $U = \{0\}$ if q_T splits over $\mathbb{F}$.
3. The minimal polynomial of $T|_{V_i}$ is $(x - \lambda_i)^{t_i}$.

Proof: By Lemma 1.23, we can write any nonzero polynomial f as

$$f(x) = g(x) \cdot \prod_{i=1}^r (x - a_i)^{t_i},$$

where $a_1, \dots, a_r$ are the roots of f , t_i is the multiplicity of the root a_i , and g has no roots in $\mathbb{F}$.

We apply this to the minimal polynomial q_T of T . The roots of q_T are precisely the eigenvalues of T . Thus,

$$q_T(x) = g(x) \cdot \prod_{i=1}^r (x - \lambda_i)^{t_i}$$

for some polynomial g with no roots in $\mathbb{F}$.

Now set

$$f_i := (x - \lambda_i)^{t_i}, \quad 1 \leq i \leq r.$$

Note that $g, f_1, \dots, f_r$ are pairwise relatively prime (using that $\lambda_1, \dots, \lambda_r$ are distinct and that g has no roots in $\mathbb{F}$).

By Theorem 4.9 (applied to $g, f_1, \dots, f_r$), we obtain

$$V = U \oplus V_1 \oplus \cdots \oplus V_r,$$

where

$$U := \ker(g(T)).$$

Also, the minimal polynomial of $T|_{V_i}$ is $f_i = (x - \lambda_i)^{t_i}$, and the minimal polynomial of $T|_U$ is g . In particular, the minimal polynomial of $T|_U$ has no roots, which means that $T|_U$ has no eigenvalues (since every eigenvalue is a root of the minimal polynomial).

Finally, if q_T splits (namely, if $\deg(g) = 0$), then clearly $U = \{0\}$. In the other direction, suppose that $U = \{0\}$, so

$$V = V_1 \oplus \cdots \oplus V_r.$$

By Lemma 4.12, setting

$$f(x) := \prod_{i=1}^r (x - \lambda_i)^{t_i},$$

we have $f(T) = 0$. But $\deg(f) = \deg(q_T) - \deg(g)$, so by minimality of q_T , this is only possible if $\deg(g) = 0$, which means that q_T splits. $\odot$

Note:-

One could go a step further here and also show that the characteristic polynomial of $T|_{V_i}$ is

$$p_{T|_{V_i}}(x) = (x - \lambda_i)^{\dim(V_i)},$$

and that p_T splits if q_T splits. We will show this indirectly using the Jordan canonical form.

Chapter 4

Week 4

4.1 Jordan Canonical Form

Note:-

Related homework problems: see [11](#), [13](#), [14](#), and [15](#) for official Jordan-form and Jordan-block computations.

Note:-

Related tutorial problems: see [9.5.1](#) for a Jordan-form computation, [9.5.2](#) for the condition $q_T = p_T$ in terms of eigenspace dimensions, [9.5.3](#) for the decomposition $T = D + N$, and [9.6.1](#) for a nilpotent Jordan example.

Note:-

Related additional exercises: see [4](#), [6](#), [8](#), [12](#), and [16](#) for past-paper Jordan computations, and [5](#), [7](#) for counting possible Jordan forms from characteristic and minimal polynomials.

As we saw, not all linear maps are diagonalizable, even if the characteristic polynomial splits. Thus, diagonal matrices are too restrictive to represent all linear maps whose characteristic polynomial splits. Here we relax the notion of diagonalizability by allowing more general matrices.

Definition 4.1.1: Jordan Block

The *Jordan block* $J(\lambda, k)$ is the $k \times k$ matrix

$$J(\lambda, k) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}.$$

Namely, the diagonal entries of $J(\lambda, k)$ are equal to λ , the entries just above the diagonal are 1's, and all other entries are 0. Note that

$$J(\lambda, k) = \lambda I_k + J(0, k).$$

A block $J(\lambda, k)$ is called a *Jordan block of type λ* .

Definition 4.1.2: Jordan Canonical Form

A matrix A is in *Jordan canonical form* if it is block-diagonal and each diagonal block is a Jordan block. Namely,

$$A = \begin{pmatrix} J(\lambda_1, k_1) & 0 & \cdots & 0 \\ 0 & J(\lambda_2, k_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J(\lambda_r, k_r) \end{pmatrix},$$

for some scalars $\lambda_1, \dots, \lambda_r$ and integers $k_1, \dots, k_r \geq 1$.

Note that if $k_1 = \cdots = k_r = 1$, then A is diagonal.

The *Jordan canonical form* of a linear operator (or matrix) is a normal form that completely describes the operator up to change of basis, even when the operator is not diagonalizable.

Why Jordan Canonical Form?

Diagonalization answers the question:

“Can I choose a basis so that T acts independently on each coordinate?”

This is possible if and only if the operator has enough linearly independent eigenvectors.

The Jordan canonical form answers the refined question:

“If T is not diagonalizable, how badly does diagonalization fail, and what is the simplest structure available?”

Jordan canonical form decomposes the space into invariant subspaces called *Jordan blocks*. Each block corresponds to a single eigenvalue and measures how that eigenvalue fails to produce enough eigenvectors.

Examples

Example 4.1.1 (Example 1 (Diagonalizable case).)

Let

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

This matrix has two distinct eigenvalues and two linearly independent eigenvectors. Hence it has two Jordan blocks of size 1, and is diagonalizable.

Example 4.1.2 (Example 2 (Non-diagonalizable case).)

Let

$$A = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix}.$$

This matrix has a single eigenvalue $\lambda = 4$ but only one linearly independent eigenvector. Its Jordan canonical form consists of a single 2×2 Jordan block. Hence A is not diagonalizable.

These examples illustrate the principle:

The larger the Jordan blocks, the further the matrix is from being diagonalizable.

Generalized Eigenvectors

When there are not enough eigenvectors to form a basis, we introduce generalized eigenvectors.

Definition 4.1.3: Generalized Eigenvector

Let $A \in \mathbb{F}^{n \times n}$ and λ be an eigenvalue of A . A nonzero vector v is called a *generalized eigenvector* of eigenvalue λ and *degree* k if

$$(A - \lambda I)^k v = 0 \quad \text{but} \quad (A - \lambda I)^{k-1} v \neq 0.$$

Generalized eigenvectors are used to construct Jordan chains, which in turn form Jordan bases.

Example 4.1.3

For

$$A = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix},$$

there is only one eigenvector. To construct a matrix M such that

$$A = M J M^{-1},$$

we must use generalized eigenvectors as the columns of M .

The main theorem we will prove is the following.

Theorem 4.1.1 Jordan Canonical Form Theorem

Let V be a finite-dimensional vector space over a field $\mathbb{F}$, and let $T : V \rightarrow V$ be a linear map.

1. There exists an ordered basis B of V such that $[T]_B$ has Jordan canonical form (over $\mathbb{F}$) if and only if the characteristic polynomial p_T splits over $\mathbb{F}$.
2. If T admits a Jordan canonical form, then this form is unique up to permuting the Jordan blocks.

In particular, every linear map can be put in Jordan canonical form over $\mathbb{F} = \mathbb{C}$. Our starting point toward JCF theorem is the following consequence of the Primary Decomposition Theorem.

Theorem 4.1.2

Let V be a finite-dimensional vector space over $\mathbb{F}$ and let $T : V \rightarrow V$. Suppose that the minimal polynomial q_T splits over $\mathbb{F}$, and write

$$q_T(x) = \prod_{i=1}^r (x - \lambda_i)^{t_i}$$

(where $\lambda_1, \dots, \lambda_r$ are the distinct eigenvalues of T). Then, setting $V_i := \text{Ker}((T - \lambda_i I)^{t_i})$ for $i = 1, \dots, r$, we have

$$V = V_1 \oplus \dots \oplus V_r,$$

and the minimal polynomial of $T|_{V_i}$ is $(x - \lambda_i)^{t_i}$.

The subspace V_i is called the *generalized eigenspace* of the eigenvalue λ_i .

Note that $V_1, \dots, V_r$ are T -invariant. Thus, in order to put T in Jordan canonical form, it suffices to put each $T|_{V_i}$ in Jordan canonical form. Hence, we will focus on the maps $T_i := T|_{V_i}$. Note that T_i has minimal polynomial $(x - \lambda_i)^{t_i}$, so $(T_i - \lambda_i I)^{t_i} \cdot v = 0$ for all $v \in V_i$. In particular, T_i has only one eigenvalue, namely λ_i (and so, the Jordan canonical form of T_i will only have λ_i on the diagonal, i.e., it will only use Jordan blocks of the form $J(\lambda_i, k)$). To keep the notation simple, we will forget about $V_1, \dots, V_r$ for the moment, and just focus on a single linear map T satisfying $(T - \lambda I)^t = 0$ for some scalar λ and $t \geq 1$ (later we will apply this to $T := T_i$).

To proceed, let us now understand what it means for a linear map T and an ordered basis $B = (v_1, \dots, v_k)$ to satisfy that $[T]_B$ is the Jordan block $J(\lambda, k)$. Namely, suppose that

$$[T]_B = \begin{pmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \lambda & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & & 0 & \lambda & 1 \\ 0 & \dots & \dots & 0 & \lambda \end{pmatrix}$$

Then

$$Tv_1 = \lambda v_1$$

(so v_1 is an eigenvector of λ),

$$Tv_2 = v_1 + \lambda v_2$$

$$Tv_3 = v_2 + \lambda v_3$$

$$\vdots$$

$$Tv_k = v_{k-1} + \lambda v_k.$$

So

$$(T - \lambda I)v_i = v_{i-1}$$

for $2 \leq i \leq k$, and

$$(T - \lambda I)v_1 = 0.$$

This means that the map $T - \lambda I$ maps $v_k \mapsto v_{k-1} \mapsto \cdots \mapsto v_2 \mapsto v_1 \mapsto 0$. In particular, setting $v := v_k$, we see that

$$(v_1, \dots, v_k) = ((T - \lambda I)^{k-1}v, (T - \lambda I)^{k-2}v, \dots, (T - \lambda I)v, v) \quad (21)$$

is a cyclic sequence generated by $v = v_k$.

To make the discussion conceptually and notationally easier, let us denote $N := T - \lambda I$ (this corresponds to setting $\lambda = 0$). As we assumed that $(T - \lambda I)^t = 0$ for some $t \geq 1$, we have $N^t = 0$. This motivates the following definition:

Definition 4.1.4: Nilpotent Map

A linear map $N : V \rightarrow V$ is *nilpotent* if there exists $t \geq 1$ such that $N^t = 0$. The minimal such t is called the *degree of N* .

We will develop most of the theory for nilpotent maps. In the end, we will apply it to the nilpotent maps

$$N := (T - \lambda_i I)|_{V_i}$$

to prove the Jordan canonical form theorem. The following lemma will be needed later.

Example 4.1.4

Let

$$A = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix}.$$

Step 1: Find an eigenvector. We compute the characteristic polynomial:

$$\det(A - \lambda I) = \det \begin{pmatrix} 4 - \lambda & 1 \\ 0 & 4 - \lambda \end{pmatrix} = (4 - \lambda)^2.$$

Hence the only eigenvalue is $\lambda = 4$.

Then

$$A - 4I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

To find an eigenvector v_1 , solve

$$(A - 4I)v_1 = 0.$$

Writing $v_1 = (v_{11}, v_{12})^T$, we obtain

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{12} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which gives $v_{12} = 0$, while v_{11} is arbitrary. We may choose

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Thus v_1 is an eigenvector corresponding to $\lambda = 4$.

Step 2: Find a generalized eigenvector. We now find a generalized eigenvector v_2 such that

$$(A - 4I)v_2 = v_1,$$

so that

$$(A - 4I)^2 v_2 = (A - 4I)v_1 = 0.$$

Let $v_2 = (v_{21}, v_{22})^T$. Then

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_{21} \\ v_{22} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

This yields $v_{22} = 1$, while v_{21} is arbitrary. We may choose

$$v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Hence v_2 is a generalized eigenvector of degree 2.

Step 3: Jordan chain and JCF. The vectors v_1, v_2 form a Jordan chain of length 2. Let

$$M = [v_1 \ v_2] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Then

$$M^{-1}AM = A,$$

so A is already in Jordan canonical form.

Note:-

The length of a Jordan chain equals the size of the corresponding Jordan block.

Lemma 4.1.1

Let $N : V \rightarrow V$ be a nilpotent map of degree t , then

$$\ker(N^k - 1) \subsetneq \ker(N^k)$$

Proof: Take any $x \in \ker(N^{k-1})$, then

$$N^{k-1}x = 0$$

and $N(N^{k-1}x) = N^k x = N(0) = 0$, meaning $x \in \ker(N^k)$, so

$$\ker(N^k - 1) \subseteq \ker(N^k)$$

Next we'll show the inclusion is strict. Since N has degree t , we know $N^{t-1} \neq 0$. Now define a vector

$$u := N^{t-k}v$$

Then,

$$N^k u = N^k(N^{t-k}v) = N^t v$$

but $N^t = 0$, so $N^t v = 0$, therefore

$$N^k u = 0$$

meaning that $u \in \ker(N^k)$. It remains to prove that $u \notin \ker(N^{k-1})$.

$$N^{k-1} u = N^{k-1}(N^{t-k} v) = N^{t-1} v$$

choose v such such $N^{t-1} v \neq 0$. Hence

$$N^{k-1} u \neq 0$$

so

$$u \notin \ker(N^{k-1})$$

☺

Definition 4.1.5: Jordan Basis

Let $T : V \rightarrow V$ be a linear map. A basis B of V is called a *Jordan basis* for T if B can be written as a disjoint union of Jordan chains for T .

Equivalently, B is a Jordan basis if, when B is ordered appropriately, the matrix $[T]_B$ is in Jordan canonical form.

Our goal is to show that every nilpotent map admits a Jordan basis. If $B = (v_1, \dots, v_k)$ and $[N]_B = J(0, k)$, then

$$(v_1, \dots, v_k) = (N^{k-1}v, N^{k-2}v, \dots, Nv, v). \quad (22)$$

Namely,

$$Nv_1 = 0, \quad Nv_i = v_{i-1} \quad (i = 2, \dots, k).$$

Such a sequence is called a *Jordan chain*. Let $N : V \rightarrow V$ be nilpotent and B a basis of V .

- $[N]_B = J(0, k)$ if and only if B is a Jordan chain.
- $[N]_B$ is in Jordan canonical form if and only if B is a disjoint union of Jordan chains.

Thus, a Jordan basis is a disjoint union of Jordan chains.

Definition 4.1.6

Let $v \in V \setminus \{0\}$ and let $k \geq 1$ be minimal such that $N^k v = 0$. Then

$$C(v) := (N^{k-1}v, N^{k-2}v, \dots, Nv, v)$$

is the Jordan chain generated by v . The number $k = |C(v)|$ is called the *length* of the chain and the *height* of v .

Note:-

If v has height k , then the elements of $C(v)$ have heights $1, 2, \dots, k$, respectively.

- $v_k = v$ is the *final vector*.
- $v_1 = N^{k-1}v$ is the *initial vector*, and $v_1 \in \ker(N)$.

Lemma 4.1.2

Let $C_1, \dots, C_m$ be Jordan chains with initial elements $u_1, \dots, u_m$. If $u_1, \dots, u_m$ are linearly independent, then

$$\bigcup_{i=1}^m C_i$$

is linearly independent, where the union is taken as a multiset.

Proof: Suppose by contradiction that

$$\sum_{i=1}^m \sum_{v \in C_i} \alpha_v v = 0,$$

with not all coefficients α_v equal to zero. Let $k \geq 1$ be maximal such that there exists $v \in \bigcup_{i=1}^m C_i$ of height k with $\alpha_v \neq 0$.

Apply N^{k-1} to both sides. If v has height at most $k-1$, then $N^{k-1}v = 0$, so the corresponding term vanishes. If v has height at least $k+1$, then $\alpha_v = 0$ by the maximality of k . Hence, only vectors of height exactly k contribute.

For each $1 \leq i \leq m$, let v_i be the unique element of C_i of height k (if such an element exists). If v_i exists, then

$$N^{k-1}v_i = u_i.$$

Thus, after applying N^{k-1} , we obtain

$$\sum_{i: |C_i| \geq k} \alpha_{v_i} u_i = 0.$$

Since $\alpha_v \neq 0$ for some vector of height k , there exists at least one index i with $|C_i| \geq k$ and $\alpha_{v_i} \neq 0$. This contradicts the linear independence of $u_1, \dots, u_m$. $\ominus$

Theorem 4.1.3

Let V be finite-dimensional and $N : V \rightarrow V$ nilpotent. Then N admits a Jordan basis.

Proof: By induction on $\dim(V)$. If $N = 0$, then the result is trivial (any basis of V will do). This in particular covers the case $\dim(V) = 1$, which is the induction base.

Suppose now that $N \neq 0$ (and so $\dim(V) \geq 2$). Let

$$V' := \text{Im}(N) \quad \text{and} \quad N' := N|_{V'}.$$

Note that V' is N -invariant, so N' is well-defined. Clearly, N' is also nilpotent. By the induction hypothesis, N' admits a Jordan basis B' .

As explained above, a Jordan basis consists of a disjoint union of Jordan chains. So let $C_1, \dots, C_m$ be Jordan chains such that

$$B' = \bigcup_{i=1}^m C_i$$

(where the union is disjoint).

For $1 \leq i \leq m$, let w_i be the final element of C_i and let u_i be the initial element of C_i . Then $C_i = C(w_i)$ and $u_i \in \ker(N)$ (recall that the initial element of a Jordan chain is always in $\ker(N)$). Also, $\{u_1, \dots, u_m\}$ is linearly independent because $u_1, \dots, u_m \in B'$ and B' is linearly independent.

Hence, we can complete $\{u_1, \dots, u_m\}$ to a basis

$$u_1, \dots, u_m, u_{m+1}, \dots, u_d$$

of $\ker(N)$, where

$$d = \dim(\ker(N)).$$

For $1 \leq i \leq m$, we have $w_i \in V' = \text{Im}(N)$, so there exists $v_i \in V$ such that $w_i = Nv_i$. This means that

$$C(v_i) = \{v_i\} \cup C(w_i) \quad \text{for } i = 1, \dots, m.$$

Also, since $u_{m+1}, \dots, u_d \in \ker(N)$, we have

$$C(u_i) = \{u_i\} \quad \text{for } i = m+1, \dots, d.$$

Now consider

$$B = \left(\bigcup_{i=1}^m C(v_i) \right) \cup \left(\bigcup_{i=m+1}^d C(u_i) \right).$$

This is a union of Jordan chains whose initial elements $u_1, \dots, u_d$ are linearly independent. Hence, by Lemma 5.9, B is linearly independent. Also,

$$|B| = |B'| + m + (d - m) = |B'| + d = \dim(V') + d = \dim(\text{Im}(N)) + \dim(\ker(N)) = \dim(V),$$

where we used that B' is a basis for V' and the rank-nullity theorem.

It follows that B is a basis for V . This completes the proof. $\odot$

Proposition 4.1.1

Let $N : V \rightarrow V$ be a nilpotent linear map of degree t . The largest length of a Jordan chain of N is t .

Proof: Let C be a Jordan chain of size k . Observe that $\{N^k v : v \in C\} = \{0\}$ while $\{N^{k-1} v : v \in C\} \neq \{0\}$. (Make sure that you see why!) Therefore, since $N^t = 0$, all Jordan chains of N have length at most t . On the other hand, since $N^{t-1} \neq 0$, some Jordan chain has length strictly larger than $t - 1$, namely exactly t . $\odot$

Visualization of Jordan Chains

v_1					
	v_2	v_3			
			v_4		
				v_5	v_6

Note:-

Each column represents a Jordan chain, ordered from bottom (initial vector) to top (final vector). The chain lengths are 5, 3, 3, 2, 1, 1. The map N shifts each vector downward, with the bottom vector mapped to 0.

Theorem 4.1.4

Let $N : V \rightarrow V$ be a nilpotent linear map of degree t , and let B be a Jordan basis for N . For $1 \leq k \leq t$, define

$$B_k := \{ v \in B : v \text{ has height } \leq k \}.$$

Then B_k is a basis for $\ker(N^k)$.

Proof: Consider the action of N^k on B_k and $B \setminus B_k$. If $v \in B_k$, then $N^k v = 0$ since the height of v is at most k . If $v \in B \setminus B_k$, then $N^k v$ is a nonzero element of B . In fact, N^k maps $B \setminus B_k$ injectively into B (applying N^k shifts each Jordan chain down by k positions).

We now use the following fact.

Note:-

Let $T : V \rightarrow W$ be a linear map and B a basis of V . Then $T(B)$ spans $\Im(T)$.

In our case,

$$N^k(B) = N^k(B \setminus B_k) \cup \{0\}$$

spans $\Im(N^k)$. Let

$$B' := N^k(B \setminus B_k) \subset B.$$

Then B' is linearly independent, hence a basis of $\Im(N^k)$. Therefore,

$$\dim \Im(N^k) = |B'| = |B| - |B_k| = \dim(V) - |B_k|.$$

By the rank-nullity theorem,

$$\dim \Im(N^k) = \dim(V) - \dim \ker(N^k).$$

Thus $|B_k| = \dim \ker(N^k)$. Since $B_k \subset \ker(N^k)$ and B_k is linearly independent, it follows that B_k is a basis for $\ker(N^k)$. ☺

Theorem 4.1.5

Let $N : V \rightarrow V$ be a nilpotent linear map of degree t , and let B be a Jordan basis for N . For $1 \leq k \leq t$, let s_k denote the number of Jordan chains in B of size exactly k . Then for every $1 \leq k \leq t$,

$$s_k + \cdots + s_t = \dim \ker(N^k) - \dim \ker(N^{k-1}), \quad (4.1)$$

and

$$s_k = 2 \dim \ker(N^k) - \dim \ker(N^{k+1}) - \dim \ker(N^{k-1}). \quad (4.2)$$

Here $\dim \ker(N^0) = \dim \ker(I) = 0$.

In particular, the numbers $s_1, \dots, s_t$ depend only on N and not on the choice of Jordan basis.

Proof: Let

$$B_k := \{v \in B : v \text{ has height } \leq k\}.$$

By the previous theorem,

$$|B_k| = \dim \ker(N^k).$$

Observe that $B_k \setminus B_{k-1}$ consists precisely of the elements of height exactly k . Each Jordan chain of length at least k contributes exactly one such element, and chains of length less than k contribute none. Hence,

$$\sum_{i=k}^t s_i = |B_k \setminus B_{k-1}| = |B_k| - |B_{k-1}| = \dim \ker(N^k) - \dim \ker(N^{k-1}),$$

which proves 4.2.

To isolate s_k , compute

$$s_k = \left(\sum_{i=k}^t s_i \right) - \left(\sum_{i=k+1}^t s_i \right),$$

and apply equation 4.1 twice to obtain equation 4.2. ☺

We now find another formula for the sum $s_k + \cdots + s_t$ appearing in equation 4.1, to be used later. To this end, we prove the following.

Lemma 4.1.3

Let $T : V \rightarrow V$ be a linear map. Then for every $k \geq 0$,

$$\dim \ker(T^k) - \dim \ker(T^{k-1}) = \dim(\operatorname{Im}(T^{k-1}) \cap \ker(T)).$$

Proof: We know that $\ker(T^{k-1}) \subseteq \ker(T^k)$. Take a subspace $W \subseteq \ker(T^k)$ such that

$$\ker(T^k) = \ker(T^{k-1}) \oplus W.$$

(We can find W by completing a basis of $\ker(T^{k-1})$ to a basis of $\ker(T^k)$ and taking W to be the span of the added elements.) Then $\dim(W) = \dim \ker(T^k) - \dim \ker(T^{k-1})$.

Consider the map

$$S := T^{k-1}|_W : W \rightarrow V,$$

i.e., we consider the restriction of T^{k-1} to W . Clearly, $\text{Im}(S) \subseteq \text{Im}(T^{k-1})$. Also, $\text{Im}(S) \subseteq \ker(T)$. Indeed, for every $w \in W \subseteq \ker(T^k)$ we have $T(T^{k-1}w) = T^kw = 0$, and so $T^{k-1}w \in \ker(T)$. We conclude that

$$\text{Im}(S) \subseteq \text{Im}(T^{k-1}) \cap \ker(T).$$

Now, we claim that S is injective. Indeed, if $w \in W$ satisfies $Sw = T^{k-1}w = 0$ then $w \in \ker(T^{k-1})$. But $\ker(T^{k-1}) \cap W = \{0\}$, so $w = 0$.

Finally, we claim that S is onto $\text{Im}(T^{k-1}) \cap \ker(T)$. So fix any $x \in \text{Im}(T^{k-1}) \cap \ker(T)$. As $x \in \text{Im}(T^{k-1})$, there is $v \in V$ such that $T^{k-1}v = x$. Also, as $x \in \ker(T)$, we have $0 = Tx = T^kv$, so $v \in \ker(T^k)$. Write $v = u + w$, where $u \in \ker(T^{k-1})$ and $w \in W$. Then

$$x = T^{k-1}v = T^{k-1}u + T^{k-1}w = T^{k-1}w,$$

using that $u \in \ker(T^{k-1})$. So $Sw = x$, as required.

We conclude that S is a bijection from W to $\text{Im}(T^{k-1}) \cap \ker(T)$. Hence,

$$\dim(\text{Im}(T^{k-1}) \cap \ker(T)) = \dim(W) = \dim \ker(T^k) - \dim \ker(T^{k-1}),$$

as required. ☺

Corollary 4.1.1

In the notation of Theorem 4.1.5, for all $1 \leq k \leq t$ we have

$$s_k + \cdots + s_t = \dim(\text{Im}(N^{k-1}) \cap \ker(N)).$$

Finally, we switch our focus back to a general (i.e., not necessarily nilpotent) linear map T whose minimal polynomial (or, equivalently, characteristic polynomial) splits. We show that T can be put in Jordan canonical form, and that this form is unique.

Theorem 4.1.6 Jordan canonical form theorem

Let V be a vector space over a field $\mathbb{F}$, $\dim(V) < \infty$. Let $T : V \rightarrow V$ be a linear map, and suppose that the minimal polynomial q_T splits (over $\mathbb{F}$) as

$$q_T(x) = \prod_{i=1}^r (x - \lambda_i)^{t_i}.$$

Then:

1. There is an ordered basis B of V such that $[T]_B$ has Jordan canonical form (i.e., B is a Jordan basis for T).
2. The Jordan canonical form of T is unique (up to permuting the blocks): For each $1 \leq i \leq r$, the size of a largest Jordan block of type λ_i is t_i . Moreover, let $s_k(\lambda_i)$, $1 \leq k \leq t_i$, denote the number of Jordan blocks of type λ_i of size exactly k . Then

$$s_k(\lambda_i) + \cdots + s_{t_i}(\lambda_i) = \dim \text{Ker}((T - \lambda_i I)^k) - \dim \text{Ker}((T - \lambda_i I)^{k-1})$$

and

$$s_k(\lambda_i) = 2 \dim \operatorname{Ker}((T - \lambda_i I)^k) - \dim \operatorname{Ker}((T - \lambda_i I)^{k+1}) - \dim \operatorname{Ker}((T - \lambda_i I)^{k-1})$$

In particular, the total number of Jordan blocks of type λ_i is $\dim \operatorname{Ker}(T - \lambda_i I) = \dim(E(\lambda_i, T))$.

3. The characteristic polynomial p_T of T equals

$$p_T(x) = \prod_{i=1}^r (x - \lambda_i)^{m_i}$$

where

$$m_i := \dim \operatorname{Ker}((T - \lambda_i I)^{t_i})$$

Also, m_i is equal to the sum of sizes of the Jordan blocks of type λ_i . In particular, p_T splits.

Proof: Set $V_i := \operatorname{Ker}((T - \lambda_i I)^{t_i})$ for $i = 1, \dots, r$. By Theorem 4.1.2, we have

$$V = V_1 \oplus \dots \oplus V_r \quad (4.3)$$

Moreover, the minimal polynomial of $T|_{V_i}$ is $(x - \lambda_i)^{t_i}$.

We now apply the theory we developed for nilpotent maps (namely, Theorem 4.1.3, Proposition 4.1.1 and Theorem 4.1.5) to the maps $N_i := (T - \lambda_i I)|_{V_i}$. The fact that the minimal polynomial of $T|_{V_i}$ is $(x - \lambda_i)^{t_i}$ exactly means that N_i is nilpotent of degree t_i (make sure that you see why!).

Hence, by Theorem 4.1.3, T_i admits a Jordan basis B_i . Then $B = \bigcup_{i=1}^r B_i$ is a Jordan basis for T . Moreover, every Jordan block $J(0, k)$ of $[N_i]_{B_i}$ becomes a Jordan block $J(\lambda_i, k)$ of $[T]_B$ because, restricted to V_i , T equals $N_i + \lambda_i I$. In particular, all Jordan blocks in $[T|_{V_i}]_{B_i}$ are of type λ_i . The assertion of Item 2 regarding the block sizes follows from Proposition 4.1.1 and Theorem 4.1.5.¹

Finally, consider the characteristic polynomial. Using (4.3), we have

$$p_T = \prod_{i=1}^r p_{T|_{V_i}}. \quad (4.4)$$

Also, $J_i := [T|_{V_i}]_{B_i}$ is in Jordan canonical form in which all Jordan blocks have type λ_i . Hence, J_i is an upper-triangular matrix with λ_i 's on the diagonal. It follows that

$$p_{T|_{V_i}} = \det(xI - J_i) = (x - \lambda_i)^{|B_i|} = (x - \lambda_i)^{\dim(V_i)} = (x - \lambda_i)^{\dim \operatorname{Ker}((T - \lambda_i I)^{t_i})},$$

as required. We also see that $m_i := \dim \operatorname{Ker}((T - \lambda_i I)^{t_i})$ is equal to the sum of sizes of Jordan blocks of type λ_i , since this sum equals $|B_i| = \dim(V_i)$. $\odot$

¹Here we are using that $\operatorname{Ker}((T - \lambda_i I)^k) = \operatorname{Ker}((T|_{V_i} - \lambda_i I)^k)$ for all $k \leq t_i$. This indeed holds because we have $\operatorname{Ker}((T - \lambda_i I)^k) \subseteq \operatorname{Ker}((T - \lambda_i I)^{t_i}) = V_i$.

Theorem 4.1.7 Jordan Canonical Form (A more elaborate proof)

Let V be a finite-dimensional vector space over a field $\mathbb{F}$, and let $T : V \rightarrow V$ be a linear map. Suppose the minimal polynomial q_T splits over $\mathbb{F}$ as:

$$q_T(x) = \prod_{i=1}^r (x - \lambda_i)^{t_i}$$

Then T admits a Jordan basis, and its matrix representation is a direct sum of Jordan blocks.

Proof: We prove this theorem in five steps, building from the decomposition of V into generalized eigenspaces down to the construction of Jordan blocks.

Step 1: Primary Decomposition into Generalized Eigenspaces

Since q_T splits, by the Primary Decomposition Theorem, the space V decomposes as a direct sum of generalized eigenspaces:

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_r$$

where each subspace is defined as

$$V_i = \text{Ker}((T - \lambda_i I)^{t_i})$$

Key observation: Each subspace V_i is T -invariant.

Proof of invariance: Let $v \in V_i$. Then $(T - \lambda_i I)^{t_i}(v) = 0$. We need to show $T(v) \in V_i$, i.e., $(T - \lambda_i I)^{t_i}(T(v)) = 0$. Since T and $(T - \lambda_i I)$ commute, we have:

$$(T - \lambda_i I)^{t_i}(T(v)) = T((T - \lambda_i I)^{t_i}(v)) = T(0) = 0$$

Therefore, we can define the restriction $T|_{V_i} : V_i \rightarrow V_i$.

Strategy: If we find a basis for each V_i that puts $T|_{V_i}$ into Jordan form, the union of these bases will provide the Jordan form for T .

Step 2: Reduction to a Nilpotent Operator

Focus on a single generalized eigenspace V_i . Define the shifted operator:

$$N_i = T|_{V_i} - \lambda_i I_{V_i}$$

Claim 4.1.1

N_i is nilpotent.

Proof: By definition of V_i , for every $v \in V_i$:

$$(T - \lambda_i I)^{t_i}(v) = 0$$

Therefore:

$$N_i^{t_i} = (T|_{V_i} - \lambda_i I_{V_i})^{t_i} = 0$$

Since a power of N_i equals the zero map, N_i is nilpotent of degree at most t_i . ☺

Step 3: Constructing the Jordan Basis for the Nilpotent Map

Since N_i is nilpotent, by Theorem 4.1.3 (Jordan canonical form for nilpotent maps), there exists a Jordan basis $\mathcal{B}_i$ for V_i with respect to N_i .

Matrix representation: With respect to such a chain ordered as $\{v_1, v_2, \dots, v_k\}$, the matrix of N_i is:

$$J_k(0) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

Why this form?

- **Column 1:** $N_i(v_1) = \mathbf{0} = 0v_1 + 0v_2 + \cdots$ gives all zeros
- **Column j (for $j \geq 2$):** $N_i(v_j) = v_{j-1} = 0v_1 + \cdots + 1v_{j-1} + 0v_j + \cdots$ gives a 1 in position $(j-1)$ and zeros elsewhere

The complete matrix $[N_i]_{\mathcal{B}_i}$ is a direct sum of such nilpotent Jordan blocks.

Step 4: Recovering the Matrix for T

From Step 2, we have:

$$T|_{V_i} = N_i + \lambda_i I_{V_i}$$

Taking matrix representations with respect to the basis $\mathcal{B}_i$:

$$[T|_{V_i}]_{\mathcal{B}_i} = [N_i]_{\mathcal{B}_i} + \lambda_i [I]_{\mathcal{B}_i}$$

Since $[I]_{\mathcal{B}_i}$ is the identity matrix, we have:

$$[T|_{V_i}]_{\mathcal{B}_i} = [N_i]_{\mathcal{B}_i} + \lambda_i I$$

Effect on Jordan blocks: Each nilpotent block transforms as:

$$\begin{aligned} J_k(0) + \lambda_i I &= \begin{pmatrix} 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \ddots & \ddots \\ 0 & 0 & \cdots & 0 \end{pmatrix} + \begin{pmatrix} \lambda_i & 0 & 0 & \cdots \\ 0 & \lambda_i & 0 & \cdots \\ \vdots & \vdots & \ddots & \ddots \\ 0 & 0 & \cdots & \lambda_i \end{pmatrix} \\ &= \begin{pmatrix} \lambda_i & 1 & 0 & \cdots \\ 0 & \lambda_i & 1 & \cdots \\ \vdots & \vdots & \ddots & \ddots \\ 0 & 0 & \cdots & \lambda_i \end{pmatrix} = J_k(\lambda_i) \end{aligned}$$

Thus, $[T|_{V_i}]_{\mathcal{B}_i}$ is a direct sum of Jordan blocks with eigenvalue λ_i .

Step 5: Synthesis and Uniqueness

Construct the complete basis for V by taking the union:

$$\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \cdots \cup \mathcal{B}_r$$

Since $V = V_1 \oplus V_2 \oplus \cdots \oplus V_r$ and each $\mathcal{B}_i$ is a basis for V_i , the set $\mathcal{B}$ is a basis for V .

The matrix $[T]_{\mathcal{B}}$ is block diagonal:

$$[T]_{\mathcal{B}} = \begin{pmatrix} [T|_{V_1}]_{\mathcal{B}_1} & & & \\ & [T|_{V_2}]_{\mathcal{B}_2} & & \\ & & \ddots & \\ & & & [T|_{V_r}]_{\mathcal{B}_r} \end{pmatrix}$$

This is the **Jordan Canonical Form** of T .

Uniqueness: The number and sizes of Jordan blocks associated with each eigenvalue λ_i are uniquely determined by the dimensions:

$$s_k(\lambda_i) = 2 \dim \operatorname{Ker}((T - \lambda_i I)^k) - \dim \operatorname{Ker}((T - \lambda_i I)^{k+1}) - \dim \operatorname{Ker}((T - \lambda_i I)^{k-1})$$

where $s_k(\lambda_i)$ is the number of blocks of size k for eigenvalue λ_i , by theorem 4.1.5. Since these kernel dimensions are basis-independent invariants of T , the Jordan form is unique up to reordering of blocks.

Also, $J_i := [T|_{V_i}]_{\mathcal{B}_i}$ is in Jordan canonical form in which all Jordan blocks have type λ_i . Hence, J_i is an upper-triangular matrix with λ_i 's on the diagonal. It follows that

$$p_{T|_{V_i}} = \det(xI - J_i) = (x - \lambda_i)^{|B_i|} = (x - \lambda_i)^{\dim(V_i)} = (x - \lambda_i)^{\dim \operatorname{Ker}((T - \lambda_i I)^{t_i})},$$

as required. We also see that $m_i := \dim \operatorname{Ker}((T - \lambda_i I)^{t_i})$ is equal to the sum of sizes of Jordan blocks of type λ_i , since this sum equals $|B_i| = \dim(V_i)$. $\odot$

Corollary 4.1.2

For a linear map $T : V \rightarrow V$ with $\dim(V) < \infty$, the following are equivalent:

1. p_T splits.
2. q_T splits.
3. T admits a Jordan canonical form.

Proof: We prove that $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$. Recall that the minimal polynomial q_T divides the characteristic polynomial p_T . Hence, if p_T splits then clearly so does q_T . Next, if q_T splits then by Theorem 5.17, T admits Jordan canonical form.

Finally, if T admits Jordan canonical form $J = [T]_B$ (for some Jordan basis B) then it is easy to see that

$$p_T(x) = \prod_{i=1}^r (x - \lambda_i)^{m_i},$$

where $\lambda_1, \dots, \lambda_r$ are the distinct eigenvalues of T and m_i is the number of appearances of λ_i on the diagonal of J .² Indeed, simply compute $\det(xI - J)$, using that J is upper-triangular. In particular, p_T splits. $\odot$

Corollary 4.1.3

Let $T : V \rightarrow V$ be a linear map, and suppose that the minimal polynomial q_T splits (over $\mathbb{F}$) as

$$q_T(x) = \prod_{i=1}^r (x - \lambda_i)^{t_i}.$$

Then T is diagonalizable iff $t_1 = \dots = t_r = 1$ (i.e., iff every root of q_T has multiplicity 1).

Proof: The fact that the Jordan canonical form is unique means that T is diagonalizable iff its Jordan canonical form is diagonal. This is equivalent to all Jordan blocks having size 1. By Item 2 of Theorem 5.17, the largest size of a Jordan block of type λ_i is t_i . Hence, all blocks have size 1 iff $t_1 = \dots = t_r = 1$. $\odot$

4.2 Finding the Jordan Canonical Form

Finally, we consider the question of how to find the Jordan canonical form and a Jordan basis for a given linear map (or matrix). These are two different questions: Finding the Jordan canonical form is usually easier; finding a Jordan basis requires more work. The Jordan canonical form can be read off Theorem 5.17. Indeed, to find the Jordan canonical form of a linear map $T : V \rightarrow V$, we can proceed as follows:

Step 1. Find the distinct eigenvalues $\lambda_1, \dots, \lambda_r$ of T . Systematically, this is done by computing the characteristic polynomial p_T , but sometimes can also be done by guessing.

²We also saw (in (26)) that $m_i = \dim \text{Ker}((T - \lambda_i I)^{t_i})$, where t_i is the multiplicity of λ_i in q_T .

Step 2. For each $1 \leq i \leq r$ and $k \geq 1$, compute $\dim \text{Ker}((T - \lambda_i I)^k)$ until the minimal k for which $\dim \text{Ker}((T - \lambda_i I)^k) = \dim \text{Ker}((T - \lambda_i I)^{k+1})$. This k equals t_i , the multiplicity of λ_i in the minimal polynomial q_T .

Step 3. For each $1 \leq i \leq r$ and $1 \leq k \leq t_i$, find the number $s_k(\lambda_i)$ of Jordan blocks of type λ_i and size k , using Item 2 of Theorem 5.17.

Note:-

For the above process, we do not actually need to compute q_T (but the process in particular finds q_T because $q_T(x) = \prod_{i=1}^r (x - \lambda_i)^{t_i}$).

We also note that in order to execute the above computations, it is usually convenient to replace T with a representing matrix. Namely, fix a basis B and do Steps 1-3 for the matrix $[T]_B$.

Claim 4.2.1

Steps 1-3 correctly compute the Jordan canonical form of T .

Proof: We only have to justify Step 2. Let $1 \leq i \leq r$, and consider the map $N_i := (T - \lambda_i I)|_{V_i}$, where V_i is the generalized eigenspace of λ_i . Let t_i be the multiplicity of λ_i in the minimal polynomial q_T . We know that N_i is nilpotent of degree t_i . Thus, by Lemma 5.6, we know that

$$\{0\} = \text{Ker}(N_i^0) \subsetneq \text{Ker}(N_i^1) \subsetneq \text{Ker}(N_i^2) \subsetneq \cdots \subsetneq \text{Ker}(N_i^{t_i-1}) \subsetneq \text{Ker}(N_i^{t_i}) = V_i.$$

Also, $N_i^k = 0$ for all $k \geq t_i$, so $\text{Ker}(N_i^k) = V_i$ for all $k \geq t_i$. Hence, $k = t_i$ is indeed the minimal k for which $\dim \text{Ker}(N_i^k) = \dim \text{Ker}(N_i^{k+1})$, namely, $\dim \text{Ker}((T - \lambda_i I)^k) = \dim \text{Ker}((T - \lambda_i I)^{k+1})$. $\odot$

Example 4.2.1 (Finding Jordan canonical form)

We find the Jordan canonical form of

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 \end{pmatrix}$$

The characteristic polynomial of A is

$$p_A(x) = \det(xI - A) = \begin{vmatrix} x-1 & 0 & 0 & 0 \\ 0 & x-1 & 0 & 0 \\ -1 & 0 & x-1 & 0 \\ -2 & 0 & -1 & x \end{vmatrix} = (x-1)^3 x.$$

Hence, the eigenvalues of A are 1 (with algebraic multiplicity 3) and 0 (with algebraic multiplicity 1). Now consider each of the eigenvalues.

For $\lambda = 0$: Since the algebraic multiplicity of $\lambda = 0$ is 1, there is exactly one Jordan block of type $\lambda = 0$, and this block has size 1.

For $\lambda = 1$: We calculate $(A - I)^k, k \geq 1$ (until we have enough information to stop^a).

$$A - I = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & 0 & 1 & -1 \end{pmatrix}$$

The above matrix has rank 2, so $\dim \text{Ker}(A - I) = 4 - 2 = 2$. Hence, there are 2 Jordan blocks of type $\lambda = 1$. Since the sum of sizes of these blocks is the algebraic multiplicity of $\lambda = 1$, namely 3, the sizes of these two blocks must be 2, 1.

^aTo find the Jordan canonical form in a systematic way, we need to find $\dim \text{Ker}((A - \lambda I)^k)$ until this sequence stabilizes, i.e., until $\dim \text{Ker}((A - \lambda I)^{k+1}) = \dim \text{Ker}((A - \lambda I)^k)$ (in this case $\lambda = 1$). Then k is the largest size of a Jordan block of type λ . But sometimes, especially for small matrices, we can stop sooner, as is the case in this example.

Thus, the Jordan canonical form of A is

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

4.3 Finding a Jordan Basis

Note:-

Related homework problem: see 14 for an official Jordan-basis computation.

Note:-

Related tutorial problem: see 9.6.1 for an explicit use of the nilpotent Jordan-basis algorithm.

To describe a method for finding a Jordan basis for a linear map $T : V \rightarrow V$, we again first consider nilpotent maps (and then apply the algorithm to the nilpotent maps $N = (T - \lambda_i I)|_{V_i}$ for each eigenvalue λ_i of T).

4.3.1 Algorithm for Finding a Jordan Basis of a Nilpotent Map

Algorithm 1: Finding a Jordan basis for nilpotent map N

Input: A nilpotent map $N : V \rightarrow V$ of degree t

Output: A Jordan basis B for N

/ Step 1: Find initial vectors for chains of maximum length */*

1 Find a basis B_t for $\text{Im}(N^{t-1}) \cap \text{Ker}(N)$;

/ Step 2: Extend to find initial vectors for all chains */*

2 **for** $k = t - 1, \dots, 1$ (in reverse order) **do**

3 Complete B_{k+1} to a basis B_k for $\text{Im}(N^{k-1}) \cap \text{Ker}(N)$;

4 *a*

5 **end**

/ Step 3: Construct Jordan chains from initial vectors */*

6 **foreach** $1 \leq k \leq t$ and $v \in B_k \setminus B_{k+1}$ (where $B_{t+1} := \emptyset$) **do**

7 Find a Jordan chain C_v of length k such that v is the initial vector of C_v ;

8 **end**

/ Step 4: Union of all chains gives Jordan basis */*

9 **return** $B := \bigcup_{v \in B_1} C_v$;

^aNote that for $t = 1$, $\text{Im}(N^0) \cap \text{Ker}(N) = V \cap \text{Ker}(N) = \text{Ker}(N)$, so B_1 is a basis of $\text{Ker}(N)$. Also, note that $B_t \subseteq B_{t-1} \subseteq \dots \subseteq B_2 \subseteq B_1$.

As before, it is convenient to replace N with a representing matrix, i.e., fix some basis $\mathcal{B}$ of V , and run the above algorithm for $[N]_{\mathcal{B}}$.³

Before proving that Algorithm 1 correctly computes a Jordan basis for N , let us explain how each of the steps is carried out.

- Steps 1 and 2 involve the calculation of $\text{Ker}(N)$ and of $\text{Im}(N^k)$ for $1 \leq k \leq t - 1$. If N is a matrix, then $\text{Ker}(N)$ is found by solving the linear system $Nx = 0$. Also, recall that for every matrix A , $\text{Im}(A)$ is the span of the columns of A (why?).⁴ We can use this when finding $\text{Im}(N^k)$.
- Let us explain why Step 3 can be executed. Fix any $1 \leq k \leq t$ and $v \in B_k \setminus B_{k+1}$. By construction, $v \in \text{Im}(N^{k-1}) \cap \text{Ker}(N)$. In particular, there is a vector w for which $v = N^{k-1}w$. Now, $(N^{k-1}w, \dots, N^2w, Nw, w)$ is a Jordan chain of length k whose initial vector is v . Indeed, $N^k w = Nv = 0$, as $v \in \text{Ker}(N)$. This shows that the required chain C_v in Item 3 indeed exists. Thus, to find C_v , we just need to find a vector w with $N^{k-1}w = v$. If N is a matrix, then this amounts to solving a system of equations.⁵

³Of course, $\mathcal{B}$ need not be a Jordan basis.

⁴Here by $\text{Im}(A)$ we mean the image of the linear map $x \mapsto Ax$.

⁵Alternatively, one can try to construct the chain C_v one element at a time: First find a vector v_1 such that $Nv_1 = v$ (again, if N is a matrix, then this is just solving a system of equations), then find a vector v_2 with $Nv_2 = v_1$ (and so $N^2v_2 = v$), and so on. If we manage to continue for $k - 1$ steps, we find a vector v_{k-1} with $N^{k-1}v_{k-1} = v$, and we can set $w := v_{k-1}$. We stress, however, that while this method would work in some cases, it is not guaranteed to work in any case. Namely, if the vectors $v_1, v_2, \dots$ are not chosen carefully, the process may get stuck. To make sure that the process does not get stuck, we have to guarantee that $v_1 \in \text{Im}(N^{k-2}), v_2 \in \text{Im}(N^{k-3})$

Theorem 4.3.1

The above algorithm correctly computes a Jordan basis for N .

Proof: By construction, B_1 is a basis for $\text{Im}(N^0) \cap \text{Ker}(N) = \text{Ker}(N)$. In particular, the vectors in B_1 are linearly independent. Hence, the chains $(C_v : v \in B_1)$ have linearly independent initial vectors. By Lemma 5.9, $B = \bigcup_{v \in B_1} C_v$ is linearly independent. Note that the number of chains of length at least k in B is exactly $|B_k| = \dim(\text{Im}(N^{k-1}) \cap \text{Ker}(N))$. By Corollary 5.16, this is also the number $s_k + \dots + s_t$ of chains of length at least k in any Jordan basis of N . It follows that B has the same number of chains of length (exactly) k as any Jordan basis for N . Indeed, the number of such chains in B is $|B_k| - |B_{k+1}|$, and as $|B_k| = s_k + \dots + s_t$, this exactly equals $(s_k + \dots + s_t) - (s_{k+1} + \dots + s_t) = s_k$. We conclude that B has the same size as any Jordan basis for N . Namely, $|B| = \dim(V)$. As B consists of Jordan chains, we see that B is a Jordan basis for N . $\oplus$

Note:-

Algorithm 5.22 is natural: it finds the initial elements of Jordan chains comprising a Jordan basis (Steps 1-2), and then works backwards from these initial elements to find the entire chains (Step 3). Observe that a vector $v \in V$ is the initial element of a Jordan chain of length at least k if and only if $v \in \text{Im}(N^{k-1}) \cap \text{Ker}(N)$. This is the reason for the definition of the sets B_k in Steps 1-2.

Example 4.3.1 (Finding a Jordan basis for a nilpotent matrix)

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

It is easy to check that

$$A^2 = \begin{pmatrix} 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

and $A^3 = 0$. So A is a nilpotent matrix of degree $t = 3$.

Write $e_1, \dots, e_6$ for the standard basis of $\mathbb{R}^6$. We have:

- $\text{Im}(A^2) = \text{Span}(\{e_1\})$.
- $\text{Im}(A) = \text{Span}(\{e_1, e_2, e_3\})$.
- $\text{Ker}(A) = \text{Span}(\{e_1, e_3, e_5\})$. (The kernel of A is found by solving the system $Ax = 0$.)

Therefore, $\text{Im}(A^2) \cap \text{Ker}(A) = \text{Span}(\{e_1\})$ and $\text{Im}(A) \cap \text{Ker}(A) = \text{Span}(\{e_1, e_3\})$. Hence, in Algorithm 5.22:

and so on; generally, we need $v_i \in \text{Im}(N^{k-1-i})$.

- $B_3 = \{e_1\}$ is a basis of $\text{Im}(A^2) \cap \text{Ker}(A)$.
- $B_2 = \{e_1, e_3\}$ is an extension of B_3 to a basis of $\text{Im}(A) \cap \text{Ker}(A)$.
- $B_1 = \{e_1, e_3, e_5\}$ is an extension of B_2 to a basis of $\text{Ker}(A)$.

Next, we need to find a Jordan chain C_{e_1} of length 3 with initial vector e_1 , and a Jordan chain C_{e_3} of length 2 with initial vector e_3 . It is easy to see that $Ae_2 = e_1$ and $A \cdot \frac{1}{2}e_4 = e_2$. Therefore, we can take $C_{e_1} = (e_1, e_2, \frac{1}{2}e_4)$. Also, it is easy to see that $Ae_6 = e_3$, so we can take $C_{e_3} = (e_3, e_6)$. (Note that we order each chain in the usual way - from the initial element to the final one.) Finally, we see that

$$B = \left\{ e_1, e_2, \frac{1}{2}e_4; e_3, e_6; e_5 \right\}$$

is a Jordan basis for A (chains are separated by semicolons). The sizes of the Jordan chains also give us the Jordan canonical form of A ; it is:

$$J = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Let C be the matrix whose columns are the elements of B , namely

$$C := \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

So $C = [I]_B^E$ is the change-of-basis matrix from basis B to the standard basis E . To recall the relation between A and J, C , it is convenient to consider the linear map T_A defined as $T_A(x) := Ax$ (we already considered this map before). Then $A = [T_A]_E^E$ and $J = [T_A]_B^B$. Therefore,

$$A = [T_A]_E^E = [I]_B^E \cdot [T_A]_B^B \cdot [I]_E^B = CJC^{-1}.$$

Example 4.3.2 (Finding a Jordan basis for a general linear map)

We find a Jordan basis for the matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

We have $p_A(x) = (x-1)^3(x-2)$. Therefore, the eigenvalues are 1, 2. We know that $\lambda = 2$

has one Jordan block of size 1. Thus, we just need to find an eigenvector of $\lambda = 2$.

$$A - 2I = \begin{pmatrix} -1 & 1 & 1 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

We see that $v_1 := (1, -1, 1, 1)$ is a solution to the above system. Thus, this is an eigenvector of $\lambda = 2$.

Now, let us consider $\lambda = 1$. We have

$$A - I = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and

$$(A - I)^2 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and $(A - I)^3 = (A - I)^2$.

Solving the system $(A - I)x = 0$, we see that

$$\text{Ker}(A - I) = \{x = (x_1, \dots, x_4) : x_4 = 0, x_2 = -x_3\}.$$

In particular, $\dim \text{Ker}(A) = 2$. Also, it is easy to see that $\dim \text{Ker}(A^2) = \dim \text{Ker}(A^3) = 3$.^a Hence, the sequence $\dim \text{Ker}(A^k)$ stabilizes at $k = 2$. This means that $A - I$ is nilpotent of degree $t = 2$.

$$\text{Im}(A - I) = \text{Span}\{(1, 0, 0, 0), (1, -1, 1, 1)\}$$

By Algorithm 5.22 (applied to the nilpotent map $N := (A - I)|_{\text{Ker}((A - I)^2)}$), we need to start by finding a basis B_2 for $\text{Im}(A - I) \cap \text{Ker}(A - I)$. It is not hard to see that

$$\text{Im}(A - I) \cap \text{Ker}(A - I) = \text{Span}(\{(1, 0, 0, 0)\}).$$

So take $v_2 := (1, 0, 0, 0)$ and $B_2 = \{v_2\}$. Now we need to complete B_2 into a basis B_1 of $\text{Ker}(A - I)$. Setting $v_3 := (0, 1, -1, 0)$, we see that $B_1 = \{v_2, v_3\}$ is such a basis. Finally, we need to find a Jordan chain of length 2 ending at v_2 . Namely, we need to find a vector v_4 with $(A - I)v_4 = v_2$. One possible choice is $v_4 := (0, 1, 0, 0)$. Thus, $(v_1; v_4, v_2; v_3)$ is a Jordan basis for A . The Jordan canonical form of A is

$$J = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Let C be the matrix whose columns are $(v_1; v_2, v_4; v_3)$, namely:

$$C = \begin{pmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 1 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Then $A = CJC^{-1}$.

^aNote that it is no coincidence that $\text{Im}((A - I)^2) = \text{Span}(\{v_1\})$, and v_1 is the eigenvector of $\lambda = 2$. Indeed, we know that the space (in this case $\mathbb{R}^4$) is the direct sum of $\text{Ker}((A - I)^2)$ and $\text{Ker}(A - 2I)$. Now, $(A - I)^2$ maps the former to 0, and acts bijectively on the latter (this follows e.g. from Problem 4 in Homework 2). Thus, the image of $(A - I)^2$ is just the eigenspace of $\lambda = 2$.

Note:-

Finally, to find a Jordan basis of a general linear map $T : V \rightarrow V$ (whose minimal polynomial splits), simply apply Algorithm 5.22 to each generalized eigenspace. Namely, writing $q_T = \prod_{i=1}^r (x - \lambda_i)^{t_i}$ and $V_i := \text{Ker}((T - \lambda_i I)^{t_i})$, apply Algorithm 5.22 to find a Jordan basis B_i (of the subspace V_i) for the nilpotent map $N_i := (T - \lambda_i I)|_{V_i}$. Then $B = \bigcup_{i=1}^r B_i$ is a Jordan basis for T .

Chapter 5

Inner product spaces

5.1 Real inner product spaces

We start with a reminder from (high-school) geometry. Consider the Euclidean plane $\mathbb{R}^2$. By Pythagoras' theorem, the length of a vector $v = (x, y)$ is

$$\|v\| = \sqrt{x^2 + y^2}.$$

Also, for two vectors $v_1 = (x_1, y_1), v_2 = (x_2, y_2)$, one can define the dot product

$$\langle v_1, v_2 \rangle := x_1 x_2 + y_1 y_2.$$

Using some trigonometry, it is then possible to show that then the angle α between v_1 and v_2 satisfies

$$\cos(\alpha) = \frac{\langle v_1, v_2 \rangle}{\|v_1\| \|v_2\|}.$$

Indeed, write v_1, v_2 in polar coordinates, namely

$$v_i = r_i(\cos(\alpha_i), \sin(\alpha_i))$$

for $i = 1, 2$. Then

$$\begin{aligned}\langle v_1, v_2 \rangle &= r_1 r_2 \cdot (\cos(\alpha_1) \cos(\alpha_2) + \sin(\alpha_1) \sin(\alpha_2)) \\ &= r_1 r_2 \cos(\alpha_1 - \alpha_2) = \|v_1\| \|v_2\| \cdot \cos(\alpha).\end{aligned}$$

The formula implies that v_1, v_2 are perpendicular iff $\langle v_1, v_2 \rangle = 0$.

Note also that we can express the length of a vector $v = (x, y)$ using the dot product:

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

We can generalize the dot product to arbitrary dimension. Namely, for $u, v \in \mathbb{R}^n$, define

$$\langle u, v \rangle := \sum_{i=1}^n u_i v_i.$$

Also, define

$$\|v\| := \sqrt{\langle v, v \rangle} = \sqrt{\sum_{i=1}^n v_i^2}$$

We will interpret $\|v\|$ as the length of the vector v . We will later show that for any two vectors $u, v \in \mathbb{R}^n$,

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

(this is the Cauchy-Schwarz inequality). Thus,

$$t := \frac{\langle u, v \rangle}{\|u\| \|v\|} \in [-1, 1].$$

This means that there is a unique $\alpha \in [0, \pi]$ such that

$$\cos(\alpha) = t,$$

and we will interpret α as the angle between u and v . Again, we will say that u, v are perpendicular (also called orthogonal) if $\langle u, v \rangle = 0$.

It turns out that there is an abstract structure that captures all the important properties of the dot product $\langle u, v \rangle$. This is the notion of a (real) inner product space.

Definition 5.1.1: Bilinear

Let V be a vector space over a field $\mathbb{F}$. A function $b : V^2 \rightarrow \mathbb{F}$ is called bilinear if:

$$b(u + v, w) = b(u, w) + b(v, w)$$

$$b(w, u + v) = b(w, u) + b(w, v)$$

$$b(\alpha u, v) = \alpha \cdot b(u, v)$$

$$b(u, \alpha v) = \alpha \cdot b(u, v).$$

In other words, b is bilinear if fixing one of the coordinates gives a linear function; namely, for every $v \in V$, the functions $u \mapsto b(v, u)$ and $u \mapsto b(u, v)$ are linear.

Definition 5.1.2: Real Inner Product Space

Let V be a vector space over $\mathbb{F} = \mathbb{R}$. An inner product is a function

$$\langle \cdot, \cdot \rangle : V^2 \rightarrow \mathbb{R}$$

satisfying the following properties:

1. **Bilinearity:**

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$$

$$\langle w, u + v \rangle = \langle w, u \rangle + \langle w, v \rangle$$

$$\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$$

$$\langle u, \alpha v \rangle = \alpha \langle u, v \rangle$$

2. **Symmetry:** $\langle u, v \rangle = \langle v, u \rangle$ for all $u, v \in V$.

3. **Positivity:** $\langle v, v \rangle \geq 0$ for all $v \in V$, with equality iff $v = 0$.

The space V together with the inner product $\langle \cdot, \cdot \rangle$ is called a real inner product space.

Definition 5.1.3: Norm Derived from an Inner Product

Given an inner product space $V, \langle \cdot, \cdot \rangle$, the corresponding norm is defined as

$$\|v\| := \sqrt{\langle v, v \rangle}.$$

By the definition of an inner product, $\|v\| \geq 0$ for every vector v , with equality iff $v = 0$. For $V = \mathbb{R}^n$ and the dot product, the norm $\|v\| = \sqrt{\langle v, v \rangle} = \sqrt{\sum_{i=1}^n v_i^2}$ is called the Euclidean norm.

Example 5.1.1 (Standard Inner Product)

$V = \mathbb{R}^n$ with the standard inner product $\langle u, v \rangle = \sum_{i=1}^n u_i v_i$.

Example 5.1.2 (Frobenius Inner Product)

Given two real matrices $A, B \in \mathbb{R}^{m \times n}$, we can treat A, B as vectors (of length mn), and multiply them using the standard inner product. Namely,

$$\langle A, B \rangle := \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij}.$$

This inner product has the following nice expression. Recall that the trace of a square matrix $M \in \mathbb{F}^{n \times n}$ is the sum of diagonal entries of M , i.e., $\text{tr}(M) = \sum_{i=1}^n M_{ii}$.

Proposition 5.1.1

For every two matrices $A, B \in \mathbb{F}^{m \times n}$, we have

$$\sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij} = \text{tr}(B^T A).$$

Proof: First, note that A is $m \times n$ and B^T is $n \times m$, so the multiplication $B^T A$ is well-defined and is an $n \times n$ matrix. By definition,

$$\text{tr}(B^T A) = \sum_{j=1}^n (B^T A)_{jj} = \sum_{j=1}^n \sum_{i=1}^m B_{ji}^T A_{ij} = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij}.$$

☺

Summarizing, for $\mathbb{F} = \mathbb{R}$, the function $\langle A, B \rangle := \text{tr}(B^T A)$ is an inner product on $\mathbb{R}^{m \times n}$ (indeed, this is simply the standard inner product on $\mathbb{R}^{mn}$, when we think of the elements of $\mathbb{R}^{m \times n}$ as vectors of length mn). This inner product is called the Frobenius inner product.

Example 5.1.3 (Inner Product on Continuous Functions)

Let $V := C([a, b])$ be the space of all continuous real functions $f : [a, b] \rightarrow \mathbb{R}$ on the interval $[a, b]$. For $f, g \in V$, define

$$\langle f, g \rangle := \int_a^b f(x)g(x)dx$$

This is an inner product on V . Indeed, bilinearity is easy to check and symmetry is immediate. We now check positivity: So suppose that $f \neq 0$. Then there is $x_0 \in [a, b]$ such that $f(x_0) \neq 0$. By continuity, there is a small interval I around x_0 such that $f(x) \neq 0$ for all $x \in I$. Hence, $f^2(x) > 0$ for all $x \in I$. It follows that

$$\langle f, f \rangle = \int_a^b f^2(x)dx > 0$$

as required.

5.2 Complex inner product spaces

Note:-

Related homework problems: see 16 for norm identities in complex inner product spaces, and 22 for detecting self-adjointness from real quadratic values.

Note:-

Related tutorial problems: see 9.6.2 for a complex inner-product consequence of $\langle Tv, v \rangle = 0$, and 9.6.3 for why the analogous real statement can fail.

We now move on to complex inner product spaces. First, some reminders on complex numbers:

- A complex number is a number of the form $z = a + bi$, where $a, b \in \mathbb{R}$.
- a is called the real part of z and b is called the imaginary part, denoted $a = \Re(z)$, $b = \Im(z)$.
- In particular, the complex numbers $\mathbb{C}$ are in bijection with $\mathbb{R}^2$, via $z \mapsto (\Re(z), \Im(z))$.
- The conjugate of $z = a + bi$ is $\bar{z} := a - bi$. The conjugate is additive and multiplicative, i.e., $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{zw} = \bar{z} \times \bar{w}$. Also, $\bar{\bar{z}} = z$ iff $z \in \mathbb{R}$, and $\bar{\bar{z}} = z$ for all $z \in \mathbb{C}$ (i.e., taking conjugate twice recovers z). Note that $z + \bar{z} = 2\Re(z)$ and $z - \bar{z} = 2\Im(z) \cdot i$.
- $z \cdot \bar{z} = |z|^2$, where $|z| := \sqrt{\Re(z)^2 + \Im(z)^2}$ is the absolute value (or length) of z . In particular, $z \cdot \bar{z}$ is real and non-negative.

We would like to define the dot product of two vectors $u, v \in \mathbb{C}^n$, as well as the norm $\|v\|$ of a vector $v \in \mathbb{C}^n$. We define

$$\|v\| = \sqrt{\sum_{i=1}^n |v_i|^2}.$$

This definition guarantees that $\|v\|$ is real and nonnegative for all $v \in \mathbb{C}^n$ (and that $\|v\| = 0$ iff $v = 0$). Also, this extends the definition of the norm on $\mathbb{R}^n$, if we consider $\mathbb{R}^n$ as a subset of $\mathbb{C}^n$.

However, we no longer have $\|v\|^2 = \sum_{i=1}^n v_i \cdot v_i$. Instead, we have $\|v\|^2 = \sum_{i=1}^n v_i \cdot \bar{v}_i$. Thus, we define the dot product on $\mathbb{C}^n$ accordingly: For $u, v \in \mathbb{C}^n$, define

$$\langle u, v \rangle = \sum_{i=1}^n u_i \bar{v}_i.$$

We then recover the identity $\|v\| = \sqrt{\langle v, v \rangle}$. Note that $\langle v, u \rangle = \overline{\langle u, v \rangle}$.

This motivates the definition of a complex inner product space.

Definition 5.2.1: Complex Inner Product Space

Let V be a vector space over $\mathbb{F} = \mathbb{C}$. An inner product is a function

$$\langle \cdot, \cdot \rangle : V^2 \rightarrow \mathbb{C}$$

satisfying the following properties:

1. **Linearity in the first coordinate:** For all $u, v, w \in V$ and scalar c ,

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$$

and

$$\langle cv, w \rangle = c \langle v, w \rangle.$$

2. **Conjugate symmetry:** $\langle v, u \rangle = \overline{\langle u, v \rangle}$ for all $u, v \in V$.

3. **Positivity:** $\langle v, v \rangle \geq 0$ for all $v \in V$, with equality iff $v = 0$.

As before, the norm derived from the inner product is

$$\|v\| := \sqrt{\langle v, v \rangle}.$$

Note:-**Remarks:**

- By conjugate symmetry (Item 2), for every $v \in V$ we have $\langle v, v \rangle = \overline{\langle v, v \rangle}$. Hence, $\langle v, v \rangle \in \mathbb{R}$, and so requiring $\langle v, v \rangle \geq 0$ in Item 3 makes sense.
- The norm $\|\cdot\|$ satisfies that $\|v\| \in \mathbb{R}$ and $\|v\| \geq 0$ for all $v \in V$, and $\|v\| = 0$ iff $v = 0$ (by Item 3).
- Items 1-2 imply that

$$\begin{aligned}\langle u, v + w \rangle &= \overline{\langle v + w, u \rangle} = \overline{\langle v, u \rangle + \langle w, u \rangle} \\ &= \overline{\langle v, u \rangle} + \overline{\langle w, u \rangle} = \langle u, v \rangle + \langle u, w \rangle\end{aligned}$$

and

$$\langle u, cv \rangle = \overline{\langle cv, u \rangle} = \overline{c \langle v, u \rangle} = \bar{c} \cdot \overline{\langle v, u \rangle} = \bar{c} \cdot \langle u, v \rangle.$$

Hence, $\langle \cdot, \cdot \rangle$ is additive in the second coordinate, but when taking a scalar out from the second coordinate, we need to take the conjugate of the scalar.

Proposition 5.2.1

For every scalar $c \in \mathbb{C}$, $\|cv\| = |c| \cdot \|v\|$.

Proof:

$$\|cv\|^2 = \langle cv, cv \rangle = c \cdot \bar{c} \cdot \langle v, v \rangle = |c|^2 \cdot \|v\|^2.$$

Taking square roots (note that these are nonnegative real numbers) gives the result. $\odot$

All examples of real inner product spaces we saw previously have complex analogues:

1. $V = \mathbb{C}^n$ with the standard dot product $\langle u, v \rangle = \sum_{i=1}^n u_i \bar{v}_i$.
2. $V = \mathbb{C}^{m \times n}$. For a matrix A , define

$$A^* := \overline{A^T}.$$

Namely, A^* is obtained from A by taking its transpose and then conjugating every entry. Now, the Frobenius inner product on $\mathbb{C}^{m \times n}$ is

$$\langle A, B \rangle := \text{tr}(B^* A) = \sum_{i=1}^m \sum_{j=1}^n A_{ij} \overline{B_{ij}}.$$

(Again, we can think of this inner product as simply the inner product on $\mathbb{C}^{nm}$, when we treat each matrix as a vector in $\mathbb{C}^{nm}$.)

3. $V = C([a, b])$ is the space of continuous functions from the interval $[a, b]$ to $\mathbb{C}$, and

$$\langle f, g \rangle := \int_a^b f(x) \overline{g(x)} dx.$$

Note:-

Note that a real inner product space satisfies all axioms of a complex inner product space, and hence can be thought of as a special case. In what follows, when we talk of an inner product

space without specifying the field, we mean a complex inner product space (thus, all results also apply to real inner product spaces).

5.3 The Cauchy-Schwarz inequality

Note:-

Related homework problem: see 17 for official Cauchy-Schwarz applications.

Note:-

Related tutorial problems: see 9.6.4 for a classic inequality proved by Cauchy-Schwarz, 9.7.1 for Bessel's inequality, and 9.11.1 for the Cauchy-Schwarz inequality induced by a PSD operator.

Note:-

Related additional exercises: see 25 for an integral Cauchy-Schwarz inequality and 27 for the equality case applied to $\langle Tv, v \rangle$.

Definition 5.3.1: Orthogonal

For an inner product space V , two vectors $u, v \in V$ are orthogonal if $\langle u, v \rangle = 0$.

Proposition 5.3.1 Pythagorean Theorem

For orthogonal vectors u, v , we have

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Proof:

$$\begin{aligned} \|u + v\|^2 &= \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} = \|u\|^2 + \|v\|^2 + 2\Re(\langle u, v \rangle). \end{aligned}$$

As u, v are orthogonal, $\langle u, v \rangle = 0$ and the result follows. $\odot$

The following is a fundamental result on inner product spaces.

Theorem 5.3.1 Cauchy-Schwarz Inequality

Let V be an inner product space. Then for all $u, v \in V$,

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|,$$

with equality if and only if u, v are linearly dependent (i.e., u, v are multiples of each other).

Proof: We may assume that $v \neq 0$. Define

$$c := \frac{\langle u, v \rangle}{\langle v, v \rangle} = \frac{\langle u, v \rangle}{\|v\|^2}$$

and put

$$w := u - cv.$$

Observe that

$$\langle w, v \rangle = \langle u - cv, v \rangle = \langle u, v \rangle - c\langle v, v \rangle = 0,$$

using the choice of c .

What we did here is subtract a multiple of v from u to obtain a vector orthogonal to v (namely, the vector w). The vector $cv = \frac{\langle u, v \rangle}{\langle v, v \rangle}v$ is called the orthogonal projection of u onto v (or onto the 1-dimensional subspace $\text{span}(\{v\})$). We will see a generalization of this (i.e., orthogonal projections onto subspaces) later on.

Returning to the proof, we have $u = cv + w$. Also, we saw that v, w are orthogonal, and hence cv, w are also orthogonal. By the Pythagorean theorem,

$$\|u\|^2 = \|cv + w\|^2 = \|cv\|^2 + \|w\|^2 \geq \|cv\|^2 = |c|^2 \cdot \|v\|^2 = \frac{|\langle u, v \rangle|^2}{\|v\|^2}.$$

Rearranging gives

$$|\langle u, v \rangle|^2 \leq \|u\|^2 \|v\|^2,$$

and taking square roots gives the result. Also, equality holds iff $w = 0$, which happens iff $u = cv$. Namely, equality holds iff u is a multiple of v . $\odot$

Summarizing, the proof of the Cauchy-Schwarz inequality works by breaking u into two components - one a multiple of v and the other orthogonal to v , and plugging this into $\langle u, v \rangle$.

Consider now a real inner product space. The Cauchy-Schwarz inequality implies that

$$-1 \leq \frac{\langle u, v \rangle}{\|u\| \|v\|} \leq 1.$$

Hence, there is a unique $\alpha \in [0, \pi]$ such that

$$\cos(\alpha) = \frac{\langle u, v \rangle}{\|u\| \|v\|}.$$

One can then define α to be the angle between u, v . (As we saw, if $V = \mathbb{R}^n$ with the standard inner product, then this is indeed the angle between u, v .)

Using the Cauchy-Schwarz inequality, we can now prove another fundamental fact about inner product spaces:

Theorem 5.3.2 Triangle Inequality

Let V be an inner product space. For all $u, v \in V$,

$$\|u + v\| \leq \|u\| + \|v\|.$$

Equality holds if and only if $u = cv$ for some scalar $c > 0$, or u, v are 0.

Proof: We compute

$$\|u + v\|^2 = \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle = \|u\|^2 + \|v\|^2 + 2 \operatorname{Re} \langle u, v \rangle.$$

For any $z \in \mathbb{C}$,

$$\operatorname{Re}(z) \leq |z|,$$

with equality if and only if z is real and $z \geq 0$.

Let

$$z := \langle u, v \rangle.$$

Then

$$\operatorname{Re} \langle u, v \rangle \leq |\langle u, v \rangle|.$$

Therefore

$$\|u + v\|^2 \leq \|u\|^2 + \|v\|^2 + 2|\langle u, v \rangle|.$$

By the Cauchy–Schwarz inequality,

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Hence

$$\|u + v\|^2 \leq \|u\|^2 + \|v\|^2 + 2\|u\| \|v\| = (\|u\| + \|v\|)^2.$$

Taking square roots gives

$$\|u + v\| \leq \|u\| + \|v\|.$$

Equality case

If $\|u + v\| = \|u\| + \|v\|$, then equality must hold in the two inequalities above. There are two implications:

1. $|\langle u, v \rangle| = \|u\| \|v\|$, which implies u, v are linearly dependent.
2. $\operatorname{Re} \langle u, v \rangle = |\langle u, v \rangle|$, which implies $\langle u, v \rangle$ is real and ≥ 0 .

In the proof of Cauchy–Schwarz we define

$$w := u - cv$$

for a suitable scalar c . Then w is orthogonal to v , so

$$\|u\|^2 = \|w + cv\|^2 = \|w\|^2 + \|cv\|^2$$

since w, v are orthogonal.

Thus

$$\|u\|^2 = \|w\|^2 + |c|^2 \|v\|^2.$$

If equality holds in Cauchy–Schwarz then $\|w\|^2 = 0$, hence $w = 0$, so

$$u = cv.$$

The condition that $\langle u, v \rangle$ is real and nonnegative implies $c > 0$.

Thus equality occurs exactly when $u = cv$ with $c > 0$, or when one of the vectors is 0. ☺

5.4 Orthonormal bases and the Gram-Schmidt process

Note:-

Related homework problems: see 18 for Gram-Schmidt on polynomial spaces, and 21 for building an orthonormal triangularizing basis over $\mathbb{C}$.

Note:-

Related tutorial problems: see 9.7.1 for the finite-dimensional Riesz representation theorem, 9.7.1 for an application of orthonormal sets, and 9.7.1 for a Gram-Schmidt and projection-matrix computation in $\mathbb{C}^3$.

Note:-

Related additional exercises: see 21, 26, and 24 for Gram-Schmidt and projection computations from past papers.

Definition 5.4.1: Orthogonal and Orthonormal Sets

Let V be an inner product space. A set $S \subseteq V$ is orthogonal if $\langle u, v \rangle = 0$ for all distinct $u, v \in S$. The set S is orthonormal if additionally, for every $v \in S$ it holds that $\langle v, v \rangle = 1$ (equivalently, $\|v\| = 1$).

Note:-

Note that for every non-zero vector $v \in V$,

$$\left\| \frac{1}{\|v\|} \cdot v \right\| = \frac{1}{\|v\|} \cdot \|v\| = 1.$$

Hence, if $S = \{v_1, \dots, v_m\}$ is orthogonal, then we can obtain an orthonormal set by normalizing each v_i ; namely, the set

$$\left\{ \frac{v_1}{\|v_1\|}, \dots, \frac{v_m}{\|v_m\|} \right\}$$

is orthonormal.

In what follows, it will be convenient to have the following (very simple) notation:

Definition 5.4.2: Kronecker Delta

The Kronecker delta δ_{ij} is defined as

$$\delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Thus, a set $\{v_1, \dots, v_n\}$ is orthonormal if and only if

$$\langle v_i, v_j \rangle = \delta_{ij}.$$

We now show that an orthogonal set must be linearly independent.

Lemma 5.4.1

An orthogonal set S of non-zero vectors is linearly independent.

Proof: Suppose that

$$\sum_{v \in S} \alpha_v \cdot v = 0.$$

For each $u \in S$, take an inner product with u of both sides. We get

$$0 = \sum_{v \in S} \langle \alpha_v \cdot v, u \rangle = \sum_{v \in S} \alpha_v \langle v, u \rangle = \alpha_u \langle u, u \rangle.$$

Here we used that $\langle v, u \rangle = 0$ for all $v \neq u$.

Since $u \neq 0$, we have $\langle u, u \rangle \neq 0$. Therefore, the equality $\alpha_u \langle u, u \rangle = 0$ gives $\alpha_u = 0$. Since this holds for every $u \in S$, we conclude that S is independent. ☺

Next, we consider orthonormal bases. We show that for an orthonormal basis $B = (v_1, \dots, v_n)$, the coordinates of any vector in basis B have a very simple form.

Proposition 5.4.1

Let V be a finite-dimensional inner product space, and let $B = (v_1, \dots, v_n)$ be an orthogonal basis for V . Then for every $x \in V$,

$$x = \sum_{j=1}^n \frac{\langle x, v_j \rangle}{\|v_j\|^2} v_j.$$

In particular, if B is orthonormal, then

$$x = \sum_{j=1}^n \langle x, v_j \rangle v_j.$$

In other words, the coefficient of v_j in the linear combination equaling x is $\frac{\langle x, v_j \rangle}{\|v_j\|^2}$.

Proof: Write $x = \sum_{i=1}^n \alpha_i v_i$. Then for every $1 \leq j \leq n$, we have

$$\langle x, v_j \rangle = \left\langle \sum_{i=1}^n \alpha_i v_i, v_j \right\rangle = \sum_{i=1}^n \alpha_i \langle v_i, v_j \rangle = \alpha_j \langle v_j, v_j \rangle,$$

Rearranging gives

$$\alpha_j = \frac{\langle x, v_j \rangle}{\|v_j\|^2},$$

as required. ☺

Proposition 5.4.2

Let $(v_1, \dots, v_n)$ be an orthonormal basis of V . Then for all $x, y \in V$,

$$\langle x, y \rangle = \sum_{i=1}^n \langle x, v_i \rangle \overline{\langle y, v_i \rangle}.$$

In particular,

$$\|x\|^2 = \langle x, x \rangle = \sum_{i=1}^n |\langle x, v_i \rangle|^2.$$

Proof: By the previous proposition, we have

$$x = \sum_{j=1}^n \langle x, v_j \rangle v_j \quad y = \sum_{j=1}^n \langle y, v_j \rangle v_j.$$

Therefore,

$$\begin{aligned} \langle x, y \rangle &= \left\langle \sum_{i=1}^n \langle x, v_i \rangle v_i, \sum_{j=1}^n \langle y, v_j \rangle v_j \right\rangle \\ &= \sum_{i,j=1}^n \langle x, v_i \rangle \overline{\langle y, v_j \rangle} \cdot \langle v_i, v_j \rangle \\ &= \sum_{i=1}^n \langle x, v_i \rangle \overline{\langle y, v_i \rangle}. \end{aligned}$$

In the second equality we used linearity in the first coordinate and conjugate linearity in the second coordinate. In the last equality we used that $\langle v_i, v_j \rangle = \delta_{ij}$. $\odot$

Corollary 5.4.1

Let V be an inner product space and let $B = (v_1, \dots, v_n)$ be an orthonormal basis for V . Then for all $x, y \in V$,

$$\langle x, y \rangle = \langle [x]_B, [y]_B \rangle$$

where on the LHS the inner product is in V , and on the RHS the inner product is in $\mathbb{C}^n$.

Example 5.4.1 (Fourier Series)

Consider the inner product space $V = C([- \pi, \pi])$ of continuous functions $f : [- \pi, \pi] \rightarrow \mathbb{R}$ with the standard inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} f(x)g(x)dx$. We claim that the functions $1, \cos(nx), \sin(nx), n \geq 1$, form an orthogonal set. Proving this requires checking several cases. For a start, note that:

$$\langle 1, \cos(nx) \rangle = \int_{-\pi}^{\pi} \cos(nx)dx = \frac{1}{n} \sin(nx) \Big|_{-\pi}^{\pi} = 0$$

and

$$\langle 1, \sin(nx) \rangle = \int_{-\pi}^{\pi} \sin(nx) dx = -\frac{1}{n} \cos(nx) \Big|_{-\pi}^{\pi} = 0.$$

(Indeed, $\sin(n\pi) = 0$ for all $n \in \mathbb{Z}$, and $\cos(n\pi)$ equals 1 if n is even and -1 if n is odd (i.e., $\cos(n\pi) = (-1)^n$), for all $n \in \mathbb{Z}$.) Using the above integrals, we now also get that for all $n, m \geq 1$ with $n \neq m$, it holds that

$$\begin{aligned} \langle \sin(nx), \sin(mx) \rangle &= \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx \\ &= \frac{1}{2} \int_{-\pi}^{\pi} [\cos((n-m)x) - \cos((n+m)x)] dx = 0. \end{aligned}$$

(Here we used the trigonometric identity $\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$.) Similarly,

$$\begin{aligned} \langle \cos(nx), \cos(mx) \rangle &= \int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx \\ &= \frac{1}{2} \int_{-\pi}^{\pi} [\cos((n-m)x) + \cos((n+m)x)] dx = 0 \end{aligned}$$

for $n, m \geq 1$ with $n \neq m$, and

$$\begin{aligned} \langle \sin(nx), \cos(mx) \rangle &= \int_{-\pi}^{\pi} \sin(nx) \cos(mx) dx \\ &= \frac{1}{2} \int_{-\pi}^{\pi} [\sin((n+m)x) + \sin((n-m)x)] dx = 0 \end{aligned}$$

for all n, m . (For the last integral, we used the trigonometric identity $\sin(\alpha + \beta) = \sin(\alpha)\cos(\beta) + \sin(\beta)\cos(\alpha)$.)

Note also that

$$\begin{aligned} \|1\|^2 &= \langle 1, 1 \rangle = \int_{-\pi}^{\pi} 1 dx = 2\pi \\ \|\sin(nx)\|^2 &= \langle \sin(nx), \sin(nx) \rangle = \int_{-\pi}^{\pi} \sin^2(nx) dx \\ &= \frac{1}{2} \int_{-\pi}^{\pi} (1 - \cos(2nx)) dx = \pi \end{aligned}$$

and

$$\begin{aligned} \|\cos(nx)\|^2 &= \langle \cos(nx), \cos(nx) \rangle = \int_{-\pi}^{\pi} \cos^2(nx) dx \\ &= \frac{1}{2} \int_{-\pi}^{\pi} (1 + \cos(2nx)) dx = \pi. \end{aligned}$$

Thus, by normalizing, we get the orthonormal set

$$\frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{\pi}} \sin(nx), \frac{1}{\sqrt{\pi}} \cos(nx), \quad n \geq 1.$$

Recall that in previous proposition, we showed that if $v_1, \dots, v_n$ is an orthonormal basis of a finite-dimension real inner product space V , then for all vectors $x \in V$, it holds that $\|x\|^2 = \sum_{j=1}^n \langle x, v_j \rangle^2$.

It turns out that the same holds for the orthonormal set, only that we need to replace the finite sum with an infinite series (we will not justify this).^a Let us apply this to the function $f(x) = x$. To this end, we need to compute the inner products of f with the functions in the orthonormal set. The interval $[-\pi, \pi]$ is symmetric around 0, so the integral of any odd function over this interval is 0. This means that $\langle x, 1 \rangle = 0$ and $\langle x, \cos(nx) \rangle = 0$ for all $n \geq 1$ (as x and $x \cos(nx)$ are odd functions). So we only need to consider the functions $\sin(nx)$. Using integration by parts, we get:

$$\langle x, \sin(nx) \rangle = \int_{-\pi}^{\pi} x \sin(nx) dx = \left. \frac{\sin(nx) - nx \cos(nx)}{n^2} \right|_{-\pi}^{\pi} = (-1)^{n+1} \cdot \frac{2\pi}{n}. \quad (5.1)$$

We conclude that

$$\langle x, x \rangle = \sum_{n=1}^{\infty} \left\langle x, \frac{1}{\sqrt{\pi}} \sin(nx) \right\rangle^2 = \sum_{n=1}^{\infty} \left((-1)^{n+1} \cdot \frac{2\sqrt{\pi}}{n} \right)^2 = 4\pi \cdot \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

On the other hand,

$$\langle x, x \rangle = \int_{-\pi}^{\pi} x^2 dx = \left. \frac{x^3}{3} \right|_{-\pi}^{\pi} = \frac{2\pi^3}{3}.$$

Combining the above, we get that

$$4\pi \cdot \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{2\pi^3}{3}$$

and therefore

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}}.$$

This is a famous result originally proved by Euler (using a different proof).

Finally, we note that instead of considering the functions $1, \sin(nx), \cos(nx)$, one can consider e^{inx} for $n \in \mathbb{Z}$; this is also an orthogonal set. It is easy to transfer between these two sets using the fact that $e^{inx} = \cos(nx) + i \sin(nx)$. We omit further details.

^aThis is a standard result in functional analysis.

So far we have not showed that every finite-dimensional inner product space has an orthonormal basis. We now prove this in a more general form: We describe a way of transforming any given linearly independent set $S = \{u_1, \dots, u_m\}$ into an orthogonal set $S' = \{v_1, \dots, v_m\}$ with $\text{span}(S') = \text{span}(S)$. This algorithm is called the Gram-Schmidt process.

Theorem 5.4.1 Gram-Schmidt Process

Let V be an inner product space, and let

$$S = \{u_1, \dots, u_m\} \subseteq V$$

be a linearly independent set. Define vectors $v_1, \dots, v_m$ inductively by setting

$$v_1 = u_1$$

and

$$v_k = u_k - \sum_{i=1}^{k-1} \frac{\langle u_k, v_i \rangle}{\|v_i\|^2} v_i$$

for $2 \leq k \leq m$. Then $S' := \{v_1, \dots, v_m\}$ is orthogonal and $\text{span}(S') = \text{span}(S)$.

Proof: We prove by induction on k that for every $1 \leq k \leq m$, the set $\{v_1, \dots, v_k\}$ is orthogonal and $\text{span}(\{v_1, \dots, v_k\}) = \text{span}(\{u_1, \dots, u_k\})$. For $k = 1$ this is clear. Indeed, $\{v_1\}$ is orthogonal (as every set of size 1 is trivially orthogonal), and $v_1 = u_1$.

For the induction step, let $k \geq 2$. By the induction hypothesis, $\{v_1, \dots, v_{k-1}\}$ is orthogonal. So let us show that $\langle v_k, v_j \rangle = 0$ for all $1 \leq j \leq k-1$. Indeed,

$$\begin{aligned} \langle v_k, v_j \rangle &= \left\langle u_k - \sum_{i=1}^{k-1} \frac{\langle u_k, v_i \rangle}{\|v_i\|^2} v_i, v_j \right\rangle \\ &= \langle u_k, v_j \rangle - \sum_{i=1}^{k-1} \frac{\langle u_k, v_i \rangle}{\|v_i\|^2} \langle v_i, v_j \rangle \\ &= \langle u_k, v_j \rangle - \frac{\langle u_k, v_j \rangle}{\|v_j\|^2} \cdot \langle v_j, v_j \rangle = 0. \end{aligned}$$

where we used that $\langle v_i, v_j \rangle = 0$ for $1 \leq i \neq j \leq k-1$.

Let us explain the idea behind the formula. To obtain the vector v_k , which is required to be orthogonal to $v_1, \dots, v_{k-1}$, we need to subtract from u_k the orthogonal projection of u_k onto $\text{span}(\{v_1, \dots, v_{k-1}\})$. It turns out that this orthogonal projection is exactly

$$\sum_{i=1}^{k-1} \frac{\langle u_k, v_i \rangle}{\|v_i\|^2} v_i.$$

It remains to show that

$$\text{span}(\{v_1, \dots, v_k\}) = \text{span}(\{u_1, \dots, u_k\}).$$

By the induction hypothesis,

$$\text{span}(\{v_1, \dots, v_{k-1}\}) = \text{span}(\{u_1, \dots, u_{k-1}\}).$$

Note that $v_k \neq 0$, because otherwise $u_k \in \text{span}(\{v_1, \dots, v_{k-1}\}) = \text{span}(\{u_1, \dots, u_{k-1}\})$, contradicting the assumption that $\{u_1, \dots, u_m\}$ is linearly independent. Now, by the definition

of v_k , we have

$$v_k \in \text{span}(\{v_1, \dots, v_{k-1}, u_k\}) \subseteq \text{span}(\{u_1, \dots, u_k\}).$$

Conversely, by rearranging the formula for v_k , we have

$$u_k \in \text{span}(\{v_1, \dots, v_{k-1}, v_k\}).$$

This proves the equality of spans. ☺

Note:-

Remark: In the above description of the Gram-Schmidt process, we first find an orthogonal set $v_1, \dots, v_k$, and then normalize $v_1, \dots, v_k$ (i.e., replacing v_i with $\frac{v_i}{\|v_i\|}$) to obtain an orthonormal set. Alternatively, we can normalize during the process. Namely, we can define $v_1 = \frac{u_1}{\|u_1\|}$, and then for each $k \geq 2$, let

$$v'_k := u_k - \sum_{i=1}^{k-1} \langle u_k, v_i \rangle v_i$$

and

$$v_k := \frac{v'_k}{\|v'_k\|}.$$

This makes sure that $v_1, \dots, v_k$ are orthonormal. Thus, when computing v'_k , we already know that $v_1, \dots, v_{k-1}$ have norm 1, and hence we do not need to divide by their norm.

The Gram-Schmidt process has the following immediate corollary:

Corollary 5.4.2

Every finite-dimensional inner product space V has an orthonormal basis.

Proof: Take any basis $B = (u_1, \dots, u_n)$ of V and apply the Gram-Schmidt process. This gives an orthogonal basis $(v_1, \dots, v_n)$. Normalizing gives an orthonormal basis $\left(\frac{v_1}{\|v_1\|}, \dots, \frac{v_n}{\|v_n\|}\right)$. ☺

Let us give an important example of the execution of the Gram-Schmidt process.

Example 5.4.2 (Legendre Polynomials)

Let $V = \mathbb{R}[x]$ be the space of real polynomials, with the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx.$$

Let us apply the Gram-Schmidt process to the standard basis $B = \{1, x, x^2, x^3, \dots\} = \{x^k : k \geq 0\}$. This gives an orthogonal basis for the above inner product. It is easy to see that the k 'th polynomial in this basis has degree exactly k . The resulting polynomials, properly normalized, are called the Legendre polynomials. Let us denote these polynomials as $\{P_0, P_1, P_2, \dots\} = \{P_k : k \geq 0\}$. The normalization here is not to require that the norm of P_k is 1, but rather to require that $P_k(1) = 1$, which can be shown to be equivalent to

$\langle P_k, P_k \rangle = \frac{2}{2k+1}$. Summarizing, the Legendre polynomials satisfy that

$$\langle P_k, P_\ell \rangle = 0$$

for all $k \neq \ell$, and $P_k(1) = 1$ for all k .

Let us compute the first few Legendre polynomials. Recall that we are applying the Gram-Schmidt process with input $B = \{x^k : k \geq 0\}$, i.e., $u_k = x^k$. Then $v_0 = u_0 = 1$, and this polynomial is already normalized since its value at $x = 1$ is 1. So $P_0(x) = 1$. Next,

$$v_1 = u_1 - \frac{\langle u_1, v_0 \rangle}{\|v_0\|^2} v_0.$$

We have

$$\|v_0\|^2 = \langle v_0, v_0 \rangle = \int_{-1}^1 1 dx = 2$$

and

$$\langle u_1, v_0 \rangle = \int_{-1}^1 x dx = 0.$$

(So in fact, u_1, u_0 are already orthogonal.) Thus, $v_1 = u_1 = x$. This element is also already normalized since its value at $x = 1$ is 1. So set $P_1(x) = x$.

Next, let us find $P_2(x)$. We have

$$v_2 = u_2 - \frac{\langle u_2, v_0 \rangle}{\|v_0\|^2} v_0 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1.$$

Let us compute the above inner products.

$$\langle u_2, v_0 \rangle = \langle x^2, 1 \rangle = \int_{-1}^1 x^2 dx = \frac{x^3}{3} \Big|_{-1}^1 = \frac{2}{3}.$$

$$\langle u_2, v_1 \rangle = \langle x^2, x \rangle = \int_{-1}^1 x^3 dx = 0.$$

Moreover, we already saw that $\|v_0\| = \|1\| = 2$. Also,

$$\|v_1\|^2 = \langle v_1, v_1 \rangle = \langle x, x \rangle = \int_{-1}^1 x^2 dx = \frac{x^3}{3} \Big|_{-1}^1 = \frac{2}{3}.$$

So

$$v_2 = x^2 - \frac{1}{3}.$$

Evaluating at $x = 1$, we have $v_2(1) = \frac{2}{3}$. To get P_2 , we let $P_2(x) := \frac{v_2(x)}{v_2(1)} = \frac{3}{2}x^2 - \frac{1}{2}$. This guarantees that $P_2(1) = \frac{v_2(1)}{v_2(1)} = 1$.

As a last example, let us find P_3 . We have

$$v_3 = u_3 - \sum_{i=0}^2 \frac{\langle u_3, v_i \rangle}{\|v_i\|^2} v_i.$$

It is not hard to check that $\langle u_3, v_0 \rangle = \langle u_3, v_2 \rangle = 0$. Indeed $u_3(x) = x^3$ is an odd function, v_0, v_2 (found above) are even functions, and therefore $u_3 \cdot v_0, u_3 \cdot v_2$ are odd functions. Moreover, the integral of any odd function over the symmetric interval $[-1, 1]$ is 0. So it remains to calculate:

$$\langle u_3, v_1 \rangle = \langle x^3, x \rangle = \int_{-1}^1 x^4 dx = \frac{2}{5}.$$

We also saw above that $\|v_1\|^2 = \frac{2}{3}$. Hence,

$$v_3 = x^3 - \frac{2/5}{2/3}x = x^3 - \frac{3}{5}x.$$

Normalizing, we get that $P_3(x) = \frac{v_3(x)}{v_3(1)} = \frac{v_3(x)}{2/5} = \frac{5}{2}x^3 - \frac{3}{2}x$.

So we see that the first few Legendre polynomials are

$$1, \quad x, \quad \frac{3}{2}x^2 - \frac{1}{2}, \quad \frac{5}{2}x^3 - \frac{3}{2}x, \quad \dots$$

Note that in order to find the Legendre polynomials, we did not have to run the Gram–Schmidt process on the standard basis $B = \{1, x, x^2, x^3, \dots\}$. We could instead run it on any basis $B = \{u_0(x), u_1(x), u_2(x), \dots\} = \{u_k(x) : k \geq 0\}$ of $\mathbb{R}^x$ where $\deg(u_k) = k$ and u_k is monic. Indeed, the Gram–Schmidt process would output the same basis $\{v_0(x), v_1(x), v_2(x), \dots\}$ as when applied with the standard basis. To see this, write $u_k(x) = x^k + w_k(x)$, where $\deg(w_k) \leq k-1$. When finding v_k , we already found $v_0, \dots, v_{k-1}$, and we know that $v_0, \dots, v_{k-1}$ span the space $\mathbb{R}_{k-1}[x]$ of polynomials of degree at most $k-1$, as $\text{span}(\{v_0, \dots, v_{k-1}\}) = \text{span}(\{u_0, \dots, u_{k-1}\})$. Thus, $w_k \in \text{span}(\{v_0, \dots, v_{k-1}\})$, and the corresponding linear combination is $w_k = \sum_{i=0}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} v_i$ (by previous prop, as $v_0, \dots, v_{k-1}$ are orthogonal). Then previous proposition gives

$$v_k = u_k - \sum_{i=0}^{k-1} \frac{\langle u_k, v_i \rangle}{\|v_i\|^2} v_i = x^k + w_k - \sum_{i=0}^{k-1} \frac{\langle x^k, v_i \rangle}{\|v_i\|^2} v_i - \sum_{i=0}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} v_i.$$

So the second and fourth terms (involving w_k) cancel. Namely, we get the same v_k as we would get by applying the process to the standard basis $\{x^k : k \geq 0\}$.

5.5 Orthogonal complements and orthogonal projections

Note:-

Related homework problems: see 19 for the identities involving $\ker(T^*)$ and $\text{Im}(T)$, and 20 for projections and orthogonal projections.

Note:-

Related tutorial problems: see 9.7.1 for an explicit projection matrix, 9.8.1 for detecting invariant subspaces using orthogonal projections, and 9.8.2 for the least-squares normal equations.

Note:-

Related additional exercises: see 20, 23, 22, 13, and 36 for projection and orthogonal-complement arguments.

Definition 5.5.1: Orthogonal Complement

Let V be an inner product space. For a subspace $W \subseteq V$, define the orthogonal complement $W^\perp$ of W as

$$W^\perp = \{v \in V : \langle v, w \rangle = 0 \text{ for all } w \in W\}.$$

Note:-

Note that $V^\perp = \{0\}$ and $\{0\}^\perp = V$.

Claim 5.5.1

(1) $W^\perp$ is a subspace.

Proof: This follows from the fact that the inner product is linear in the first coordinate. Let $v_1, v_2 \in W^\perp$ and $\langle cv_1, w \rangle = c \langle v_1, w \rangle = 0$. So $v_1 + v_2$ and cv_1 are in $W^\perp$, as required. ☺

(2) $W \cap W^\perp = \{0\}$.

Proof: Let $w \in W \cap W^\perp$. Then, by the definition of $W^\perp$, we have $\langle w, w \rangle = 0$. Since the inner product is positive definite, this implies that $w = 0$. ☺

(3) $W \subseteq (W^\perp)^\perp$.

Proof: Let $w \in W$ and $v \in W^\perp$. Then $\langle w, v \rangle = 0$. So $w \in (W^\perp)^\perp$, as required. ☺

Theorem 5.5.1

Let V be a finite-dimensional inner product space. Then for every subspace $W \subseteq V$, it holds that

$$V = W \oplus W^\perp.$$

Proof: We already know that $W \cap W^\perp = \{0\}$. So it suffices to show that $V = W + W^\perp$. Let $\{w_1, \dots, w_m\}$ be an orthonormal basis for W . Let $v \in V$, and set

$$w := \sum_{i=1}^m \langle v, w_i \rangle w_i.$$

We claim that $v - w \in W^\perp$. Thus, we need to show that $v - w$ is orthogonal to all elements of W . Since $w_1, \dots, w_m$ is a basis for W , it suffices to show that $\langle v - w, w_j \rangle = 0$ for all $1 \leq j \leq m$. And indeed:

$$\langle v - w, w_j \rangle = \langle v, w_j \rangle - \langle w, w_j \rangle$$

and

$$\langle w, w_j \rangle = \left\langle \sum_{i=1}^m \langle v, w_i \rangle w_i, w_j \right\rangle = \sum_{i=1}^m \langle v, w_i \rangle \cdot \langle w_i, w_j \rangle = \langle v, w_j \rangle,$$

using that $\langle w_i, w_j \rangle = \delta_{ij}$. Combining the above, we see that indeed $\langle v - w, w_j \rangle = 0$.

So $v - w \in W^\perp$ and

$$v = w + (v - w) \in W + W^\perp.$$

☺

Corollary 5.5.1

If V is finite-dimensional then for every subspace $W \subseteq V$ it holds that $(W^\perp)^\perp = W$.

Proof: We already know that $W \subseteq (W^\perp)^\perp$. Also, by the previous theorem, we have

$$\dim(W) = \dim(V) - \dim(W^\perp)$$

and

$$\dim((W^\perp)^\perp) = \dim(V) - \dim(W^\perp).$$

So $\dim(W) = \dim((W^\perp)^\perp)$, and hence $W = (W^\perp)^\perp$.

☺

If V is infinite-dimensional, then it is no longer always true that $W = (W^\perp)^\perp$ (i.e., the containment $W \subseteq (W^\perp)^\perp$ may be strict). As an example, let $V = \mathcal{C}([a, b])$ be the set of all continuous functions $f : [a, b] \rightarrow \mathbb{C}$, with the standard inner product $\langle f, g \rangle = \int_a^b f(x) \overline{g(x)} dx$. Let $W = \text{Span}(\{x^n : n \geq 0\})$ be the subspace generated by all polynomial functions. One can show that $W^\perp = \{0\}$. This can be proved using the Weierstrass approximation theorem, which states that any continuous function $f \in \mathcal{C}([a, b])$ can be uniformly approximated by a polynomial. So we see that $(W^\perp)^\perp = \{0\}^\perp = V \neq W$. What happened here is that the (topological) *closure* $\overline{W}$ of W equals V ; this again follows from the Weierstrass approximation theorem. (Here, the topology is induced by the norm derived from the inner product.) Generally, it is true that $(W^\perp)^\perp = \overline{W}$ for every subspace W . This is beyond the material for this course.

We now define the important notion of orthogonal projection.

Definition 5.5.2: Orthogonal Projection

Let V be a finite-dimensional inner product space, and let $W \subseteq V$ be a subspace. For each $v \in V$, there are unique $w \in W$ and $u \in W^\perp$ such that $v = w + u$ (by the theorem above). Define

$$P_W(v) := w.$$

The map $P_W : V \rightarrow W$ is called the orthogonal projection on W .

Note:-

Note that $P_W w = w$ for all $w \in W$ and $P_W u = 0$ for all $u \in W^\perp$.

Lemma 5.5.1

P_W is a linear map.

Proof: Let $\{w_1, w_2, \dots, w_k\}$ be an orthogonal basis for W . By definition,

$$P_W(v) = \sum_{i=1}^k \frac{\langle v, w_i \rangle}{\|w_i\|^2} w_i$$

We need to show that for all $u, v \in V$ and $c \in \mathbb{R}$:

(i) **Additivity:** $P_W(u + v) = P_W(u) + P_W(v)$

$$\begin{aligned} P_W(u + v) &= \sum_{i=1}^k \frac{\langle u + v, w_i \rangle}{\|w_i\|^2} w_i \\ &= \sum_{i=1}^k \frac{\langle u, w_i \rangle + \langle v, w_i \rangle}{\|w_i\|^2} w_i \quad (\text{by linearity of } \langle \cdot, \cdot \rangle \text{ in the first slot}) \\ &= \sum_{i=1}^k \frac{\langle u, w_i \rangle}{\|w_i\|^2} w_i + \sum_{i=1}^k \frac{\langle v, w_i \rangle}{\|w_i\|^2} w_i \\ &= P_W(u) + P_W(v) \end{aligned}$$

(ii) **Homogeneity:** $P_W(cv) = c P_W(v)$

$$\begin{aligned} P_W(cv) &= \sum_{i=1}^k \frac{\langle cv, w_i \rangle}{\|w_i\|^2} w_i \\ &= \sum_{i=1}^k \frac{c \langle v, w_i \rangle}{\|w_i\|^2} w_i \quad (\text{by linearity of } \langle \cdot, \cdot \rangle \text{ in the first slot}) \\ &= c \sum_{i=1}^k \frac{\langle v, w_i \rangle}{\|w_i\|^2} w_i \\ &= c P_W(v) \end{aligned}$$

Since both additivity and homogeneity hold, P_W is a linear map. ☺

Lemma 5.5.2

$$P_{W^\perp} = I - P_W.$$

Proof: For each $v \in V$, consider the unique way to write v as $v = w + u$ with $w \in W$ and $u \in W^\perp$. This is also the unique way to write v as a sum of an element of $W^\perp$ and an element of $(W^\perp)^\perp = W$. So $w = P_W(v)$ and $u = P_{W^\perp}(v)$. Hence,

$$P_{W^\perp}(v) = u = v - w = (I - P_W)(v).$$

☺

Proposition 5.5.1

If $\{w_1, \dots, w_m\}$ is an orthonormal basis for W , then

$$P_W(v) = \sum_{i=1}^m \langle v, w_i \rangle w_i.$$

for all $v \in V$.

Note:-**Matrices of Orthogonal Projections in $\mathbb{R}^n, \mathbb{C}^n$**

For $V = \mathbb{R}^n$ or $\mathbb{C}^n$ with the standard inner product, the orthogonal projection P_W onto a subspace W can be expressed using matrices.

For a column vector w , denote by w^* the conjugate-transpose of w , namely

$$w^* = \overline{w}^T.$$

The standard inner product between vectors $x, y \in \mathbb{F}^n$ (where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$) can be written as

$$\langle x, y \rangle = y^* x.$$

Note that the right-hand side is ordinary matrix multiplication: a row vector ($1 \times n$ matrix) multiplied by a column vector ($n \times 1$ matrix), giving a 1×1 matrix (a scalar).

Now let W be a subspace and let $w_1, \dots, w_m$ be an orthonormal basis for W . Then the orthogonal projection onto W is

$$P_W(v) = \sum_{i=1}^m w_i \langle v, w_i \rangle = \sum_{i=1}^m w_i w_i^* v.$$

Consider the $n \times n$ matrix

$$M := \sum_{i=1}^m w_i w_i^*.$$

For a vector w , the matrix ww^* satisfies

$$(ww^*)_{ij} = w_i \overline{w_j}.$$

With this notation,

$$P_W(v) = Mv.$$

Thus M is the representing matrix of the orthogonal projection P_W in the standard basis.

Lemma 5.5.3

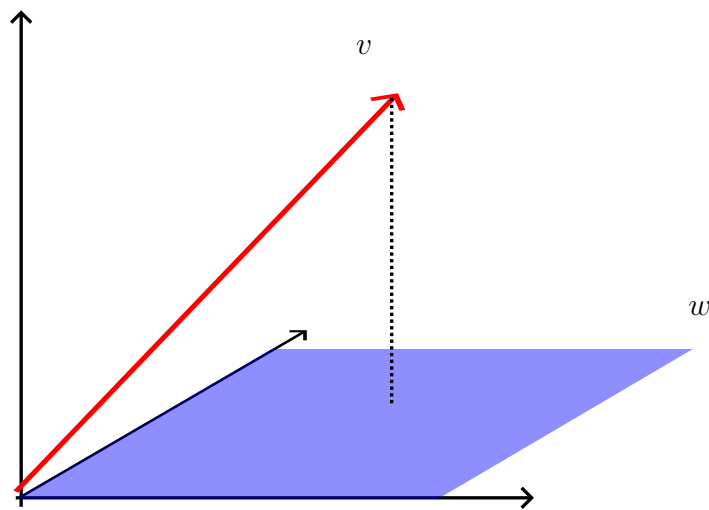
For every $v \in V$, we have $\|P_W v\| \leq \|v\|$, with equality iff $v \in W$.

Proof: Write $v = w + u$, where $w \in W$ and $u \in W^\perp$. So $P_W v = w$. Since w, u are orthogonal, by the Pythagorean theorem we have

$$\|v\|^2 = \|w\|^2 + \|u\|^2 \geq \|w\|^2 = \|P_W v\|^2,$$

with equality iff $u = 0$, i.e., iff $v = w$. $\odot$

For some geometric intuition, consider an orthogonal projection on a plane in $\mathbb{R}^3$: We can see that the orthogonal projection of v on the plane W is the point in W closest to v . Namely, the minimum in $\min_{w \in W} \|v - w\|$ is obtained for $w := P_W v$.



We now show that this is true in general:

Theorem 5.5.2

Let V be a finite-dimensional inner product space, and let $W \subseteq V$ be a subspace. For every $v \in V$, it holds that

$$\min_{w \in W} \|v - w\| = \|v - P_W v\|,$$

and the only minimizer is $w = P_W v$.

Proof: Write $v = w^* + u$, where $w^* \in W$ and $u \in W^\perp$. So $P_W v = w^*$. For every $w \in W$, we have

$$\|v - w\|^2 = \|(w^* - w) + u\|^2 = \|w^* - w\|^2 + \|u\|^2 \geq \|u\|^2,$$

with equality if and only if $w = w^*$. This proves the claim. Note that the equality holds by Pythagoras, since $w^* - w \in W$ is orthogonal to $u \in W^\perp$. $\odot$

5.6 The Adjoint Operator

Note:-

Related homework problems: see 19 for fundamental subspace identities involving adjoints, and 21 for an adjoint-based proof of Schur triangularization.

Note:-

Related tutorial problems: see 9.7.1 for the representation result behind adjoints, 9.8.2 for the role of T^*T in least squares, and 9.8.3 for the relationship between eigenvalues of T and T^* .

Note:-

Related additional exercises: see 30 for characteristic polynomials after complexification and adjoints, 34 for kernels of T and T^* , and 44 for a rank proof using T^*T .

Our starting point is the following calculation:

Proposition 5.6.1

Let $A \in \mathbb{C}^{m \times n}$, and let $x \in \mathbb{C}^n$, $y \in \mathbb{C}^m$. Then

$$\langle Ax, y \rangle = \sum_{i=1}^m \sum_{j=1}^n A_{ij} x_j \overline{y_i}$$

Proof:

$$\langle Ax, y \rangle = \sum_{i=1}^m (Ax)_i \overline{y_i} = \sum_{i=1}^m \left(\sum_{j=1}^n A_{ij} x_j \right) \overline{y_i} = \sum_{i=1}^m \sum_{j=1}^n A_{ij} x_j \overline{y_i}$$

☺

Recall that we use the notation A^* for the matrix $A^* := \overline{A^T}$. The matrix A^* is called the *conjugate transpose* or the *adjoint* of A . We have the following important identity:

Proposition 5.6.2

Let $A \in \mathbb{C}^{m \times n}$. Then for all $x \in \mathbb{C}^n$, $y \in \mathbb{C}^m$, it holds that

$$\langle Ax, y \rangle = \langle x, A^*y \rangle$$

Note:-

A^* is $n \times m$, so the multiplication A^*y makes sense.

Proof:

$$\begin{aligned} \langle x, A^*y \rangle &= \sum_{j=1}^n x_j \cdot \overline{(A^*y)_j} = \sum_{j=1}^n x_j \overline{\sum_{i=1}^m A_{ji}^* y_i} \stackrel{(*)}{=} \sum_{i=1}^m \sum_{j=1}^n \overline{(A^T)_{ji}} x_j \overline{y_i} \\ &= \sum_{i=1}^m \sum_{j=1}^n A_{ij} x_j \overline{y_i} = \langle Ax, y \rangle. \end{aligned}$$

In (*) we use that $A^* = \overline{A^T}$, and so taking conjugate again gives A^T . $\ominus$

The above motivates obtaining analogous statements for linear maps. And indeed, we prove the following:

Theorem 5.6.1 Existence and Uniqueness of the Adjoint

Let V, W be finite-dimensional inner product spaces. Then for every linear map $T : V \rightarrow W$, there is a unique linear map $T^* : W \rightarrow V$ satisfying that

$$\langle Tv, w \rangle = \langle v, T^*w \rangle \quad (5.2)$$

for all $v \in V$ and $w \in W$. Furthermore, if $B = (v_1, \dots, v_n)$ and $C = (w_1, \dots, w_m)$ are orthonormal bases for V and W , respectively, then

$$[T^*]_C^B = [T]_B^C^*. \quad (5.3)$$

In other words, (5.3) states that the representing matrix of T^* in the bases C, B is precisely the conjugate transpose of the representing matrix of T in bases B, C .

Note:-

In (5.2), on the LHS the inner product is in W and on the RHS it is in V . To stress this, we could write $\langle Tv, w \rangle_W = \langle v, T^*w \rangle_V$, where $\langle \cdot, \cdot \rangle_V$ denotes the inner product on V and $\langle \cdot, \cdot \rangle_W$ the inner product on W .

Definition 5.6.1: Adjoint Map

The linear map T^* given by 5.6.1 is called the *adjoint* of the linear map T .

Proof of 5.6.1: We first prove existence. Fix an orthonormal basis $B = (v_1, \dots, v_n)$ of V and an orthonormal basis $C = (w_1, \dots, w_m)$ of W . Let $A := [T]_B^C$. We know that there is a unique linear map $T^* : W \rightarrow V$ such that

$$[T^*]_C^B = A^*.$$

(Indeed, here we use the fact that $S \mapsto [S]_C^B$ is a linear isomorphism from the space $\mathcal{L}(W, V)$ of linear maps from W to V to the space of matrices $\mathbb{C}^{n \times m}$.)

Now let $v \in V, w \in W$. We have

$$[Tv]_C = [T]_B^C \cdot [v]_B = A \cdot [v]_B.$$

By Corollary 5.4.1,

$$\langle Tv, w \rangle = \langle [Tv]_C, [w]_C \rangle = \langle [T]_B^C[v]_B, [w]_C \rangle = \langle A \cdot [v]_B, [w]_C \rangle$$

and similarly

$$\langle v, T^*w \rangle = \langle [v]_B, [T^*w]_B \rangle = \langle [v]_B, [T^*]_C^B[w]_C \rangle = \langle [v]_B, A^* \cdot [w]_C \rangle.$$

Finally, by 5.6.2, we have

$$\langle A \cdot [v]_B, [w]_C \rangle = \langle [v]_B, A^* \cdot [w]_C \rangle.$$

Combining all of the above gives $\langle Tv, w \rangle = \langle v, T^*w \rangle$, as required.

We now prove uniqueness. So suppose that T_1^*, T_2^* are both adjoint to T . Then for all $v \in V, w \in W$,

$$\langle v, T_1^*w \rangle = \langle Tv, w \rangle = \langle v, T_2^*w \rangle.$$

Hence,

$$\langle v, T_1^*w - T_2^*w \rangle = 0 \quad (5.4)$$

for all $v \in V, w \in W$.

Claim 5.6.1

In an inner product space V , if a vector $u \in V$ satisfies $\langle v, u \rangle = 0$ for all $v \in V$, then $u = 0$.

Proof: If $u \neq 0$ then $\langle u, u \rangle \neq 0$ (so the assumption fails for $v := u$). $\ominus$

By 5.6.1 and (5.4), we have $T_1^*w - T_2^*w = 0$ and hence $T_1^*w = T_2^*w$ for all $w \in W$. Hence, $T_1^* = T_2^*$, as required.

Finally, note that the adjoint T^* can be defined from any orthonormal bases B and C (of V and W , respectively) as done above, i.e., by requiring that $[T^*]_C^B = [T_B^C]^*$. Since the adjoint is unique, it follows that (5.3) holds for any two orthonormal bases B and C . $\ominus$

Example 5.6.1

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear map defined by $T(x_1, x_2) = (x_1 + 2x_2, 3x_1, x_1 - x_2)$, where $\mathbb{R}^2$ and $\mathbb{R}^3$ are equipped with their standard inner products. We use the standard (orthonormal) bases $B = (e_1, e_2)$ of $\mathbb{R}^2$ and $C = (e_1, e_2, e_3)$ of $\mathbb{R}^3$. The representing matrix of T in these bases is

$$A = [T]_B^C = \begin{pmatrix} 1 & 2 \\ 3 & 0 \\ 1 & -1 \end{pmatrix}.$$

By 5.6.1, the adjoint $T^* : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ satisfies $[T^*]_C^B = A^*$. Since we are over $\mathbb{R}$, the conjugate transpose is just the transpose:

$$[T^*]_C^B = A^T = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 0 & -1 \end{pmatrix}.$$

Hence $T^*(y_1, y_2, y_3) = (y_1 + 3y_2 + y_3, 2y_1 - y_3)$. We can verify the defining property (5.2) directly. For example, taking $v = (1, 2)$ and $w = (1, 0, -1)$:

$$\begin{aligned} \langle Tv, w \rangle &= \langle (5, 3, -1), (1, 0, -1) \rangle = 5 \cdot 1 + 3 \cdot 0 + (-1) \cdot (-1) = 6, \\ \langle v, T^*w \rangle &= \langle (1, 2), (0, 3) \rangle = 1 \cdot 0 + 2 \cdot 3 = 6. \end{aligned}$$

Let us now establish some basic properties of adjoints. Recall that $\mathcal{L}(V, W)$ denotes the space of all linear maps from V to W . For the rest of this section, all vector spaces are finite-dimensional inner product spaces.

Proposition 5.6.3 Basic Properties of the Adjoint

1. For all $T \in \mathcal{L}(V, W)$ it holds that $(T^*)^* = T$.
2. For all $S, T \in \mathcal{L}(V, W)$ and scalars $\lambda \in \mathbb{C}$,

$$(S + T)^* = S^* + T^*$$

and

$$(\lambda T)^* = \bar{\lambda} \cdot T^*$$

3. For all $S \in \mathcal{L}(U, V)$ and $T \in \mathcal{L}(V, W)$,

$$(TS)^* = S^*T^*.$$

4. If $T \in \mathcal{L}(V, W)$ is invertible, then so is $T^* \in \mathcal{L}(W, V)$, and

$$(T^*)^{-1} = (T^{-1})^*.$$

Proof: In each case, we just have to check that the defining property of the adjoint, namely the equality (5.2), is satisfied.

1. We have

$$\langle T^*w, v \rangle = \overline{\langle v, T^*w \rangle} = \overline{\langle Tv, w \rangle} = \langle w, Tv \rangle$$

for all $w \in W$, $v \in V$. Here, the second equality uses the defining property of T^* . The above implies, by definition, that T is the adjoint of T^* , namely, $(T^*)^* = T$.

2. For all $v \in V$, $w \in W$, we have

$$\langle (S + T)v, w \rangle = \langle Sv, w \rangle + \langle Tv, w \rangle = \langle v, S^*w \rangle + \langle v, T^*w \rangle = \langle v, (S^* + T^*)w \rangle.$$

Therefore, by definition, $S^* + T^*$ is the adjoint of $(S + T)$, namely, $(S + T)^* = S^* + T^*$.

Similarly,

$$\langle (\lambda T)v, w \rangle = \lambda \langle Tv, w \rangle = \lambda \langle v, T^*w \rangle = \langle v, (\bar{\lambda} \cdot T^*)w \rangle.$$

Again, this implies by definition that $\bar{\lambda} \cdot T^*$ is the adjoint of λT , namely $(\lambda T)^* = \bar{\lambda} \cdot T^*$.

3. For all $u \in U$, $w \in W$, we have

$$\langle TSu, w \rangle = \langle T(Su), w \rangle = \langle Su, T^*w \rangle = \langle u, S^*T^*w \rangle.$$

Hence, $(TS)^* = S^*T^*$.

4. We will derive this from 3. First, we observe that the adjoint of the identity map $I_V : V \rightarrow V$ is itself, namely, $(I_V)^* = I_V$. Indeed, for all $v, v' \in V$ we trivially have $\langle I_V v, v' \rangle = \langle v, v' \rangle = \langle v, I_V v' \rangle$. Now, $T^{-1}T = I_V$. Therefore, by 3, we have

$$I_V = (I_V)^* = (T^{-1}T)^* = T^* \cdot (T^{-1})^*.$$

Similarly, $TT^{-1} = I_W$, and therefore

$$I_W = (I_W)^* = (TT^{-1})^* = (T^{-1})^* \cdot T^*.$$

It follows that $(T^*)^{-1} = (T^{-1})^*$, as required.

☺

1 of 5.6.3 implies that $\langle T^*v, w \rangle = \langle v, Tw \rangle$ for all v, w . Also, by the definition of the adjoint, we have $\langle Tv, w \rangle = \langle v, T^*w \rangle$ for all v, w . Thus, informally speaking, in an inner product we can move a linear map T from the first coordinate to the second coordinate or vice versa, by taking its adjoint.

Next, we compute the adjoint of $f(T)$ for a polynomial f .

Lemma 5.6.1 Adjoint of a Polynomial in T

Let $T : V \rightarrow V$ and let $f(x) = \sum_{i=0}^d a_i x^i$ be a polynomial. Then

$$(f(T))^* = \sum_{i=0}^d \overline{a_i} \cdot (T^*)^i.$$

Proof: We expand $(f(T))^*$ using 5.6.3, as follows:

$$(f(T))^* = \left(\sum_{i=0}^d a_i T^i \right)^* \stackrel{(*)}{=} \sum_{i=0}^d \overline{a_i} \cdot (T^i)^* \stackrel{(**)}{=} \sum_{i=0}^d \overline{a_i} \cdot (T^*)^i,$$

as required. Here, in $(*)$ we used that $(S + T)^* = S^* + T^*$ and $(\lambda T)^* = \overline{\lambda} \cdot T^*$. In $(**)$ we used that $(ST)^* = T^*S^*$, and therefore $(T^i)^* = (T^*)^i$ for all $i \geq 1$. ☺

Finally, we prove the following lemma, showing that if W is T -invariant, then its orthogonal complement $W^\perp$ is T^* -invariant. This will be used later on.

Lemma 5.6.2 T^* -Invariance of Orthogonal Complement

Let V be an inner product space, let $T : V \rightarrow V$ be a linear map, and let $W \subseteq V$ be a T -invariant subspace. Then $W^\perp$ is T^* -invariant.

Proof: Let $u \in W^\perp$. Then for all $w \in W$, we have

$$\langle T^*u, w \rangle = \langle u, Tw \rangle \stackrel{(*)}{=} 0,$$

where $(*)$ uses that $Tw \in W$, as W is T -invariant. Since the above holds for all $w \in W$, it follows that $T^*u \in W^\perp$. Hence, $W^\perp$ is T^* -invariant. ☺

Corollary 5.6.1

If W is both T -invariant and T^* -invariant, then $W^\perp$ is both T -invariant and T^* -invariant.

Proof: Since W is T -invariant, 5.6.2 (applied to T) gives that $W^\perp$ is T^* -invariant.

Since W is T^* -invariant, we apply 5.6.2 to T^* : this gives that $W^\perp$ is $(T^*)^*$ -invariant. By 5.6.31, $(T^*)^* = T$, so $W^\perp$ is T -invariant.

Hence, $W^\perp$ is both T -invariant and T^* -invariant. ☺

Chapter 6

Spectral Decomposition in Inner Product Spaces

6.1 Normal Operators

Note:-

Related homework problems: see [23](#) for invariant subspaces of normal operators, and [24](#) for the minimax principle for self-adjoint operators.

Note:-

Related tutorial problems: see [9.9.1](#) for a converse to the normal-operator norm identity, [9.9.2](#) for the numerical range of a normal operator, [9.9.3](#) for maps commuting with spectral projections, and [9.10.2](#) for the matrix form of the spectral theorem.

Note:-

Related additional exercises: see [29](#), [37](#), and [38](#) for normal, commuting, and skew-adjoint operator arguments.

Consider a finite-dimensional (complex) inner product space V . We start by considering the following natural question:

Question 1

Which linear maps $T : V \rightarrow V$ admit an orthogonal basis of eigenvectors?

We will be able to precisely characterize such maps. To this end, we need the following definition.

Definition 6.1.1: Normal Operator

A linear map $T : V \rightarrow V$ is *normal* if $TT^* = T^*T$.
Namely, T is normal if it commutes with its adjoint.

Note:-

$T : V \rightarrow V$, let B be an orthonormal basis of V , and let $A := [T]_B$. By 5.6.1, we have $[T^*]_B = A^*$. Hence, $[TT^*]_B = AA^*$ and $[T^*T]_B = A^*A$. Therefore, T is normal iff $AA^* = A^*A$.

The following is the main theorem we will prove:

Theorem 6.1.1 Orthogonal Diagonalization in Complex Inner Product Spaces

Let V be a finite-dimensional complex inner product space and let $T : V \rightarrow V$. Then TFAE:

1. T is normal.
2. There is an orthonormal basis of V consisting of eigenvectors of T .

In particular, over $\mathbb{C}$, every normal map is diagonalizable.

We stress that it is crucial for this theorem that we work over $\mathbb{C}$. The reason is that over $\mathbb{R}$, a normal map does not necessarily have real eigenvalues, and thus normality is not sufficient for diagonalizability. We will consider the real case later on.

Later on, we will also study the following two important special cases of normal maps:

- T is *Hermitian* (or *self-adjoint*) if $T = T^*$.
- T is *unitary* if $TT^* = T^*T = I$.

(Over $\mathbb{F} = \mathbb{R}$, self-adjoint operators are also called *symmetric*, and unitary operators are also called *orthogonal*.)

Towards 6.1.1, we now establish some properties of normal operators. We begin with the following lemma.

Lemma 6.1.1 Normality is Preserved by Polynomials and Inversion

Let $T : V \rightarrow V$ be a normal map. Then:

1. For all polynomials f , it holds that $f(T)$ is normal. In particular, λT and $T - \lambda I$ are normal for all scalars λ .
2. If T is invertible then T^{-1} is normal.

Proof: **Idea:** For 1, we use the adjoint formula for polynomials and the fact that powers of T and T^* commute whenever T and T^* commute. For 2, we use the identity $(T^{-1})^* = (T^*)^{-1}$ and rewrite the product using the normality of T .

Proof of 1. Write $f(x) = \sum_{i=0}^d a_i x^i$, so that $f(T) = \sum_{i=0}^d a_i T^i$. By 5.6.1,

$$(f(T))^* = \sum_{j=0}^d \overline{a_j} \cdot (T^*)^j.$$

Step 1: powers of T and T^ commute.* We claim that $T^i (T^*)^j = (T^*)^j T^i$ for all $i, j \geq 0$. Fix j and induct on i . The case $i = 0$ is immediate. Assuming $T^i (T^*)^j = (T^*)^j T^i$, we compute

$$T^{i+1} (T^*)^j = T \cdot T^i (T^*)^j = T \cdot (T^*)^j T^i.$$

, where the second equality applies the induction hypothesis. It now suffices to show that $T(T^*)^j = (T^*)^j T$, i.e., that T commutes with every power of T^* . Inducting on j : for $j = 0$ this is trivial, and assuming $T(T^*)^j = (T^*)^j T$, we get

$$T(T^*)^{j+1} = T(T^*)^j \cdot T^* = (T^*)^j T \cdot T^* \stackrel{(\dagger)}{=} (T^*)^j T^* T = (T^*)^{j+1} T,$$

where $(\dagger)$ uses normality $TT^* = T^*T$. This proves the claim.

Step 2: $f(T)$ commutes with $(f(T))^$.* Using the bilinearity of composition and the commutation $T^i(T^*)^j = (T^*)^j T^i$ from Step 1,

$$\begin{aligned} f(T) \cdot (f(T))^* &= \left(\sum_{i=0}^d a_i T^i \right) \left(\sum_{j=0}^d \overline{a_j} (T^*)^j \right) = \sum_{i,j} a_i \overline{a_j} T^i (T^*)^j \\ &= \sum_{i,j} a_i \overline{a_j} (T^*)^j T^i = \left(\sum_{j=0}^d \overline{a_j} (T^*)^j \right) \left(\sum_{i=0}^d a_i T^i \right) = (f(T))^* \cdot f(T). \end{aligned}$$

Hence $f(T)$ is normal.

The “in particular” statements follow by taking $f(x) = \lambda x$ and $f(x) = x - \lambda$, respectively.

Proof of 2. First we verify the identity $(T^{-1})^* = (T^*)^{-1}$. Starting from $TT^{-1} = I = T^{-1}T$ and taking adjoints (using $(AB)^* = B^*A^*$ and $I^* = I$),

$$(T^{-1})^* T^* = I^* = I = I^* = T^* (T^{-1})^*,$$

which says exactly that $(T^{-1})^*$ is the inverse of T^* , i.e., $(T^{-1})^* = (T^*)^{-1}$. In particular, T^* is invertible.

Since T and T^* are both invertible, we can apply the identity $(AB)^{-1} = B^{-1}A^{-1}$ for invertible A, B . We compute

$$T^{-1}(T^{-1})^* = T^{-1}(T^*)^{-1} \stackrel{(*)}{=} (T^*T)^{-1} \stackrel{(**)}{=} (TT^*)^{-1} \stackrel{(*)}{=} (T^*)^{-1}T^{-1} = (T^{-1})^*T^{-1},$$

where $(*)$ uses $(AB)^{-1} = B^{-1}A^{-1}$ (applied with $A = T^*, B = T$ and then $A = T, B = T^*$), and $(**)$ uses normality $T^*T = TT^*$. Hence T^{-1} commutes with its adjoint, so T^{-1} is normal. $\odot$

Lemma 6.1.2 Normal Maps Preserve Norms Under Adjoint

If $T : V \rightarrow V$ is normal then $\|Tv\| = \|T^*v\|$ for all $v \in V$.

Proof: **Idea:** We express $\|Tv\|^2$ and $\|T^*v\|^2$ in terms of inner products and use the normality condition $T^*T = TT^*$ to equate them.

By assumption, $TT^* = T^*T$. Now, for all $v \in V$,

$$\|Tv\|^2 = \langle Tv, Tv \rangle \stackrel{(*)}{=} \langle v, T^*Tv \rangle \stackrel{(**)}{=} \langle v, TT^*v \rangle = \langle T^*v, T^*v \rangle = \|T^*v\|^2.$$

Hence, $\|Tv\| = \|T^*v\|$. $\odot$

Note:-

The converse of 6.1.2 is also true: if $\|Tv\| = \|T^*v\|$ for all $v \in V$, then T is normal. This can be proved as an exercise.

We can now prove the following important facts about normal maps.

Lemma 6.1.3 Eigenvector Properties of Normal Maps

Let $T : V \rightarrow V$ be a normal map. Then:

1. If v is an eigenvector of T with eigenvalue λ , then v is an eigenvector of T^* with eigenvalue $\bar{\lambda}$.
2. If u, v are non-zero eigenvectors of T with distinct eigenvalues, then $\langle u, v \rangle = 0$ (i.e., u, v are orthogonal).

Proof: Idea: For 1, the key insight is that $T - \lambda I$ is itself normal, so its norm-equality (6.1.2) forces $(T^* - \bar{\lambda}I)v = 0$ whenever $(T - \lambda I)v = 0$. For 2, we compute $\langle Tu, v \rangle$ in two ways—once using $Tu = \lambda u$ and once using 1 and the adjoint—and the two distinct eigenvalues force $\langle u, v \rangle = 0$.

1. Since v is an eigenvector of T with eigenvalue λ , we have $Tv = \lambda v$, i.e., $(T - \lambda I)v = 0$.

By 6.1.1 (with $f(x) = x - \lambda$), the operator $T - \lambda I$ is normal. By 5.6.1, its adjoint is

$$(T - \lambda I)^* = T^* - \bar{\lambda}I.$$

Applying 6.1.2 to the normal operator $T - \lambda I$ and the vector v gives

$$\|(T - \lambda I)v\| = \|(T - \lambda I)^*v\| = \|(T^* - \bar{\lambda}I)v\|.$$

The left side equals $\|0\| = 0$ since $(T - \lambda I)v = 0$. Hence,

$$\|(T^* - \bar{\lambda}I)v\| = 0.$$

Because a norm is zero if and only if the vector is zero (positive-definiteness of the inner product), this gives $(T^* - \bar{\lambda}I)v = 0$, i.e., $T^*v = \bar{\lambda}v$. Thus v is an eigenvector of T^* with eigenvalue $\bar{\lambda}$.

2. Let λ be the eigenvalue of u and let μ be the eigenvalue of v ; so $\lambda \neq \mu$. We compute $\langle Tu, v \rangle$, as follows:

$$\lambda \langle u, v \rangle = \langle \lambda u, v \rangle = \langle Tu, v \rangle = \langle u, T^*v \rangle \stackrel{(*)}{=} \langle u, \bar{\mu}v \rangle = \bar{\mu} \langle u, v \rangle. \quad (6.1)$$

The key step here is the equality $(*)$, where we use 1. Also, the last equality in (6.1) uses that $\bar{\bar{z}} = z$ for all complex z .

By (6.1), we now see that if $\langle u, v \rangle \neq 0$ then $\lambda = \mu$, a contradiction. Hence, $\langle u, v \rangle = 0$.

☺

Finally, we are ready to prove 6.1.1:

Proof of 6.1.1: Idea: The direction $2 \rightarrow 1$ is a direct computation: given an orthonormal eigenbasis, one explicitly verifies $T^*T = TT^*$ by evaluating both sides on each basis vector. The direction $1 \rightarrow 2$ proceeds by induction on $\dim(V)$: since we work over $\mathbb{C}$, T has an eigenvalue λ with eigenvector v ; by 6.1.3, v is also an eigenvector of T^* , so $W = \text{Span}(\{v\})$ is both T - and T^* -invariant; by 5.6.1, $W^\perp$ is invariant as well, and the restriction $T|_{W^\perp}$ is normal of smaller dimension, so the induction hypothesis applies.

2 $\rightarrow$ 1: Suppose that T admits an orthonormal basis of eigenvectors $v_1, \dots, v_n$. Let λ_i be the eigenvalue of v_i , $1 \leq i \leq n$. So $Tv_i = \lambda_i v_i$. We claim that

$$T^*v_i = \overline{\lambda_i}v_i. \quad (6.2)$$

To see this, recall that since $(v_1, \dots, v_n)$ is an orthonormal basis, for every $x \in V$ we have $x = \sum_{j=1}^n \langle x, v_j \rangle v_j$. Applying this to $x = T^*v_i$, we have

$$\begin{aligned} T^*v_i &= \sum_{j=1}^n \langle T^*v_i, v_j \rangle v_j = \sum_{j=1}^n \langle v_i, Tv_j \rangle v_j = \sum_{j=1}^n \langle v_i, \lambda_j v_j \rangle v_j \\ &= \sum_{j=1}^n \overline{\lambda_j} \langle v_i, v_j \rangle v_j = \overline{\lambda_i}v_i, \end{aligned}$$

where the last equality uses that $\langle v_i, v_j \rangle = \delta_{ij}$. This proves (6.2).

We now use the above to show that $T^*T = TT^*$. It is enough to show that the maps T^*T and TT^* agree on a basis. And indeed, for every $1 \leq i \leq n$ we have

$$T^*Tv_i = T^*\lambda_i v_i = \lambda_i(T^*v_i) = \lambda_i \cdot \overline{\lambda_i}v_i = |\lambda_i|^2 v_i$$

and

$$TT^*v_i = T\overline{\lambda_i}v_i = \overline{\lambda_i}(Tv_i) = \overline{\lambda_i} \cdot \lambda_i v_i = |\lambda_i|^2 v_i,$$

as required.

1 $\rightarrow$ 2: We prove by induction on $n := \dim(V)$ that if $T : V \rightarrow V$ is normal then T admits an orthonormal basis of eigenvectors. The base case $n = 0$ is trivial.

For the induction step, suppose that $n \geq 1$. Since we are working over $\mathbb{C}$, every linear map T has an eigenvalue; indeed, this is because the characteristic polynomial p_T splits, and in particular has a root. So let λ be an eigenvalue of T and let v be a non-zero eigenvector of T . By 6.1.3, v is also an eigenvector of T^* . Now put

$$W := \text{Span}(\{v\}).$$

Since v is an eigenvector of T and T^* , it follows that W is T -invariant and T^* -invariant. By 5.6.1, $W^\perp$ is T -invariant and T^* -invariant.

So let us consider the restriction

$$S := T|_{W^\perp} : W^\perp \rightarrow W^\perp.$$

We claim that S is normal. First, let us identify its adjoint. Since $W^\perp$ is T^* -invariant, the restriction

$$T^*|_{W^\perp} : W^\perp \rightarrow W^\perp$$

is well-defined. For all $x, y \in W^\perp$, using the inner product inherited from V , we have

$$\langle Sx, y \rangle = \langle Tx, y \rangle = \langle x, T^*y \rangle = \langle x, (T^*|_{W^\perp})y \rangle.$$

By the uniqueness of the adjoint, this proves

$$S^* = (T|_{W^\perp})^* = T^*|_{W^\perp}.$$

Now compute the two products on $W^\perp$:

$$SS^* = (T|_{W^\perp})(T^*|_{W^\perp}) = (TT^*)|_{W^\perp},$$

and

$$S^*S = (T^*|_{W^\perp})(T|_{W^\perp}) = (T^*T)|_{W^\perp}.$$

Since T is normal, $TT^* = T^*T$, so the two restrictions are equal. Hence $SS^* = S^*S$, and therefore $S = T|_{W^\perp}$ is normal. Also, $\dim(W^\perp) = n - 1$.

By the induction hypothesis, $T|_{W^\perp}$ admits an orthonormal basis of eigenvectors $\{v_1, \dots, v_{n-1}\}$. Finally, set

$$v_n := \frac{v}{\|v\|}.$$

Then $\{v_1, \dots, v_{n-1}, v_n\}$ is an orthonormal basis of V (since $v_1, \dots, v_{n-1} \in W^\perp$ are orthogonal to $v_n \in W$), and every vector in this basis is an eigenvector of T . This completes the proof. $\odot$

6.1.1 can be restated in the following form:

Theorem 6.1.2 Spectral Decomposition of Normal Operators

Let V be a finite-dimensional complex inner product space. Let $T : V \rightarrow V$ be a normal map, and let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of T . For each $1 \leq i \leq r$, let P_i be the orthogonal projection on the eigenspace $E(\lambda_i, T)$. Then:

1. $P_i P_j = 0$ for $i \neq j$.
2. $\sum_{i=1}^r P_i = I$.
3. $\sum_{i=1}^r \lambda_i P_i = T$.

Proof: **Idea:** Since T is normal, V decomposes as a direct sum of eigenspaces. The orthogonal projections onto distinct eigenspaces annihilate each other because eigenvectors with different eigenvalues are orthogonal (6.1.3). The resolution of the identity and the spectral formula then follow by decomposing any $v \in V$ along the eigenspaces.

Proof of 1: Let $1 \leq i \neq j \leq r$. By 6.1.3, any two eigenvectors of T belonging to different eigenvalues are orthogonal. Hence, $E(\lambda_j, T) \subseteq (E(\lambda_i, T))^\perp$.

Now, the image of P_j is $E(\lambda_j, T)$. Also, P_i maps $(E(\lambda_i, T))^\perp$ to 0. It follows that $P_i P_j = 0$.

Proof of 2 and 3: By 6.1.1, T is diagonalizable, and therefore $V = \bigoplus_{i=1}^r E(\lambda_i, T)$.

Fix any $v \in V$, and write $v = v_1 + \cdots + v_r$ with $v_i \in E(\lambda_i, T)$. Then $Tv_i = \lambda_i v_i$. Also,

$$P_i v_j = \begin{cases} v_j & i = j, \\ 0 & i \neq j. \end{cases} \quad (6.3)$$

Indeed, P_j fixes any element in $E(\lambda_j, T)$, so $P_j v_j = v_j$. For $i \neq j$, we have $P_i v_j = 0$ because $v_j \in E(\lambda_j, T) \subseteq (E(\lambda_i, T))^\perp$.

Now, we have:

$$\left(\sum_{i=1}^r P_i \right) v = \sum_{i=1}^r P_i \left(\sum_{j=1}^r v_j \right) \stackrel{(6.3)}{=} \sum_{i=1}^r P_i v_i = \sum_{i=1}^r v_i = v.$$

This shows that $\sum_{i=1}^r P_i = I$.

Similarly,

$$\left(\sum_{i=1}^r \lambda_i P_i \right) v \stackrel{(6.3)}{=} \sum_{i=1}^r \lambda_i v_i = \sum_{i=1}^r T v_i = T v,$$

which shows that $\sum_{i=1}^r \lambda_i P_i = T$. ☺

Theorem 6.1.3 Orthogonal Projections are Self-Adjoint

Let V be a finite-dimensional inner product space, and let $W \subseteq V$ be a subspace. Then the orthogonal projection $P_W : V \rightarrow V$ satisfies $P_W^* = P_W$.

Proof: We verify the defining property of the adjoint: we show that $\langle P_W v_1, v_2 \rangle = \langle v_1, P_W v_2 \rangle$ for all $v_1, v_2 \in V$.

Write $v_1 = w_1 + u_1$ and $v_2 = w_2 + u_2$, where $w_1, w_2 \in W$ and $u_1, u_2 \in W^\perp$. Then $P_W v_1 = w_1$ and $P_W v_2 = w_2$. We compute:

$$\langle P_W v_1, v_2 \rangle = \langle w_1, w_2 + u_2 \rangle = \langle w_1, w_2 \rangle + \langle w_1, u_2 \rangle = \langle w_1, w_2 \rangle,$$

since $w_1 \in W$ and $u_2 \in W^\perp$ are orthogonal. Similarly,

$$\langle v_1, P_W v_2 \rangle = \langle w_1 + u_1, w_2 \rangle = \langle w_1, w_2 \rangle + \langle u_1, w_2 \rangle = \langle w_1, w_2 \rangle,$$

since $u_1 \in W^\perp$ and $w_2 \in W$ are orthogonal. Therefore,

$$\langle P_W v_1, v_2 \rangle = \langle w_1, w_2 \rangle = \langle v_1, P_W v_2 \rangle$$

for all $v_1, v_2 \in V$, and so $P_W^* = P_W$ by uniqueness of the adjoint. ☺

Lemma 6.1.4 Adjoint of a Normal Operator via Spectral Decomposition

In the notation of 6.1.2, we have

$$T^* = \sum_{i=1}^r \overline{\lambda_i} P_i.$$

Proof: By 6.1.3, $P^* = P$ for every orthogonal projection P . Hence, $P_i^* = P_i$ for all $1 \leq i \leq r$. Now, by 5.6.3, we have

$$T^* = \left(\sum_{i=1}^r \lambda_i P_i \right)^* = \sum_{i=1}^r \overline{\lambda_i} P_i^* = \sum_{i=1}^r \overline{\lambda_i} P_i.$$



6.2 Self-Adjoint Operators

Note:-

Related homework problems: see 22 for a self-adjointness criterion, 24 for the Rayleigh quotient minimax formula, and 26 for negative eigenvalues under low-rank perturbations.

Note:-

Related tutorial problems: see 9.10.1 for characterizing self-adjoint maps by normality and real eigenvalues, and 9.6.2 for a related zero quadratic form criterion over $\mathbb{C}$.

Note:-

Related additional exercises: see 31 for functional calculus on a real self-adjoint matrix and 33 for maps that are simultaneously self-adjoint and unitary.

We now consider an important subclass of normal operators.

Definition 6.2.1: Self-Adjoint Operator

Let V be an inner product space. A linear map $T : V \rightarrow V$ is *self-adjoint* (or *Hermitian*) if $T^* = T$.

Clearly, if T is self-adjoint then T is normal.

Let us recall again the connection to matrices. Fix an orthonormal basis B for V and let $A = [T]_B$. Then $[T^*]_B = A^*$ by 5.6.1. Therefore, $T = T^*$ if and only if $A = A^*$.

Definition 6.2.2: Self-Adjoint Matrix

A matrix $A \in \mathbb{C}^{n \times n}$ is *self-adjoint* (or *Hermitian*) if $A^* = A$.

Note:-

If A is a real matrix then $A^* = A^T$ (since $\bar{z} = z$ for all real numbers z). A matrix $A \in \mathbb{R}^{n \times n}$ satisfying $A^T = A$ is called *symmetric*.

We start by showing that self-adjoint maps can only have real eigenvalues.

Lemma 6.2.1 Eigenvalues of Self-Adjoint Maps are Real

Let $T : V \rightarrow V$ be a self-adjoint map, and let λ be an eigenvalue of T . Then $\lambda \in \mathbb{R}$.

Proof: **Idea:** Compute $\langle Tv, v \rangle$ two ways: directly as $\lambda \langle v, v \rangle$, and via the adjoint as $\bar{\lambda} \langle v, v \rangle$. Since $\langle v, v \rangle \neq 0$, we conclude $\lambda = \bar{\lambda}$.

Let v be a non-zero eigenvector of λ . Then

$$\lambda \langle v, v \rangle = \langle \lambda v, v \rangle = \langle Tv, v \rangle = \langle v, T^*v \rangle = \langle v, Tv \rangle = \bar{\lambda} \langle v, v \rangle.$$

Since $v \neq 0$, we may divide by $\langle v, v \rangle$ and obtain $\lambda = \bar{\lambda}$. This implies that $\lambda \in \mathbb{R}$. ☺

The main theorem we prove regarding self-adjoint operators is the following:

Theorem 6.2.1 Orthogonal Diagonalization in Real Inner Product Spaces

Let V be a finite-dimensional real inner product space and let $T : V \rightarrow V$. Then TFAE:

1. T is self-adjoint.
2. There is an orthogonal (or orthonormal) basis of V consisting of eigenvectors of T .

Thus, self-adjoint operators play a similar role in real inner product spaces as the role played by normal operators in complex inner product spaces; that is, they are exactly the linear maps which admit an orthogonal basis of eigenvectors.

The proof of 6.2.1 is similar to that of 6.1.1. The only missing ingredient is that p_T has a root over $\mathbb{R}$. We show that if T is self-adjoint then p_T splits over $\mathbb{R}$.

Lemma 6.2.2 Characteristic Polynomial of Self-Adjoint Map Splits over $\mathbb{R}$

Let V be a finite-dimensional real inner product space, and let $T : V \rightarrow V$ be self-adjoint. Then p_T splits over $\mathbb{R}$.

Proof: **Idea:** We “complexify” the problem by passing to coordinate vectors in $\mathbb{R}^n$ and then considering $\mathbb{C}^n$. The representing matrix A satisfies $A = A^*$, so the corresponding linear map on $\mathbb{C}^n$ is also self-adjoint. Its characteristic polynomial splits over $\mathbb{C}$, but since all eigenvalues are real (6.2.1), the polynomial actually splits over $\mathbb{R}$.

Fix an orthonormal basis B of V . Then $A := [T]_B$ satisfies $A = A^*$.

Now let $T_A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be the linear map given by $T_A v = Av$. Letting E denote the standard basis of $\mathbb{C}^n$, we have $[T_A]_E = A$. Note that E is an orthonormal basis for the standard inner product on $\mathbb{C}^n$. Hence, T_A is self-adjoint; indeed, $[(T_A)^*]_E = A^* = A = [T_A]_E$, so $(T_A)^* = T_A$.

We know that p_{T_A} splits over $\mathbb{C}$. But by 6.2.1, all eigenvalues of T_A are real (since T_A is self-adjoint), hence p_{T_A} splits over $\mathbb{R}$.

Finally, $p_{T_A} = p_A = p_T$ (since A is a representing matrix of both T_A and T). Hence, p_T splits over $\mathbb{R}$. ☺

We are now ready to prove 6.2.1.

Proof of 6.2.1: Idea: The argument mirrors the complex case (6.1.1). Direction $2 \rightarrow 1$: if all eigenvalues are real and the eigenbasis is orthonormal, then $T^*v_i = \lambda_i v_i = Tv_i$ on each basis vector, so $T^* = T$. Direction $1 \rightarrow 2$: by 6.2.2, p_T splits over $\mathbb{R}$ so we get a real eigenvalue; we restrict to the orthogonal complement and induct.

$2 \rightarrow 1$: Suppose that T admits an orthonormal basis of eigenvectors $v_1, \dots, v_n$. Let $\lambda_i \in \mathbb{R}$ be the eigenvalue of v_i . So $Tv_i = \lambda_i v_i$.

As in the proof of 6.1.1, one can show that $T^*v_i = \overline{\lambda_i}v_i$. In our case $\lambda_i \in \mathbb{R}$, so in fact $T^*v_i = \lambda_i v_i$. This means that $T^*v_i = Tv_i$ for all i .

Since T, T^* agree on a basis, we have $T^* = T$, i.e., T is self-adjoint.

$1 \rightarrow 2$: We prove by induction on $n := \dim(V)$. The base case $n = 0$ is trivial.

For the induction step, suppose $n \geq 1$. By 6.2.2, p_T splits over $\mathbb{R}$. So let $\lambda \in \mathbb{R}$ be an eigenvalue of T and let v be a non-zero eigenvector. Put $W := \text{Span}(\{v\})$.

Since W is T -invariant (and hence also T^* -invariant, as $T^* = T$), we get by 5.6.1 that $W^\perp$ is T -invariant. Also, $T|_{W^\perp}$ is self-adjoint.

By the induction hypothesis, $T|_{W^\perp}$ admits an orthogonal basis of eigenvectors $\{v_1, \dots, v_{n-1}\}$, and then $\{v_1, \dots, v_{n-1}, v\}$ is an orthogonal basis of eigenvectors of T . $\odot$

Just like 6.1.1, 6.2.1 can also be stated in the language of orthogonal projections. Namely, for a self-adjoint operator T with distinct eigenvalues $\lambda_1, \dots, \lambda_r \in \mathbb{R}$, we can write $T = \sum_{i=1}^r \lambda_i P_i$, where P_i is the orthogonal projection onto $E(\lambda_i, T)$.

Finally, we prove the following lemma, which will be used later on.

Lemma 6.2.3 Zero Inner Product Criterion for Self-Adjoint Maps

Let V be a finite-dimensional (real or complex) inner product space, and let $T : V \rightarrow V$ be self-adjoint. If $\langle Tv, v \rangle = 0$ for all $v \in V$, then $T = 0$.

Note:-

Over $\mathbb{C}$, 6.2.3 holds without the assumption that T is self-adjoint. Indeed, in the tutorials we proved that if $\langle Tv, v \rangle = 0$ for all $v \in V$ for a linear map T over a finite-dimensional complex inner product space V , then $T = 0$.

Proof: Idea: Since T is self-adjoint (hence diagonalizable), we evaluate $\langle Tv_i, v_i \rangle = \lambda_i \langle v_i, v_i \rangle = 0$ on each eigenvector to force all eigenvalues to vanish.

Let $v_1, \dots, v_n$ be a basis of eigenvectors for T , and let $\lambda_1, \dots, \lambda_n$ be the corresponding eigenvalues. Then

$$\langle Tv_i, v_i \rangle = \langle \lambda_i v_i, v_i \rangle = \lambda_i \langle v_i, v_i \rangle.$$

On the other hand, $\langle Tv_i, v_i \rangle = 0$ by assumption. Since $\langle v_i, v_i \rangle \neq 0$, we get $\lambda_i = 0$ for all i . Hence, $T = 0$. ☺

6.3 Unitary Operators

Note:-

Related tutorial problem: see 9.10.1 for characterizing unitary maps by normality and eigenvalues of absolute value 1.

Note:-

Related additional exercises: see 33 and 39 for past-paper characterizations of unitary operators.

We now consider another important class of normal operators.

Definition 6.3.1: Unitary Operator

Let V be an inner product space. A linear map $T : V \rightarrow V$ is *unitary* if $TT^* = T^*T = I$. In other words, T is unitary if T is invertible and $T^{-1} = T^*$. Note that if T is unitary then so is T^* . Over $\mathbb{R}$, unitary maps are also called *orthogonal*.

Definition 6.3.2: Unitary Matrix

A matrix $A \in \mathbb{C}^{n \times n}$ is *unitary* if $AA^* = A^*A = I$.

Note:-

A unitary operator is in particular normal, so 6.1.1 and 6.1.2 apply to unitary operators.

We now observe that unitary maps form a group.

Lemma 6.3.1 Unitary Maps Form a Group

1. If $S, T : V \rightarrow V$ are unitary then ST is unitary.
2. If $T : V \rightarrow V$ is unitary then T^{-1} is unitary.

Proof: We use 5.6.3.

For 1,

$$ST \cdot (ST)^* = ST \cdot T^*S^* = SS^* = I,$$

and similarly $(ST)^*ST = T^*S^*ST = T^*T = I$.

For 2, we have $(T^{-1})^* = (T^*)^{-1} = (T^{-1})^{-1}$, which shows that the adjoint of T^{-1} equals its inverse. ☺

It turns out that unitary maps have many equivalent characterizations.

Theorem 6.3.1 Equivalent Characterizations of Unitary Maps

Let V be a finite-dimensional (real or complex) inner product space, and let $T : V \rightarrow V$ be a linear map. Then TFAE:

1. T is unitary.
2. $\langle Tu, Tv \rangle = \langle u, v \rangle$ for all $u, v \in V$.
3. $\|Tv\| = \|v\|$ for all $v \in V$.
4. There exists an orthonormal basis $\{v_1, \dots, v_n\}$ of V such that $\{Tv_1, \dots, Tv_n\}$ is also an orthonormal basis.
5. For every orthonormal basis $\{v_1, \dots, v_n\}$ of V , the set $\{Tv_1, \dots, Tv_n\}$ is also an orthonormal basis.

Proof: **Idea:** We prove the chains $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ and $2 \rightarrow 5 \rightarrow 4 \rightarrow 2$. The most interesting step is $3 \rightarrow 1$: norm preservation gives $\langle (T^*T - I)v, v \rangle = 0$ for all v , and since $T^*T - I$ is self-adjoint, 6.2.3 forces $T^*T = I$.

1 $\rightarrow$ 2: For all $u, v \in V$,

$$\langle Tu, Tv \rangle = \langle u, T^*Tv \rangle = \langle u, v \rangle,$$

using that $T^*T = I$.

2 $\rightarrow$ 3: Immediate — apply 2 with $u = v$:

$$\|Tv\|^2 = \langle Tv, Tv \rangle = \langle v, v \rangle = \|v\|^2.$$

3 $\rightarrow$ 1: Suppose 3 holds. Then for all $v \in V$,

$$\langle T^*Tv, v \rangle = \langle Tv, Tv \rangle = \langle v, v \rangle.$$

Therefore, $\langle (T^*T - I)v, v \rangle = 0$ for all $v \in V$.

Now observe that $S := T^*T - I$ is self-adjoint:

$$(T^*T - I)^* = (T^*T)^* - I^* = T^*(T^*)^* - I = T^*T - I.$$

By 6.2.3, $S = 0$, meaning $T^*T = I$. Hence T is unitary.

5 $\rightarrow$ 4: Immediate.

2 $\rightarrow$ 5: If 2 holds then $\langle Tv_i, Tv_j \rangle = \langle v_i, v_j \rangle = \delta_{ij}$ for any orthonormal basis $\{v_1, \dots, v_n\}$, so $\{Tv_1, \dots, Tv_n\}$ is also orthonormal.

4 $\rightarrow$ 2: There is an orthonormal basis $B = \{v_1, \dots, v_n\}$ such that $C = \{Tv_1, \dots, Tv_n\}$ is also an orthonormal basis. Note that $[Tu]_C = [u]_B$ and $[Tv]_C = [v]_B$.

Since B is orthonormal, Corollary 6.19 gives $\langle u, v \rangle = \langle [u]_B, [v]_B \rangle$. Similarly, since C is orthonormal, $\langle Tu, Tv \rangle = \langle [Tu]_C, [Tv]_C \rangle$. But $[Tu]_C = [u]_B$ and $[Tv]_C = [v]_B$, so $\langle Tu, Tv \rangle = \langle u, v \rangle$. $\odot$

Next, we show that all eigenvalues of a unitary map lie on the unit circle.

Proposition 6.3.1 Eigenvalues of Unitary Maps Lie on the Unit Circle

If T is unitary then every eigenvalue λ of T satisfies $|\lambda| = 1$.

Proof: Let v be a nonzero eigenvector of λ . Then

$$\langle Tv, Tv \rangle = \langle \lambda v, \lambda v \rangle = |\lambda|^2 \langle v, v \rangle.$$

By 6.3.1, $\langle Tv, Tv \rangle = \langle v, v \rangle$. Hence, $|\lambda|^2 \langle v, v \rangle = \langle v, v \rangle$. Since $\langle v, v \rangle > 0$, we get $|\lambda| = 1$. $\odot$

Finally, we prove the following characterization of unitary matrices.

Theorem 6.3.2 Characterization of Unitary Matrices

For a matrix $A \in \mathbb{C}^{n \times n}$, TFAE:

1. A is unitary.
2. The columns of A form an orthonormal basis (of $\mathbb{C}^n$).
3. The rows of A form an orthonormal basis.

Proof: Denote the rows of A by $r_1, \dots, r_n$ and the columns by $c_1, \dots, c_n$. Note that

$$(AA^*)_{ij} = \sum_{k=1}^n A_{ik} \cdot \overline{A_{jk}} = \langle r_i, r_j \rangle$$

and similarly

$$(A^*A)_{ij} = \sum_{k=1}^n \overline{A_{ki}} \cdot A_{kj} = \langle c_j, c_i \rangle.$$

By definition, A is unitary means $AA^* = A^*A = I$. It suffices to require only one of these (since each implies the other).

If A is unitary then $(AA^*)_{ij} = (A^*A)_{ij} = \delta_{ij}$, so both $(r_1, \dots, r_n)$ and $(c_1, \dots, c_n)$ are orthonormal. Conversely, if either set is orthonormal then $AA^* = I$ or $A^*A = I$, implying A is unitary. $\odot$

6.4 Positive (Semi-)Definite Operators

Note:-

Related homework problems: see 25 for PSD matrices as Gram matrices, and 26 for a PSD perturbation argument.

Note:-

Related tutorial problems: see 9.10.3 for preservation of PSD maps under S^*TS , 9.10.4 for the Frobenius inner product of PSD matrices, and 9.11.1 for the PSD Cauchy-Schwarz inequality.

Note:-

Related additional exercises: see 31, 35, and 35 for positive operators and functional-calculus style arguments.

Recall that if T is self-adjoint, then all eigenvalues of T are real. Here we consider a special class of self-adjoint operators: those whose eigenvalues are all nonnegative.

Definition 6.4.1: Positive Semi-Definite Operator

Let V be an inner product space. A map $T : V \rightarrow V$ is *positive semi-definite* (PSD) if T is self-adjoint and

$$\langle Tv, v \rangle \geq 0$$

for all $v \in V$.

Note:-

Similarly, T is called *positive definite* if T is self-adjoint and $\langle Tv, v \rangle > 0$ for all non-zero $v \in V$. One can similarly define *negative semi-definite* and *negative definite* operators. Note that T is negative semi-definite iff $-T$ is positive semi-definite.

Definition 6.4.2: Positive Semi-Definite Matrix

A matrix $A \in \mathbb{C}^{n \times n}$ is *positive semi-definite* if $A^* = A$ and $\langle Ax, x \rangle \geq 0$ for all $x \in \mathbb{C}^n$.

Claim 6.4.1 PSD Equivalence for Operators and Matrices

Let V be a finite-dimensional inner product space, and let $T : V \rightarrow V$. If B is an orthonormal basis of V , then T is PSD if and only if $A := [T]_B$ is PSD.

Proof: Since B is orthonormal, $[T^*]_B = A^*$ by 5.6.1, so $T^* = T$ iff $A^* = A$.

Also, by Corollary 6.19, for all $v \in V$ we have

$$\langle Tv, v \rangle = \langle [Tv]_B, [v]_B \rangle = \langle A[v]_B, [v]_B \rangle.$$

This shows that $\langle Tv, v \rangle \geq 0$ for all v iff $\langle Ax, x \rangle \geq 0$ for all $x \in \mathbb{C}^n$. ⊙

Proposition 6.4.1 PSD Characterization via Eigenvalues

Let V be a finite-dimensional (real or complex) inner product space, and let $T : V \rightarrow V$ be self-adjoint. Then TFAE:

1. T is PSD.
2. All eigenvalues of T are non-negative.

Proof: Idea: Since T is self-adjoint, it has an orthonormal eigenbasis with real eigenvalues. Expanding $\langle Tv, v \rangle$ in this basis yields $\sum_i \lambda_i |\alpha_i|^2$. This is nonnegative for all v iff every $\lambda_i \geq 0$.

Let $\{v_1, \dots, v_n\}$ be an orthonormal eigenbasis with eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{R}$. For any $v = \sum_{i=1}^n \alpha_i v_i$, we have

$$\langle Tv, v \rangle = \left\langle \sum_{i=1}^n \alpha_i \lambda_i v_i, \sum_{i=1}^n \alpha_i v_i \right\rangle = \sum_{i,j=1}^n \lambda_i \alpha_i \overline{\alpha_j} \langle v_i, v_j \rangle \stackrel{(*)}{=} \sum_{i=1}^n \lambda_i |\alpha_i|^2,$$

where $(*)$ uses $\langle v_i, v_j \rangle = \delta_{ij}$.

If $\lambda_1, \dots, \lambda_n \geq 0$ then $\langle Tv, v \rangle = \sum_i \lambda_i |\alpha_i|^2 \geq 0$ for all v , so T is PSD.

Conversely, if some $\lambda_i < 0$, then $\langle Tv_i, v_i \rangle = \lambda_i \langle v_i, v_i \rangle < 0$, so T is not PSD. $\odot$

We now observe that for every linear map T , the map T^*T is PSD.

Proposition 6.4.2 T^*T is PSD

Let V, W be inner product spaces and let $T : V \rightarrow W$ be a linear map. Then $T^*T : V \rightarrow V$ is PSD.

Proof: First, $(T^*T)^* = T^*(T^*)^* = T^*T$, so T^*T is self-adjoint.

Second, for every $v \in V$, $\langle T^*Tv, v \rangle = \langle Tv, Tv \rangle \geq 0$. $\odot$

A key property of nonnegative real numbers is that they have square roots. We now show that PSD operators behave analogously.

Proposition 6.4.3 PSD Square Root

Let V be a finite-dimensional (real or complex) inner product space, and let $T : V \rightarrow V$ be a PSD map. Then there is a unique PSD map $S : V \rightarrow V$ such that $S^2 = T$.

Proof: Idea: Since T is PSD, it has an orthonormal eigenbasis with nonnegative eigenvalues $\lambda_1, \dots, \lambda_n$. Define S by $Sv_i := \sqrt{\lambda_i} v_i$. Then $S^2v_i = \lambda_i v_i = Tv_i$, so $S^2 = T$. For uniqueness, any PSD square root must send each $v \in E(\lambda, T)$ to $\sqrt{\lambda} v$.

Existence: Let $\{v_1, \dots, v_n\}$ be an orthonormal eigenbasis of T with eigenvalues $\lambda_1, \dots, \lambda_n \geq 0$. Define $S : V \rightarrow V$ by $Sv_i := \sqrt{\lambda_i} v_i$.

Then $v_1, \dots, v_n$ are eigenvectors of S with eigenvalues $\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}$. Crucially,

$$S^2v_i = (\sqrt{\lambda_i})^2 v_i = \lambda_i v_i = Tv_i,$$

so $S^2 = T$.

S is PSD: it has an orthonormal eigenbasis, hence is normal by 6.1.1. A normal operator with all real eigenvalues is self-adjoint. Since the eigenvalues $\sqrt{\lambda_i} \geq 0$, S is PSD by 6.4.1.

Uniqueness: Suppose S is a PSD map with $S^2 = T$. Let $\{v_1, \dots, v_n\}$ be an orthonormal eigenbasis for S with eigenvalues $\mu_1, \dots, \mu_n \geq 0$.

Then $Tv_i = S^2v_i = \mu_i^2v_i$, so $\lambda_i = \mu_i^2$, meaning $\mu_i = \sqrt{\lambda_i}$ (using $\mu_i \geq 0$).

It follows that $Sv = \sqrt{\lambda}v$ for every $v \in E(\lambda, T)$. Therefore, S equals the PSD square root constructed above. $\odot$

We will denote the (unique) PSD square root of T by $T^{1/2}$ or $\sqrt{T}$.

The following gives a converse to 6.4.2.

Corollary 6.4.1 Every PSD Map is of the Form S^*S

Let V be a finite-dimensional (real or complex) inner product space, and let $T : V \rightarrow V$ be a PSD map. Then there is a map $S : V \rightarrow V$ such that $T = S^*S$.

Proof: Let $S = T^{1/2}$. Then $S^2 = T$. Also, S is self-adjoint, so $S^* = S$. Thus, $S^*S = S^2 = T$. $\odot$

6.5 The Singular Value Decomposition and Polar Decomposition

Note:-

Related homework problems: see 27 for an official SVD computation, and 28 for the largest-singular-value operator norm bound.

Note:-

Related tutorial problems: see 9.11.1 for a complete SVD and polar decomposition computation, and 9.11.2 for an application of SVD to maps with equal output norms.

Note:-

Related additional exercises: see 40, 42, 41, 43, and 44 for SVD, polar decomposition, and T^*T rank computations.

We saw (6.1.1) that every normal operator has an orthonormal basis of eigenvectors. Here we prove a theorem, called the *singular value decomposition* (SVD), which extends orthogonal diagonalization to *all* linear maps (not necessarily normal), and holds equally over both $\mathbb{C}$ and $\mathbb{R}$.

In fact, the SVD applies to maps between any two finite-dimensional inner product spaces V, W , not just to maps from V to itself.

To prepare the SVD, we need the following lemma.

Lemma 6.5.1 Kernel and Rank of T^*T

Let V, W be finite-dimensional inner product spaces, and let $T : V \rightarrow W$ be a linear map.

Then:

1. $\text{Ker}(T^*T) = \text{Ker}(T)$.
2. $\text{rank}(T^*T) = \text{rank}(T)$.

Proof: **Idea:** $\text{Ker}(T) \subseteq \text{Ker}(T^*T)$ is trivial. For the reverse, $T^*Tx = 0$ implies $\|Tx\|^2 = \langle T^*Tx, x \rangle = 0$. 2 follows by rank-nullity.

For 1: if $x \in \text{Ker}(T)$ then $T^*Tx = T^*0 = 0$. Conversely, if $x \in \text{Ker}(T^*T)$, then

$$0 = \langle T^*Tx, x \rangle = \langle Tx, Tx \rangle = \|Tx\|^2,$$

so $Tx = 0$.

For 2: by rank-nullity,

$$\text{rank}(T^*T) = \dim(V) - \dim \text{Ker}(T^*T) = \dim(V) - \dim \text{Ker}(T) = \text{rank}(T).$$

☺

We now present the SVD construction. The setting is:

- V, W are finite-dimensional inner product spaces, $n = \dim(V)$, $m = \dim(W)$.
- $T : V \rightarrow W$ is a linear map.

The idea is to consider $T^*T : V \rightarrow V$. By 6.4.2, T^*T is PSD, so by 6.2.1 it has an orthonormal eigenbasis $B = (v_1, \dots, v_n)$ with nonnegative eigenvalues $\lambda_1, \dots, \lambda_n$ (where $T^*Tv_i = \lambda_i v_i$).

Let $r := \text{rank}(T)$. By 6.5.1, $r = \text{rank}(T^*T)$.

Claim 6.5.1 Rank via Nonzero Eigenvalues

If $S : V \rightarrow V$ is diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_n$ (with multiplicities), then

$$\text{rank}(S) = \#\{1 \leq i \leq n : \lambda_i \neq 0\}.$$

Proof: Let $v_1, \dots, v_n$ be an eigenbasis with $Sv_i = \lambda_i v_i$. Then $[S]_B$ is diagonal with entries $\lambda_1, \dots, \lambda_n$, and $\text{rank}(S)$ equals the number of non-zero entries. ☺

Applying ?? to T^*T , exactly r of the λ_i are non-zero. WLOG, $\lambda_1, \dots, \lambda_r > 0$ and $\lambda_i = 0$ for $i > r$. For $1 \leq i \leq r$, put

$$\sigma_i := \sqrt{\lambda_i}.$$

Now, crucially, define

$$w_i := \frac{1}{\sigma_i} T v_i \quad \text{for } 1 \leq i \leq r.$$

Thus $T v_i = \sigma_i w_i$.

Claim 6.5.2 Orthonormality of w_i

$\{w_1, \dots, w_r\} \subseteq W$ is an orthonormal set.

Proof:

$$\langle w_i, w_j \rangle = \frac{1}{\sigma_i \sigma_j} \langle Tv_i, Tv_j \rangle = \frac{1}{\sigma_i \sigma_j} \langle T^* T v_i, v_j \rangle = \frac{1}{\sigma_i \sigma_j} \langle \lambda_i v_i, v_j \rangle = \frac{\lambda_i}{\sigma_i \sigma_j} \delta_{ij} = \delta_{ij},$$

using that $\sigma_i, \sigma_j \in \mathbb{R}$, that v_i is an eigenvector of T^*T , that $\{v_i\}$ is orthonormal, and that $\sigma_i^2 = \lambda_i$. $\odot$

Complete $\{w_1, \dots, w_r\}$ to an orthonormal basis $C = (w_1, \dots, w_r, w_{r+1}, \dots, w_m)$ of W by taking an orthonormal basis of $\text{Span}(\{w_1, \dots, w_r\})^\perp$.

Claim 6.5.3 Action of T on the Basis

$$Tv_i = \begin{cases} \sigma_i w_i & 1 \leq i \leq r, \\ 0 & i > r. \end{cases}$$

Proof: The case $1 \leq i \leq r$ is by definition. For $i > r$: $T^*T v_i = \lambda_i v_i = 0$, so by 1 of 6.5.1, $T v_i = 0$. $\odot$

Claim 6.5.4 Action of T^* on the Basis

$$T^* w_i = \begin{cases} \sigma_i v_i & 1 \leq i \leq r, \\ 0 & i > r. \end{cases}$$

Proof: Idea: For $i \leq r$, substitute the definition of w_i and use $T^*T v_i = \sigma_i^2 v_i$. For $i > r$, note that $\text{Im}(T) = \text{Span}(\{w_1, \dots, w_r\})$, so $w_i \in (\text{Im}(T))^\perp = \text{Ker}(T^*)$.

For $1 \leq i \leq r$:

$$T^* w_i = T^* \left(\frac{1}{\sigma_i} \cdot Tv_i \right) = \frac{1}{\sigma_i} \cdot T^* T v_i = \frac{1}{\sigma_i} \cdot \lambda_i v_i = \sigma_i v_i.$$

For $i > r$: since $\{v_1, \dots, v_n\}$ is a basis of V , we have $\text{Im}(T) = \text{Span}(\{w_1, \dots, w_r\})$. So $w_i \perp w_1, \dots, w_r$, meaning $w_i \in (\text{Im}(T))^\perp = \text{Ker}(T^*)$, hence $T^* w_i = 0$. $\odot$

Let us now summarize all of the above:

Theorem 6.5.1 Singular Value Decomposition (SVD) for Linear Maps

Let V, W be finite-dimensional (real or complex) inner product spaces, $n = \dim(V)$, $m = \dim(W)$. Let $T : V \rightarrow W$ have $\text{rank}(T) = r$. Then there are real numbers $\sigma_1, \dots, \sigma_r > 0$

and orthonormal bases $(v_1, \dots, v_n)$ and $(w_1, \dots, w_m)$ of V and W such that

$$Tv_i = \begin{cases} \sigma_i w_i & 1 \leq i \leq r, \\ 0 & i > r, \end{cases} \quad T^*w_i = \begin{cases} \sigma_i v_i & 1 \leq i \leq r, \\ 0 & i > r. \end{cases}$$

Moreover, $v_1, \dots, v_n$ are eigenvectors of T^*T with eigenvalues $\lambda_i = \sigma_i^2$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$.

Definition 6.5.1: Singular Values

The numbers $\sigma_1, \dots, \sigma_r$ are called the *singular values* of T .

Note:-

Often one pads the sequence $(\sigma_1, \dots, \sigma_r)$ with $\min(m, n) - r$ zeros, so that there are exactly $\min(m, n)$ singular values.

Next, we prove the matrix version.

Theorem 6.5.2 Singular Value Decomposition (SVD) for Matrices

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $A \in \mathbb{F}^{m \times n}$ with $r := \text{rank}(A)$. Then there exist $\sigma_1, \dots, \sigma_r > 0$ and unitary matrices $U_1 \in \mathbb{F}^{n \times n}$, $U_2 \in \mathbb{F}^{m \times m}$, and $\Sigma \in \mathbb{F}^{m \times n}$ such that:

1. $A = U_2 \cdot \Sigma \cdot U_1^*$.
2. U_1, U_2 are unitary.
3. $\Sigma_{ii} = \sigma_i$ for $1 \leq i \leq r$, and all other entries of Σ are 0.

Proof: Idea: Apply 6.5.1 to $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ defined by $Tx := Ax$. The orthonormal bases give unitary change-of-basis matrices, and the change-of-basis formula yields $A = U_2 \Sigma U_1^*$.

Consider $\mathbb{F}^n, \mathbb{F}^m$ with the standard inner product. Define $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ by $Tx := Ax$, so $[T]_{E_n}^{E_m} = A$ and $\text{rank}(T) = r$.

By 6.5.1, there are $\sigma_1, \dots, \sigma_r > 0$ and orthonormal bases $B = (v_1, \dots, v_n)$ of $\mathbb{F}^n$ and $C = (w_1, \dots, w_m)$ of $\mathbb{F}^m$ such that $Tv_i = \sigma_i w_i$ for $i \leq r$ and $Tv_i = 0$ for $i > r$.

Hence $[T]_B^C = \Sigma$. By the change-of-basis formula,

$$A = [I]_C^{E_m} \cdot \Sigma \cdot [I]_{E_n}^B.$$

Put $U_1 := [I]_B^{E_n}$ and $U_2 := [I]_C^{E_m}$. Their columns are the v_i and w_j respectively, which form orthonormal bases, so U_1, U_2 are unitary by 6.3.2. Since $U_1^{-1} = U_1^*$, we get $A = U_2 \cdot \Sigma \cdot U_1^*$. ☺

Example 6.5.1 (SVD Computation)

We compute the SVD of

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 4 \\ 0 & 3 \end{pmatrix}.$$

The columns of A are orthogonal, so A^*A is diagonal:

$$A^*A = \begin{pmatrix} 1 & 0 \\ 0 & 25 \end{pmatrix}.$$

Take $B = (v_1, v_2) := (e_2, e_1)$ with eigenvalues $\lambda_1 = 25$, $\lambda_2 = 1$. Then $\sigma_1 = 5$, $\sigma_2 = 1$.

Next,

$$w_1 := \frac{1}{5}Ae_2 = \frac{1}{5} \begin{pmatrix} 0 \\ 4 \\ 3 \end{pmatrix}, \quad w_2 := Ae_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Complete to an orthonormal basis of $\mathbb{R}^3$ by taking $w_3 = \frac{1}{5} \begin{pmatrix} 0 \\ -3 \\ 4 \end{pmatrix}$.

Thus,

$$U_2 = \begin{pmatrix} 0 & 1 & 0 \\ \frac{4}{5} & 0 & -\frac{3}{5} \\ \frac{3}{5} & 0 & \frac{4}{5} \end{pmatrix}, \quad U_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 5 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

and $A = U_2 \cdot \Sigma \cdot U_1^*$.

The SVD has an important consequence: the *polar decomposition*. This shows that every linear map $V \rightarrow V$ can be written as the composition of a unitary map and a PSD map.

Theorem 6.5.3 Polar Decomposition of Matrices

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For every matrix $A \in \mathbb{F}^{n \times n}$ there are matrices $U, R \in \mathbb{F}^{n \times n}$ such that U is unitary, R is PSD, and $A = UR$.

Proof: **Idea:** From the SVD $A = U_2 \Sigma U_1^*$, define $R := U_1 \Sigma U_1^*$ and $U := U_2 U_1^*$. Then $UR = U_2 \Sigma U_1^* = A$, U is unitary (product of unitaries), and R is PSD (conjugation of Σ by U_1 preserves PSD-ness).

By 6.5.2, $A = U_2 \cdot \Sigma \cdot U_1^*$ with U_1, U_2 unitary and Σ diagonal with nonnegative entries. Define

$$R := U_1 \Sigma U_1^* \quad \text{and} \quad U := U_2 U_1^*.$$

$UR = A$: Since $U_1^* U_1 = I$,

$$UR = (U_2 U_1^*)(U_1 \Sigma U_1^*) = U_2 \Sigma U_1^* = A.$$

U is unitary: U_2, U_1^* are unitary, and the product of unitaries is unitary.

R is PSD: First, Σ is PSD (diagonal with nonneg. entries). Then

$$R^* = (U_1 \Sigma U_1^*)^* = U_1 \Sigma^* U_1^* = U_1 \Sigma U_1^* = R,$$

so R is self-adjoint. Also, for all $x \in \mathbb{F}^n$,

$$\langle Rx, x \rangle = \langle U_1 \Sigma U_1^* x, x \rangle = \langle \Sigma(U_1^* x), U_1^* x \rangle \geq 0.$$

☺

Theorem 6.5.4 Polar Decomposition of Linear Maps

Let V be a finite-dimensional (real or complex) inner product space, and let $T : V \rightarrow V$. Then there are maps $U, R : V \rightarrow V$ such that U is unitary, R is PSD, and $T = UR$.

Proof 1: Fix an orthonormal basis B and let $A := [T]_B$. By 6.5.3, $A = U'R'$ with U' unitary and R' PSD. Let $U, R \in \mathcal{L}(V)$ with $[U]_B = U'$ and $[R]_B = R'$. Then $T = UR$, U is unitary, and R is PSD. ☺

Proof 2: Idea: Using the SVD (6.5.1), define R by $Rv_i = \sigma_i v_i$ (PSD) and U by $Uv_i = w_i$ (unitary). Then $URv_i = \sigma_i w_i = Tv_i$.

Let $\sigma_1, \dots, \sigma_r > 0$, $(v_1, \dots, v_n)$, $(w_1, \dots, w_n)$ be given by 6.5.1. Set $\sigma_i = 0$ for $i > r$.

- Define R via $Rv_i = \sigma_i v_i$. Since $v_1, \dots, v_n$ is an orthonormal eigenbasis of R with eigenvalues $\sigma_i \geq 0$, R is PSD by 6.4.1.
- Define U via $Uv_i = w_i$. Since U sends one orthonormal basis to another, U is unitary by 6.3.1.

Now, $Tv_i = \sigma_i w_i$ for all i (by 6.5.1), and $(UR)v_i = U(\sigma_i v_i) = \sigma_i w_i$. Hence $T = UR$. ☺

At this point it is worth pointing out some analogies between linear maps and complex numbers:

- **Self-adjoint operators** are analogous to **real numbers**. (A normal operator is self-adjoint iff all eigenvalues are real.) In particular, PSD operators are analogous to nonnegative reals.
- **Unitary operators** are analogous to **complex numbers on the unit circle**, i.e., z with $|z| = 1$. (A normal operator is unitary iff all eigenvalues have absolute value 1.)
- 6.5.4 is analogous to **polar coordinates**: for $z \in \mathbb{C}$, we can write $z = ur$ with $r = |z| \geq 0$ and $|u| = 1$. Analogously, every linear map decomposes as a unitary map times a PSD map.

Chapter 7

Quadratic and Bilinear Forms

Quadratic forms are homogeneous polynomials of degree 2. For instance,

$$q(x_1, x_2) = x_1^2 + 3x_2^2, \quad q(x_1, x_2, x_3) = 2x_1^2 - 2x_1x_2 + x_1x_3 + 5x_3^2.$$

In general, a quadratic form in n variables over a field $\mathbb{F}$ has the shape

$$q(x_1, \dots, x_n) = \sum_{i=1}^n c_i x_i^2 + \sum_{1 \leq i < j \leq n} c_{ij} x_i x_j. \quad (7.1)$$

When $\text{char}(\mathbb{F}) \neq 2$, this can be represented by a symmetric matrix A :

$$A_{ii} = c_i, \quad A_{ij} = A_{ji} = \frac{c_{ij}}{2} \quad (i < j).$$

Writing $x = (x_1, \dots, x_n)^T$ and

$$\langle u, v \rangle_0 := \sum_{i=1}^n u_i v_i = u^T v,$$

we then have $q(x) = \langle Ax, x \rangle_0 = x^T A x$.

Note:-

The notation $\langle u, v \rangle_0$ in this chapter denotes the standard dot product without conjugation. Over an arbitrary field it is not an inner product in the sense of Chapter 5, since positivity need not make sense.

Lemma 7.0.1 Matrix Expansion of the Dot Product

Let $A \in \mathbb{F}^{n \times n}$ and let $x, y \in \mathbb{F}^n$. Then

$$\langle Ay, x \rangle_0 = \sum_{i,j=1}^n A_{ij} x_i y_j. \quad (7.2)$$

Proof: Since $(Ay)_i = \sum_{j=1}^n A_{ij} y_j$,

$$\langle Ay, x \rangle_0 = \sum_{i=1}^n (Ay)_i x_i = \sum_{i=1}^n x_i \sum_{j=1}^n A_{ij} y_j = \sum_{i,j=1}^n A_{ij} x_i y_j.$$

**Proposition 7.0.1** Quadratic Forms from Symmetric Matrices

Let q be as in 7.1, and assume $\text{char}(\mathbb{F}) \neq 2$. If A is the symmetric matrix defined above, then

$$q(x_1, \dots, x_n) = \langle Ax, x \rangle_0 = x^T Ax.$$

Proof: Using 7.2 with $y = x$,

$$\langle Ax, x \rangle_0 = \sum_{i,j=1}^n A_{ij}x_i x_j = \sum_{i=1}^n A_{ii}x_i^2 + \sum_{1 \leq i < j \leq n} (A_{ij} + A_{ji})x_i x_j.$$

By the definition of A , this is exactly 7.1. ☺

7.1 Bilinear Forms

Definition 7.1.1: Bilinear Form

Let V be a vector space over a field $\mathbb{F}$. A *bilinear form* on V is a function

$$b : V \times V \rightarrow \mathbb{F}$$

such that b is linear in each variable separately. Equivalently, for all $u, v, w \in V$ and $\alpha \in \mathbb{F}$,

$$\begin{aligned} b(u + v, w) &= b(u, w) + b(v, w), & b(w, u + v) &= b(w, u) + b(w, v), \\ b(\alpha u, v) &= \alpha b(u, v), & b(u, \alpha v) &= \alpha b(u, v). \end{aligned}$$

Definition 7.1.2: Representing Matrix of a Bilinear Form

Let V be finite-dimensional, let b be a bilinear form on V , and let $B = (v_1, \dots, v_n)$ be a basis of V . The *representing matrix* of b in the basis B is

$$[b]_B = (b(v_i, v_j))_{i,j}.$$

Lemma 7.1.1 Computing a Bilinear Form from Coordinates

Let $A := [b]_B$. For all $x, y \in V$,

$$b(x, y) = \langle A[y]_B, [x]_B \rangle_0 = [x]_B^T A [y]_B. \quad (7.3)$$

Proof: Write $x = \sum_{i=1}^n \alpha_i v_i$ and $y = \sum_{j=1}^n \beta_j v_j$. Then

$$b(x, y) = \sum_{i,j=1}^n \alpha_i \beta_j b(v_i, v_j) = \sum_{i,j=1}^n A_{ij} \alpha_i \beta_j.$$

By 7.2, this equals $\langle A[y]_B, [x]_B \rangle_0$. ☺

Corollary 7.1.1 Bilinear Forms on $\mathbb{F}^n$

If b is a bilinear form on $\mathbb{F}^n$, E is the standard basis, and $A = [b]_E$, then

$$b(x, y) = \langle Ay, x \rangle_0 = x^T Ay.$$

Conversely, every matrix $A \in \mathbb{F}^{n \times n}$ defines a bilinear form by this formula.

Proposition 7.1.1 Bilinear Forms and Matrices

Let V be an n -dimensional vector space over $\mathbb{F}$ and let B be a basis of V . The map

$$b \mapsto [b]_B$$

from the vector space of bilinear forms on V to $\mathbb{F}^{n \times n}$ is a linear isomorphism.

Proof: Linearity follows from the definition of $[b]_B$. If $[b]_B = 0$, then 7.3 gives $b(x, y) = 0$ for all $x, y \in V$, so $b = 0$.

For surjectivity, let $A \in \mathbb{F}^{n \times n}$. Define

$$b_A(x, y) := \langle A[y]_B, [x]_B \rangle_0.$$

This is bilinear, and a direct check on the basis vectors gives $[b_A]_B = A$. Hence $b \mapsto [b]_B$ is a linear isomorphism. $\odot$

Lemma 7.1.2 Change of Basis for Bilinear Forms

Let b be a bilinear form on V , let B, C be bases of V , and put

$$X = [b]_B, \quad Y = [b]_C, \quad Q = [I]_C^B.$$

Then

$$Y = Q^T X Q. \tag{7.4}$$

Proof: Write $C = (w_1, \dots, w_n)$. Since $Q = [I]_C^B$, we have $[w_k]_B = Qe_k$. Hence

$$Y_{ij} = b(w_i, w_j) = \langle X[w_j]_B, [w_i]_B \rangle_0 = (Qe_i)^T X(Qe_j) = e_i^T (Q^T X Q) e_j.$$

Since $e_i^T A e_j = A_{ij}$ for every matrix A , we get $Y_{ij} = (Q^T X Q)_{ij}$ for all i, j . $\odot$

Definition 7.1.3: Congruent Matrices

Matrices $X, Y \in \mathbb{F}^{n \times n}$ are *congruent* if there exists an invertible matrix $Q \in \mathbb{F}^{n \times n}$ such that

$$Y = Q^T X Q.$$

Proposition 7.1.2 Congruence is an Equivalence Relation

Matrix congruence is an equivalence relation on $\mathbb{F}^{n \times n}$.

Proposition 7.1.3 Congruence and Bilinear Forms

Let V be an n -dimensional vector space over $\mathbb{F}$. Two matrices $X, Y \in \mathbb{F}^{n \times n}$ represent the same bilinear form on V in possibly different bases if and only if X and Y are congruent.

Proof: If $X = [b]_B$ and $Y = [b]_C$, then 7.1.2 gives $Y = Q^T X Q$ for $Q = [I]_C^B$, so X and Y are congruent.

Conversely, suppose $Y = Q^T X Q$ with Q invertible. Fix a basis B and let b be the unique bilinear form with $[b]_B = X$, using 7.1.1. Choose the unique basis C such that $Q = [I]_C^B$. Then 7.1.2 gives $[b]_C = Y$. $\odot$

Lemma 7.1.3 Rank is Preserved Under Congruence

If $X, Y \in \mathbb{F}^{n \times n}$ are congruent, then $\text{rank}(X) = \text{rank}(Y)$.

Proof: Let $Y = Q^T X Q$ with Q invertible. Then

$$\ker(Y) = Q^{-1}(\ker(X)).$$

Indeed, $Xv = 0$ implies $YQ^{-1}v = 0$, and $Yw = 0$ implies $XQw = 0$ because Q^T is invertible. Thus $\dim \ker(Y) = \dim \ker(X)$, and rank-nullity gives $\text{rank}(Y) = \text{rank}(X)$. $\odot$

7.2 Symmetric Bilinear Forms and Quadratic Forms

Definition 7.2.1: Quadratic Form Derived from a Bilinear Form

Let b be a bilinear form on V . The map $q : V \rightarrow \mathbb{F}$ defined by

$$q(v) = b(v, v)$$

is called the *quadratic form derived from b* .

Definition 7.2.2: Symmetric Bilinear Form

A bilinear form b on V is *symmetric* if

$$b(u, v) = b(v, u)$$

for all $u, v \in V$.

Lemma 7.2.1 Polarization Identity

Assume $\text{char}(\mathbb{F}) \neq 2$. Let b be a symmetric bilinear form and let $q(v) = b(v, v)$. Then for all $u, v \in V$,

$$b(u, v) = \frac{1}{2}(q(u+v) - q(u) - q(v)). \quad (7.5)$$

Proof: By bilinearity and symmetry,

$$q(u + v) = b(u + v, u + v) = q(u) + 2b(u, v) + q(v).$$

Rearrange and divide by 2. ☺

Note:-

When $\text{char}(\mathbb{F}) \neq 2$, 7.5 shows that a symmetric bilinear form is determined by its quadratic form. Thus quadratic forms and symmetric bilinear forms contain the same information in this setting.

Note:-

Every bilinear form b can be decomposed as

$$b = b_{\text{sym}} + b_{\text{alt}}, \quad b_{\text{sym}}(u, v) = \frac{b(u, v) + b(v, u)}{2}, \quad b_{\text{alt}}(u, v) = \frac{b(u, v) - b(v, u)}{2},$$

provided $\text{char}(\mathbb{F}) \neq 2$. The alternating part satisfies $b_{\text{alt}}(v, v) = 0$, so only the symmetric part contributes to the derived quadratic form.

Lemma 7.2.2 Symmetry and Representing Matrices

Let B be a basis of a finite-dimensional vector space V . A bilinear form b on V is symmetric if and only if $[b]_B$ is a symmetric matrix.

Proof: Let $A = [b]_B$ and $B = (v_1, \dots, v_n)$. If b is symmetric, then

$$A_{ij} = b(v_i, v_j) = b(v_j, v_i) = A_{ji}.$$

Conversely, if A is symmetric, then for $x = \sum_i \alpha_i v_i$ and $y = \sum_j \beta_j v_j$,

$$b(x, y) = \sum_{i,j} \alpha_i \beta_j b(v_i, v_j) = \sum_{i,j} \alpha_i \beta_j b(v_j, v_i) = b(y, x).$$

☺

7.3 Diagonalization of Quadratic Forms

Lemma 7.3.1 Nonzero Symmetric Forms Have a Nonzero Square

Let $\text{char}(\mathbb{F}) \neq 2$, and let b be a symmetric bilinear form on V . If $b \neq 0$, then there exists $v \in V$ such that $b(v, v) \neq 0$.

Proof: We prove the contrapositive. If $b(v, v) = 0$ for all $v \in V$, then 7.5 gives $b(u, v) = 0$ for all $u, v \in V$. Hence $b = 0$. ☺

Theorem 7.3.1 Diagonalization of Symmetric Bilinear Forms

Let $\text{char}(\mathbb{F}) \neq 2$, let V be finite-dimensional over $\mathbb{F}$, and let b be a symmetric bilinear form on V . Then there exists an ordered basis B of V such that $[b]_B$ is diagonal.

Proof: We induct on $n = \dim(V)$. The case $n = 1$ is immediate. If $b = 0$, then $[b]_B = 0$ in every basis. Otherwise, by 7.3.1, choose $v \in V$ with $b(v, v) \neq 0$.

Let

$$U := \{u \in V : b(u, v) = 0\}.$$

This is the kernel of the nonzero linear functional $u \mapsto b(u, v)$, and since $b(v, v) \neq 0$, this functional has image $\mathbb{F}$. Hence $\dim(U) = n - 1$. The restriction $b|_U$ is again symmetric, so by induction there is a basis $(v_1, \dots, v_{n-1})$ of U for which $[b|_U]$ is diagonal.

Then $B = (v_1, \dots, v_{n-1}, v)$ is a basis of V . Since each $v_i \in U$, we have $b(v_i, v) = 0$, and therefore $[b]_B$ is diagonal. $\odot$

Corollary 7.3.1 Symmetric Matrices are Congruent to Diagonal Matrices

Let $\text{char}(\mathbb{F}) \neq 2$. Every symmetric matrix $A \in \mathbb{F}^{n \times n}$ is congruent to a diagonal matrix.

Proof: Define $b(x, y) = \langle Ay, x \rangle_0$ on $\mathbb{F}^n$. Then $A = [b]_E$ in the standard basis. Since A is symmetric, 7.2.2 says b is symmetric. By 7.3.1, there is a basis B such that $D = [b]_B$ is diagonal. Now 7.1.2 shows A and D are congruent. $\odot$

Example 7.3.1 (Characteristic Two Obstruction)

Over $\mathbb{F}_2$, define

$$b\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = x_1y_2 + x_2y_1.$$

This bilinear form is symmetric, but it is not diagonalizable. Indeed, for every $v = (x, y)^T$,

$$b(v, v) = xy + xy = 2xy = 0$$

in $\mathbb{F}_2$. If $[b]_B$ were diagonal in some basis, its diagonal entries would all be zero, so $[b]_B = 0$, forcing $b = 0$, a contradiction since $b(e_1, e_2) = 1$.

Gaussian Elimination for Symmetric Matrices

The diagonalization in 7.3.1 can be done algorithmically. The key idea is that every row operation is paired with the corresponding column operation.

Proposition 7.3.1 Elementary Row and Column Operations

Let E be an elementary row-operation matrix.

1. Multiplying A on the left by E performs the corresponding row operation on A .
2. Multiplying A on the right by E^T performs the corresponding column operation.

3. Every elementary row-operation matrix is invertible.

Algorithm 2: Symmetric Gaussian elimination

Input: A symmetric matrix $A \in \mathbb{F}^{n \times n}$ with $\text{char}(\mathbb{F}) \neq 2$

Output: A diagonal matrix congruent to A

```

1 while  $A$  has an off-diagonal or later block not yet diagonalized do
2   Use simultaneous row and column swaps, if needed, to place a nonzero pivot in the
   active  $(1, 1)$  position;
3   If all diagonal entries in the active block are zero but some  $A_{ij} \neq 0$  with  $i \neq j$ , first
   replace  $R_1 \leftarrow R_1 + R_i$  and  $C_1 \leftarrow C_1 + C_i$  to create a nonzero diagonal pivot;
4   Use the pivot to clear all other entries in its row and column by paired operations

       
$$R_i \leftarrow R_i - \frac{A_{1i}}{A_{11}}R_1, \quad C_i \leftarrow C_i - \frac{A_{1i}}{A_{11}}C_1.$$


   Continue with the remaining lower-right block;
5 end
```

Lemma 7.3.2 Symmetry is Preserved During the Algorithm

If A is symmetric, then every paired operation in 2 keeps the matrix symmetric.

Proof: Each paired operation has the form $A \mapsto EAE^T$. If $A^T = A$, then

$$(EAE^T)^T = (E^T)^T A^T E^T = EAE^T.$$

☺

Theorem 7.3.2 Symmetric Gaussian Elimination Diagonalizes

Let $\text{char}(\mathbb{F}) \neq 2$ and let $A \in \mathbb{F}^{n \times n}$ be symmetric. Then 2 transforms A into a diagonal matrix.

Proof: Induct on n . If $A = 0$ or $n = 1$, there is nothing to prove. Otherwise choose a nonzero entry A_{ij} . If $A_{11} = 0$ and $i = j$, swap row/column 1 with row/column i . If $A_{11} = 0$ and $i \neq j$, perform $R_1 \leftarrow R_1 + R_i$ and $C_1 \leftarrow C_1 + C_i$; the new $(1, 1)$ entry is $A_{11} + A_{1i} + A_{i1} = 2A_{1i} \neq 0$ after relabeling.

Now use the nonzero pivot to clear all other entries in the first row and first column. By 7.3.2, the remaining lower-right block is symmetric, so induction diagonalizes it. ☺

Lemma 7.3.3 Computing the Congruence Matrix

During 2, start with the augmented matrix $(A \mid I)$. Apply the paired row and column operations to the left block A , but apply only the row operations to the right block I . If the left block becomes D , then the right block becomes an invertible matrix Q satisfying

$$D = QAQ^T. \tag{7.6}$$

Proof: Suppose the row operations are represented by elementary matrices $E_1, \dots, E_k$ in order. Then the left block becomes

$$E_k \cdots E_1 A E_1^T \cdots E_k^T.$$

The right block becomes

$$E_k \cdots E_1 I = E_k \cdots E_1 =: Q.$$

Thus the diagonal left block is $D = Q A Q^T$, and Q is invertible as a product of invertible elementary matrices. $\odot$

Example 7.3.2 (A Symmetric Elimination Computation)

Let

$$A = \begin{pmatrix} 2 & -2 & -1 \\ -2 & 3 & 2 \\ -1 & 2 & -1 \end{pmatrix}.$$

Symmetric Gaussian elimination gives

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -\frac{5}{2} \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -\frac{1}{2} & -1 & 1 \end{pmatrix},$$

and these satisfy $D = Q A Q^T$, as in 7.6.

Note:-

If b is a symmetric bilinear form and $A = [b]_B$, then finding $D = Q A Q^T$ diagonal gives a diagonalizing basis for b . Set $R := Q^T$. Since R is invertible, there is a basis C such that $R = [I]_C^B$, and 7.1.2 gives $[b]_C = D$.

7.4 Quadratic Forms over the Reals

Theorem 7.4.1 Real Symmetric Matrices are Orthogonally Diagonalizable

Every symmetric matrix $A \in \mathbb{R}^{n \times n}$ has an orthonormal basis of eigenvectors. Equivalently, there exist a diagonal matrix D and an orthogonal matrix Q such that

$$D = Q^T A Q. \tag{7.7}$$

Proof: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be $T(x) = Ax$. Since $A^T = A$, the map T is self-adjoint with respect to the standard inner product. By 6.2.1, T has an orthonormal basis of eigenvectors. Put these eigenvectors as the columns of Q , and put the corresponding eigenvalues on the diagonal of D . Then Q is orthogonal by 6.3.2, and $D = Q^T A Q$. $\odot$

Proposition 7.4.1 Real Symmetric Bilinear Forms Have Orthonormal Diagonalizing Bases

Let b be a symmetric bilinear form on $\mathbb{R}^n$. Then there is an orthonormal basis B of $\mathbb{R}^n$ such that $[b]_B$ is diagonal.

Proof: Let $A = [b]_E$. By 7.2.2, A is symmetric. Apply 7.4.1 to get $D = Q^T A Q$ with Q orthogonal. If B is the ordered basis given by the columns of Q , then $Q = [I]_B^E$, so 7.1.2 gives $[b]_B = D$. $\odot$

Theorem 7.4.2 Sylvester's Law of Inertia

Let V be a finite-dimensional real vector space and let b be a symmetric bilinear form on V . Suppose B and B' are bases such that

$$D = [b]_B, \quad D' = [b]_{B'}$$

are diagonal. Then D and D' have the same number of positive diagonal entries and the same number of negative diagonal entries.

Proof: Let p, q be the numbers of positive and negative diagonal entries of D , and define p', q' similarly for D' . By 7.1.3, diagonal representing matrices for the same form have the same rank, so

$$p + q = p' + q'.$$

It is enough to prove $p = p'$. Suppose, toward a contradiction, that $p < p'$.

Reorder the bases so that

$$D_{ii} > 0 \ (1 \leq i \leq p), \quad D_{ii} \leq 0 \ (i > p),$$

and

$$D'_{ii} > 0 \ (1 \leq i \leq p'), \quad D'_{ii} \leq 0 \ (i > p').$$

Write $B = (v_1, \dots, v_n)$ and $B' = (v'_1, \dots, v'_n)$, and define

$$U = \text{span}(v_{p+1}, \dots, v_n), \quad W = \text{span}(v'_1, \dots, v'_{p'}).$$

Then $\dim(U) + \dim(W) = n - p + p' > n$, so $U \cap W$ contains a nonzero vector x .

Since $x \in U$, we can write $x = \sum_{i=p+1}^n \alpha_i v_i$, and diagonality of D gives

$$b(x, x) = \sum_{i=p+1}^n \alpha_i^2 D_{ii} \leq 0.$$

Since $x \in W$, we can also write $x = \sum_{i=1}^{p'} \beta_i v'_i$ with at least one $\beta_i \neq 0$, and diagonality of D' gives

$$b(x, x) = \sum_{i=1}^{p'} \beta_i^2 D'_{ii} > 0.$$

This contradiction proves $p = p'$, and hence also $q = q'$. $\odot$

Corollary 7.4.1 Diagonal Congruence over $\mathbb{R}$

If two diagonal matrices $D, D' \in \mathbb{R}^{n \times n}$ are congruent, then they have the same number of positive diagonal entries and the same number of negative diagonal entries.

Definition 7.4.1: Signature

Let V be a finite-dimensional real vector space and let b be a symmetric bilinear form on V . The *signature* of b is the pair

$$(p, q),$$

where p is the number of positive diagonal entries and q is the number of negative diagonal entries in any diagonal representing matrix of b . This is well-defined by [7.4.2](#).

The signature of a quadratic form is the signature of its corresponding symmetric bilinear form.

Chapter 8

Linear-Algebraic Methods in Combinatorics

Note:-

This chapter is extra material. The main idea is a common linear-algebraic strategy: encode combinatorial objects as vectors, prove that the vectors are linearly independent or lie in a small subspace, and then use dimension.

8.1 Oddtown and Eventown

For $n \in \mathbb{N}$, write

$$[n] := \{1, \dots, n\}.$$

If $A \subseteq [n]$, its *characteristic vector* is the vector $v_A \in \{0, 1\}^n$ defined by

$$v_A(i) = \begin{cases} 1 & i \in A, \\ 0 & i \notin A. \end{cases}$$

For characteristic vectors, the standard dot product counts intersections:

$$\langle v_A, v_B \rangle_0 = |A \cap B|. \quad (8.1)$$

Indeed, the i th summand is 1 exactly when $i \in A \cap B$.

Theorem 8.1.1 Oddtown Bound

Let $A_1, \dots, A_m \subseteq [n]$ satisfy the *Oddtown rules*:

1. $|A_i|$ is odd for every i .
2. $|A_i \cap A_j|$ is even for every $i \neq j$.

Then $m \leq n$.

Proof: Let $v_i := v_{A_i}$, but view these vectors as elements of $\mathbb{F}_2^n$. By 8.1, over $\mathbb{F}_2$ we have

$$\langle v_i, v_j \rangle_0 = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

We claim that $v_1, \dots, v_m$ are linearly independent. Suppose

$$\sum_{i=1}^m \alpha_i v_i = 0.$$

Taking the dot product with v_j gives

$$0 = \left\langle \sum_{i=1}^m \alpha_i v_i, v_j \right\rangle_0 = \sum_{i=1}^m \alpha_i \langle v_i, v_j \rangle_0 = \alpha_j.$$

Thus all $\alpha_j = 0$, so the vectors are linearly independent. Since they lie in the n -dimensional vector space $\mathbb{F}_2^n$, we get $m \leq n$. $\odot$

Note:-

The Oddtown bound is sharp: take $A_i = \{i\}$ for $1 \leq i \leq n$.

Theorem 8.1.2 Eventown Bound

Let $A_1, \dots, A_m \subseteq [n]$ be distinct and suppose

$$|A_i \cap A_j|$$

is even for all $1 \leq i < j \leq m$. Then

$$m \leq 2^{\lfloor n/2 \rfloor}.$$

Proof: Let $v_i := v_{A_i} \in \mathbb{F}_2^n$, and set

$$W := \text{span}(v_1, \dots, v_m), \quad d := \dim(W).$$

Choose a basis $w_1, \dots, w_d$ of W , and let $M \in \mathbb{F}_2^{d \times n}$ be the matrix whose rows are $w_1, \dots, w_d$.

For every k , we have $Mv_k = 0$. Indeed, each coordinate of Mv_k is a dot product $\langle w_i, v_k \rangle_0$, and since w_i is a linear combination of $v_1, \dots, v_m$, the parity condition and 8.1 imply $\langle w_i, v_k \rangle_0 = 0$ in $\mathbb{F}_2$.

Thus $v_1, \dots, v_m \in \ker(M)$, so $W \subseteq \ker(M)$ and $\dim \ker(M) \geq d$. Also $\text{rank}(M) = d$ because the rows of M are linearly independent. By rank-nullity,

$$n = \dim \ker(M) + \text{rank}(M) \geq 2d,$$

so $d \leq \lfloor n/2 \rfloor$.

Since W is a d -dimensional vector space over $\mathbb{F}_2$, it has 2^d elements. The sets A_i are distinct, so the vectors v_i are distinct, and hence

$$m \leq |W| = 2^d \leq 2^{\lfloor n/2 \rfloor}.$$

$\odot$

Note:-

The Eventown bound is also sharp. If n is even, pair the elements as $\{1, 2\}, \{3, 4\}, \dots, \{n-1, n\}$ and take all sets which contain either both elements or neither element from each pair. There are $2^{n/2}$ such sets, and all pairwise intersections have even size.

8.2 Subsystems with Equal Unions

Theorem 8.2.1 Equal-Union Subsystems

Let $A_1, \dots, A_m \subseteq [n]$ be nonempty sets with $m = n + 1$. Then there exist disjoint subsets $I, J \subseteq [m]$ such that

$$\bigcup_{i \in I} A_i = \bigcup_{i \in J} A_i.$$

Proof: Let $v_i \in \{0, 1\}^n \subseteq \mathbb{R}^n$ be the characteristic vector of A_i . Since $m = n + 1$, the vectors $v_1, \dots, v_m$ are linearly dependent in $\mathbb{R}^n$. Hence there exist real numbers $\alpha_1, \dots, \alpha_m$, not all zero, such that

$$\sum_{i=1}^m \alpha_i v_i = 0. \quad (8.2)$$

Define

$$I := \{i : \alpha_i > 0\}, \quad J := \{i : \alpha_i < 0\}.$$

After moving the negative terms in 8.2 to the other side, we obtain

$$\sum_{i \in I} \alpha_i v_i = \sum_{i \in J} (-\alpha_i) v_i. \quad (8.3)$$

The sets I and J are disjoint by definition.

For $k \in [n]$, the k th coordinate of the left side of 8.3 is positive exactly when $k \in A_i$ for some $i \in I$. Similarly, the k th coordinate of the right side is positive exactly when $k \in A_i$ for some $i \in J$. Since the two vectors in 8.3 are equal, these two conditions are equivalent for every k . Hence the two unions are equal. $\ominus$

8.3 Equiangular Lines

Definition 8.3.1: Equiangular Lines

Lines $\ell_1, \dots, \ell_m$ through the origin in $\mathbb{R}^d$ are *equiangular* if there is an angle $\theta \in (0, \pi/2]$ such that every pair of distinct lines makes angle θ .

Theorem 8.3.1 Gerzon Bound for Equiangular Lines

There are at most

$$\binom{d+1}{2}$$

equiangular lines through the origin in $\mathbb{R}^d$.

Proof: Let $\ell_1, \dots, \ell_m$ be equiangular lines in $\mathbb{R}^d$. Choose a unit vector v_i in the direction of each ℓ_i . If θ is the common angle between the lines, set

$$\alpha := \cos(\theta).$$

Since the angle between v_i and v_j as vectors is either θ or $\pi - \theta$, we have

$$\langle v_i, v_j \rangle = \begin{cases} 1 & i = j, \\ \pm\alpha & i \neq j. \end{cases} \quad (8.4)$$

Also $0 \leq \alpha < 1$.

For each i , define the symmetric matrix

$$A_i := v_i v_i^T.$$

Let $\mathcal{S}_d$ be the vector space of real symmetric $d \times d$ matrices. Then

$$\dim(\mathcal{S}_d) = d + \binom{d}{2} = \binom{d+1}{2}.$$

It therefore suffices to show that $A_1, \dots, A_m$ are linearly independent.

Use the Frobenius inner product on matrices:

$$\langle A, B \rangle_F := \sum_{k,\ell=1}^d A_{k\ell} B_{k\ell}.$$

For $A_i = v_i v_i^T$ and $A_j = v_j v_j^T$,

$$\langle A_i, A_j \rangle_F = \langle v_i, v_j \rangle^2. \quad (8.5)$$

Indeed,

$$\langle A_i, A_j \rangle_F = \sum_{k,\ell=1}^d v_i(k) v_i(\ell) v_j(k) v_j(\ell) = \left(\sum_{k=1}^d v_i(k) v_j(k) \right)^2.$$

By 8.4, this gives

$$\langle A_i, A_j \rangle_F = \begin{cases} 1 & i = j, \\ \alpha^2 & i \neq j. \end{cases}$$

Suppose $\sum_{i=1}^m c_i A_i = 0$. Taking the Frobenius inner product with A_j gives

$$c_j + \sum_{i \neq j} \alpha^2 c_i = 0$$

for every j . In matrix form, this system is

$$(\alpha^2 J + (1 - \alpha^2) I) c = 0,$$

where J is the all-ones $m \times m$ matrix and $c = (c_1, \dots, c_m)^T$.

We now use the following fact: if $a, b \in \mathbb{R}$, $b \neq 0$, and $am + b \neq 0$, then $aJ + bI$ is invertible. Indeed, J has eigenvalues $m, 0, \dots, 0$, so $aJ + bI$ has eigenvalues $am + b, b, \dots, b$.

Here $a = \alpha^2$ and $b = 1 - \alpha^2$. Since $0 \leq \alpha < 1$, both b and $\alpha^2 m + (1 - \alpha^2)$ are nonzero. Thus the matrix above is invertible, so $c = 0$. Hence $A_1, \dots, A_m$ are linearly independent, and therefore

$$m \leq \dim(\mathcal{S}_d) = \binom{d+1}{2}.$$

☺

Note:-

For $d = 2$, 8.3.1 gives at most $\binom{3}{2} = 3$ equiangular lines, so the familiar picture of three equally spaced lines through the origin is best possible.

Chapter 9

Tutorial Problems and Solutions

9.1 Tutorial 1: Diagonalization

Example 9.1.1 (Similar Matrices Have the Same Characteristic Polynomial)

If A and B are similar, say $B = CAC^{-1}$, then

$$p_B(x) = \det(xI - B) = \det(C(xI - A)C^{-1}) = \det(C) \det(xI - A) \det(C^{-1}) = p_A(x).$$

Example 9.1.2 (Distinct Eigenvalues Imply Diagonalizable)

Let $\dim(V) = n$ and suppose $T : V \rightarrow V$ has n distinct eigenvalues. Then each algebraic multiplicity is 1, while each geometric multiplicity is at least 1 and at most the corresponding algebraic multiplicity. Hence every geometric multiplicity equals the algebraic multiplicity, so T is diagonalizable.

Example 9.1.3 (A Diagonalization Computation)

For

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix},$$

we have

$$p_A(x) = \det(xI - A) = (x - 1)^2(x - 2).$$

For $\lambda = 2$, an eigenvector is $(1, 0, 1)^T$. For $\lambda = 1$, two independent eigenvectors are $(0, 1, -1)^T$ and $(1, 0, 0)^T$. Thus

$$C = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{pmatrix}$$

has eigenvectors as columns, and

$$C^{-1}AC = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Example 9.1.4 (Cyclic Shift Matrix and Roots of Unity)

Let

$$A = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

If $z^n = 1$, then

$$v_z = (1, z, z^2, \dots, z^{n-1})^T$$

satisfies $Av_z = zv_z$. Hence every n th root of unity is an eigenvalue. Since A is $n \times n$ and there are n distinct n th roots of unity, these are all eigenvalues.

9.2 Tutorial 2: Direct Sums

Theorem 9.2.1 Equivalent Conditions for Direct Sums

Let $V_1, \dots, V_r$ be subspaces and $V = V_1 + \cdots + V_r$. The following are equivalent:

1. $V = V_1 \oplus \cdots \oplus V_r$.
2. If $v_i \in V_i$ and $v_1 + \cdots + v_r = 0$, then $v_i = 0$ for all i .
3. $V_i \cap \sum_{j \neq i} V_j = \{0\}$ for every i .
4. For all bases B_i of V_i , the union $\bigcup_i B_i$ is a basis of V .
5. If V is finite-dimensional, then $\dim(V) = \sum_i \dim(V_i)$.

Proof: The equivalence of the first three statements is the uniqueness of the representation $v = v_1 + \cdots + v_r$. If the zero vector has only the trivial representation, then subtracting two representations of the same vector gives uniqueness, and conversely uniqueness applied to 0 gives the zero-representation criterion. The intersection condition is exactly the statement that no nonzero vector in one summand can be represented using the others.

The zero-representation criterion is equivalent to the union-of-bases condition: spanning is clear, and linear independence of $\bigcup_i B_i$ is exactly the zero-representation criterion after grouping coefficients by subspace. Finally, in finite dimension, a spanning set of size $\dim(V)$ is a basis, so the basis condition is equivalent to the dimension formula. ☺

Example 9.2.1 (Characteristic Polynomial over an Invariant Direct Sum)

If $V = V_1 \oplus \cdots \oplus V_r$ and each V_i is T -invariant, then

$$p_T(x) = \prod_{i=1}^r p_{T|_{V_i}}(x).$$

Indeed, using bases of the V_i , the matrix of T is block diagonal with blocks $[T|_{V_i}]$, so the determinant of $xI - [T]$ is the product of the block determinants.

Example 9.2.2 (Idempotents Split as Kernel Plus Image)

If $T^2 = T$, then

$$V = \ker(T) \oplus \operatorname{Im}(T).$$

For any $v \in V$, set $w = Tv$. Then $w \in \operatorname{Im}(T)$ and

$$T(v - w) = Tv - T^2v = 0,$$

so $v = (v - w) + w \in \ker(T) + \operatorname{Im}(T)$. Rank-nullity gives the dimension sum, hence the sum is direct.

9.3 Tutorial 3: Minimal Polynomials and Cyclic Vectors

Example 9.3.1 (Minimal Polynomial of a Diagonalizable Map)

If T is diagonalizable with distinct eigenvalues $\lambda_1, \dots, \lambda_k$, then

$$q_T(x) = \prod_{i=1}^k (x - \lambda_i).$$

The product annihilates every eigenvector, hence annihilates V . Conversely, every eigenvalue is a root of q_T , so the product divides q_T and q_T divides the product.

Example 9.3.2 (Cyclic Vectors for Diagonalizable Maps)

Let T be diagonalizable.

1. If the eigenvalues are not distinct, then T has no cyclic vector. By 9.3.1, the minimal polynomial has one factor for each distinct eigenvalue, while the characteristic polynomial counts multiplicity, so $q_T \neq p_T$. A cyclic vector would force $q_T = p_T$.
2. If $\dim(V) = n$ and T has distinct eigenvalues $\lambda_1, \dots, \lambda_n$, with eigenvectors $v_1, \dots, v_n$, then $v = v_1 + \cdots + v_n$ is cyclic. Indeed,

$$T^k v = \sum_{i=1}^n \lambda_i^k v_i,$$

and the Vandermonde matrix $(\lambda_i^k)_{1 \leq i \leq n, 0 \leq k \leq n-1}$ is invertible because the λ_i are distinct.

Example 9.3.3 (Invariant Subspaces with No Real Eigenvalues)

Let V be finite-dimensional over $\mathbb{R}$ and suppose $T : V \rightarrow V$ has no real eigenvalues. If W is T -invariant, then $\dim(W)$ is even. Otherwise $p_{T|_W}$ has odd degree and therefore a real root. This root is an eigenvalue of $T|_W$, hence also of T , a contradiction.

9.4 Tutorial 4: Cayley-Hamilton and Primary Decomposition

Example 9.4.1 (Span of Powers of a Matrix)

For $A \in \mathbb{F}^{n \times n}$,

$$W = \text{span}\{A^i : i \geq 0\}$$

has dimension at most n . By Cayley-Hamilton, A^n is a linear combination of $I, A, \dots, A^{n-1}$. Multiplying by powers of A inductively shows every A^m with $m \geq n$ lies in $\text{span}(I, A, \dots, A^{n-1})$.

Example 9.4.2 (Nilpotent-Invertible Decomposition)

Let $T : V \rightarrow V$ be finite-dimensional. Write the minimal polynomial as

$$q_T(x) = x^k f(x),$$

where $f(0) \neq 0$. By primary decomposition,

$$V = \ker(T^k) \oplus \ker(f(T)).$$

On $U = \ker(T^k)$, the restriction $T|_U$ is nilpotent. On $W = \ker(f(T))$, the minimal polynomial has no factor x , so 0 is not an eigenvalue of $T|_W$; hence $T|_W$ is invertible.

Example 9.4.3 (Image Equals Kernel for Coprime Factors)

Suppose $q_T = fg$ with $\gcd(f, g) = 1$. Then

$$\text{Im}(g(T)) = \ker(f(T)).$$

The inclusion $\text{Im}(g(T)) \subseteq \ker(f(T))$ follows from $f(T)g(T) = q_T(T) = 0$. Primary decomposition gives

$$\dim \ker(f(T)) + \dim \ker(g(T)) = \dim(V),$$

while rank-nullity for $g(T)$ gives

$$\dim \text{Im}(g(T)) + \dim \ker(g(T)) = \dim(V).$$

Thus the two subspaces have equal dimension, so they are equal.

9.5 Tutorial 5: Jordan Form

Example 9.5.1 (A Jordan Form Computation)

For

$$A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 3 \end{pmatrix},$$

we get

$$p_A(x) = (x - 2)^2(x - 3)^2.$$

For $\lambda = 3$, $\text{rank}(A - 3I) = 2$, so $\dim \ker(A - 3I) = 2$ and there are two Jordan blocks of type 3, both of size 1. For $\lambda = 2$, $\text{rank}(A - 2I) = 3$, so there is one Jordan block of type 2, necessarily of size 2. Therefore

$$J = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

The minimal polynomial is

$$q_A(x) = (x - 2)^2(x - 3).$$

Example 9.5.2 (When $q_T = p_T$)

Assume p_T splits. Write

$$q_T(x) = \prod_{i=1}^r (x - \lambda_i)^{t_i}, \quad p_T(x) = \prod_{i=1}^r (x - \lambda_i)^{m_i},$$

where t_i is the largest Jordan block size for λ_i and m_i is the sum of all Jordan block sizes for λ_i . Then $q_T = p_T$ iff $t_i = m_i$ for all i , which happens iff there is exactly one Jordan block for each eigenvalue. Since the number of Jordan blocks for λ_i is $\dim E(\lambda_i, T)$, this is equivalent to $\dim E(\lambda, T) = 1$ for every eigenvalue.

Example 9.5.3 (Jordan-Chevalley Decomposition)

If p_T splits, then

$$T = D + N$$

where D is diagonalizable, N is nilpotent, and $DN = ND$. Choose a Jordan basis. On every Jordan chain for eigenvalue λ_i , define D to act by multiplication by λ_i , and set $N = T - D$. Then D is diagonal in the Jordan basis, while N is the nilpotent part of the Jordan blocks. On a chain $(v_1, \dots, v_k)$,

$$Nv_1 = 0, \quad Nv_j = v_{j-1} \quad (j \geq 2),$$

and D acts as $\lambda_i I$, so $DN = ND$ on every chain.

9.6 Tutorial 6: Nilpotents and Inner Products

Example 9.6.1 (A Nilpotent Jordan Basis)

Let

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & -1 \end{pmatrix}.$$

One checks that $A^2 = 0$ and $\text{rank}(A) = 2$, so $\dim \ker(A) = 3$. Hence the Jordan block sizes are 2, 2, 1,

$$J = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad p_A(x) = x^5, \quad q_A(x) = x^2.$$

A Jordan basis is

$$\left(\begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, e_1, \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, e_2, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right),$$

so the change-of-basis matrix with these columns is

$$C = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix},$$

and $A = CJC^{-1}$.

Example 9.6.2 (Zero Quadratic Form over $\mathbb{C}$)

Let V be a finite-dimensional complex inner product space and suppose $\langle Tv, v \rangle = 0$ for all v . Then $T = 0$. First, every eigenvalue is 0, since an eigenvector v with eigenvalue $\lambda \neq 0$ would give $\langle Tv, v \rangle = \lambda \langle v, v \rangle \neq 0$. In Jordan form, if a block had size at least 2, choose u, v in the chain with $Tv = u$ and $Tu = 0$. Then $\langle u, v \rangle = \langle Tv, v \rangle = 0$, but

$$\langle T(u+v), u+v \rangle = \langle u, u+v \rangle = \langle u, u \rangle \neq 0,$$

a contradiction.

Example 9.6.3 (Real Counterexample)

Over $\mathbb{R}^2$, the rotation

$$T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

satisfies $\langle Tv, v \rangle = 0$ for all $v = (x, y)^T$, since $Tv = (y, -x)^T$ and $xy - yx = 0$.

Example 9.6.4 (Cauchy-Schwarz AM-HM Inequality)

For $x_1, \dots, x_n > 0$,

$$\left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n \frac{1}{x_i} \right) \geq n^2.$$

Apply Cauchy-Schwarz to $u_i = \sqrt{x_i}$ and $v_i = 1/\sqrt{x_i}$.

9.7 Tutorial 7: Riesz, Bessel, and Projections

Theorem 9.7.1 Finite-Dimensional Riesz Representation

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For every linear map $T : V \rightarrow \mathbb{F}$, there exists $v \in V$ such that

$$Tu = \langle u, v \rangle$$

for all $u \in V$.

Proof: Let $v_1, \dots, v_n$ be an orthonormal basis and set $\alpha_i = T(v_i)$. Since our inner product is linear in the first coordinate, take

$$v = \sum_{i=1}^n \overline{\alpha_i} v_i.$$

If $u = \sum_i \beta_i v_i$, then $Tu = \sum_i \beta_i \alpha_i = \langle u, v \rangle$. ☺

Proposition 9.7.1 Bessel's Inequality

If $v_1, \dots, v_m$ are orthonormal vectors in an inner product space V , then for all $x \in V$,

$$\|x\|^2 \geq \sum_{i=1}^m |\langle x, v_i \rangle|^2.$$

Proof: Let $y = \sum_i \langle x, v_i \rangle v_i$, the orthogonal projection of x onto $\text{span}(v_1, \dots, v_m)$. Then $x - y \perp y$, so $\|x\|^2 = \|y\|^2 + \|x - y\|^2 \geq \|y\|^2$. Also $\|y\|^2 = \sum_i |\langle x, v_i \rangle|^2$. ☺

Example 9.7.1 (Projection onto a Subspace of $\mathbb{C}^3$)

Let

$$W = \text{span} \left\{ \begin{pmatrix} 1 \\ -1 \\ -1+i \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\} \subseteq \mathbb{C}^3.$$

Gram-Schmidt gives the orthonormal basis

$$w_1 = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ -1+i \end{pmatrix}, \quad w_2 = \frac{1}{2\sqrt{3}} \begin{pmatrix} 2+i \\ 2-i \\ 1+i \end{pmatrix}.$$

The orthogonal projection matrix is

$$M = w_1 w_1^* + w_2 w_2^*,$$

where

$$w_1 w_1^* = \frac{1}{4} \begin{pmatrix} 1 & -1 & -1-i \\ -1 & 1 & 1+i \\ -1+i & 1-i & 2 \end{pmatrix},$$

and

$$w_2 w_2^* = \frac{1}{12} \begin{pmatrix} 5 & 3+4i & 3-i \\ 3-4i & 5 & 1-3i \\ 3+i & 1+3i & 2 \end{pmatrix}.$$

9.8 Tutorial 8: Adjoints and Least Squares

Example 9.8.1 (Invariant Subspaces via Orthogonal Projections)

Let $W \subseteq V$ and let P_W be the orthogonal projection onto W . Then W is T -invariant if and only if

$$(I - P_W)TP_W = 0.$$

The operator P_W extracts the W -component, T is applied, and $I - P_W$ extracts the $W^\perp$ -component. Thus the expression vanishes exactly when T sends every vector of W back into W .

Example 9.8.2 (Normal Equations for Least Squares)

Let $T : V \rightarrow W$ and fix $w_0 \in W$. If v_0 minimizes $\|Tv - w_0\|$, then

$$T^*Tv_0 = T^*w_0.$$

Indeed, Tv_0 is the orthogonal projection of w_0 onto $\text{Im}(T)$, so $w_0 - Tv_0 \perp \text{Im}(T)$, and hence

$$0 = \langle Tv, w_0 - Tv_0 \rangle = \langle v, T^*w_0 - T^*Tv_0 \rangle$$

for all v .

Also,

$$\ker(T^*T) = \ker(T),$$

since $\langle T^*Tv, v \rangle = \|Tv\|^2$. Therefore, if $\ker(T) = \{0\}$, then T^*T is invertible and

$$v_0 = (T^*T)^{-1}T^*w_0.$$

Example 9.8.3 (Eigenvalues of the Adjoint)

For $A \in \mathbb{C}^{n \times n}$,

$$\det(A^*) = \overline{\det(A)}.$$

This follows from the Leibniz formula and $\det(A^T) = \det(A)$. If T is a linear map on a finite-dimensional complex inner product space and λ is an eigenvalue of T , then $\bar{\lambda}$ is an eigenvalue of T^* . With an orthonormal basis and $A = [T]$, we have

$$0 = \det(\lambda I - A)$$

and therefore

$$0 = \det((\lambda I - A)^*) = \det(\bar{\lambda} I - A^*).$$

9.9 Tutorial 9: Normal Operators

Example 9.9.1 (Norm Equality Characterizes Normality)

If $\|Tv\| = \|T^*v\|$ for all v , then T is normal. Indeed,

$$\langle (T^*T - TT^*)v, v \rangle = \|Tv\|^2 - \|T^*v\|^2 = 0.$$

The map $T^*T - TT^*$ is self-adjoint, so by 6.2.3, it is zero.

Example 9.9.2 (Numerical Range of a Normal Operator)

Let V be finite-dimensional complex, let T be normal, and let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues. Then

$$\left\{ \frac{\langle Tv, v \rangle}{\langle v, v \rangle} : v \neq 0 \right\} = \text{conv}\{\lambda_1, \dots, \lambda_k\}.$$

Write $v = v_1 + \dots + v_k$ with $v_i \in E(\lambda_i, T)$. Orthogonality of distinct eigenspaces gives

$$\frac{\langle Tv, v \rangle}{\langle v, v \rangle} = \sum_{i=1}^k \frac{\|v_i\|^2}{\|v\|^2} \lambda_i,$$

a convex combination. Conversely, for any convex combination $\sum_i \alpha_i \lambda_i$, take unit eigenvectors $u_i \in E(\lambda_i, T)$ and set $v = \sum_i \sqrt{\alpha_i} u_i$.

Example 9.9.3 (Commuting with Spectral Projections)

Let $T = \sum_{i=1}^r \lambda_i P_i$ be the spectral decomposition of a normal operator. If S commutes with T , then S commutes with every P_i and hence with T^* . Indeed, $ST = TS$ implies each eigenspace $E(\lambda_i, T)$ is S -invariant. Since P_i acts as the identity on $E(\lambda_i, T)$ and as 0 on the other eigenspaces, $SP_i = P_iS$. Then

$$ST^* = S \sum_i \overline{\lambda_i} P_i = \sum_i \overline{\lambda_i} P_i S = T^* S.$$

9.10 Tutorial 10: Spectral Theorems and PSD Maps

Example 9.10.1 (Self-Adjoint and Unitary via Normality)

For a finite-dimensional complex inner product space:

1. T is self-adjoint if and only if T is normal and all eigenvalues of T are real.
2. T is unitary if and only if T is normal and every eigenvalue λ satisfies $|\lambda| = 1$.

Use an orthonormal eigenbasis for the normal operator. In that basis, T^* has eigenvalues $\overline{\lambda_i}$ on the same eigenvectors. Thus $T = T^*$ exactly when $\lambda_i = \overline{\lambda_i}$, and $TT^* = I$ exactly when $|\lambda_i|^2 = 1$ for every i .

Example 9.10.2 (Normal Matrix Spectral Theorem)

For $A \in \mathbb{C}^{n \times n}$, the following are equivalent:

1. A is normal.
2. There are a unitary matrix U and diagonal matrix D such that $A = UDU^*$.

If $A = UDU^*$, then $A^* = UD^*U^*$ and $AA^* = A^*A$ because diagonal matrices commute. Conversely, if A is normal, the operator $x \mapsto Ax$ has an orthonormal eigenbasis; put these eigenvectors as columns of U .

Example 9.10.3 (PSD is Preserved by Congruence)

If T is PSD and $S : V \rightarrow V$, then S^*TS is PSD:

$$(S^*TS)^* = S^*T^*S = S^*TS, \quad \langle S^*TSv, v \rangle = \langle TSv, Sv \rangle \geq 0.$$

Example 9.10.4 (Frobenius Inner Product of PSD Matrices)

If $A, B \in \mathbb{C}^{n \times n}$ are PSD, then

$$\langle A, B \rangle_F = \text{tr}(B^*A) \geq 0.$$

Let $A^{1/2}$ and $B^{1/2}$ be PSD square roots. Since $B^* = B$ and $\text{tr}(XY) = \text{tr}(YX)$,

$$\text{tr}(BA) = \text{tr}(A^{1/2}BA^{1/2}) = \text{tr}((A^{1/2}B^{1/2})(B^{1/2}A^{1/2})).$$

With $M = B^{1/2}A^{1/2}$, this becomes $\text{tr}(M^*M) = \|M\|_F^2 \geq 0$.

9.11 Tutorial 11: SVD and Polar Decomposition

Proposition 9.11.1 Cauchy-Schwarz for PSD Forms

If T is PSD on a finite-dimensional inner product space, then for all $x, y \in V$,

$$|\langle Tx, y \rangle|^2 \leq \langle Tx, x \rangle \langle Ty, y \rangle.$$

Proof: Let $T^{1/2}$ be the PSD square root of T . Then

$$\langle Tx, y \rangle = \langle T^{1/2}x, T^{1/2}y \rangle, \quad \langle Tx, x \rangle = \|T^{1/2}x\|^2, \quad \langle Ty, y \rangle = \|T^{1/2}y\|^2.$$

Apply Cauchy-Schwarz. ☺

Example 9.11.1 (SVD and Polar Decomposition of a Matrix)

For

$$A = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix},$$

we have

$$A^*A = \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix},$$

with eigenvalues 8 and 2. Thus $\sigma_1 = \sqrt{8}$ and $\sigma_2 = \sqrt{2}$. Take

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then

$$w_1 = \frac{1}{\sqrt{8}}Av_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad w_2 = \frac{1}{\sqrt{2}}Av_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Hence

$$A = U_2 \Sigma U_1^*, \quad U_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}.$$

The polar decomposition $A = UR$ is

$$R = U_1 \Sigma U_1^* = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}, \quad U = U_2 U_1^* = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Example 9.11.2 (Equal Norms Imply a Unitary Intertwiner)

Let $S, T : V \rightarrow V$ satisfy $\|Sv\| = \|Tv\|$ for all v .

1. Then $S^*S = T^*T$. Indeed,

$$\langle (S^*S - T^*T)v, v \rangle = \|Sv\|^2 - \|Tv\|^2 = 0,$$

and $S^*S - T^*T$ is self-adjoint, so 6.2.3 applies.

2. There is a unitary map U such that $T = US$. Use the SVD construction for S and

T from the common PSD operator $S^*S = T^*T$. If

$$Sv_i = \sigma_i w_i, \quad Tv_i = \sigma_i w'_i$$

for orthonormal bases (w_i) and (w'_i) , define $Uw_i = w'_i$. Then U is unitary and $US = T$ on the basis (v_i) .

Chapter 10

Homework Problems and Official Solutions

Note:-

This chapter transcribes the official homework questions and solutions from MAT247 HW* questions.pdf and Solution HW*.pdf. The user's own answer PDFs (MAT247 HW* ans.pdf) were not used.

10.1 Homework 1

Question 1: HW1 Q1: Diagonalization

Let

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Show that A is diagonalizable and find an invertible matrix C such that $C^{-1}AC$ is diagonal.

Solution: **Characteristic polynomial.** Expanding $\det(xI - A)$ along the first row,

$$p_A(x) = \det \begin{pmatrix} x-1 & 0 & 0 \\ -1 & x-2 & 1 \\ 0 & 0 & x-1 \end{pmatrix} = (x-1) \cdot \begin{vmatrix} x-2 & 1 \\ 0 & x-1 \end{vmatrix} = (x-1)^2(x-2).$$

So the eigenvalues are $\lambda = 1$ (algebraic multiplicity 2) and $\lambda = 2$ (algebraic multiplicity 1).

Eigenvectors for $\lambda = 2$. We solve $(A - 2I)v = 0$. Now

$$A - 2I = \begin{pmatrix} -1 & 0 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & -1 \end{pmatrix},$$

so the equations $-x = 0$ and $-z = 0$ give $x = z = 0$ with y free. An eigenvector is

$$v_1 = (0, 1, 0)^T.$$

Eigenvectors for $\lambda = 1$. We solve $(A - I)v = 0$. Now

$$A - I = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix},$$

which gives the single equation $x + y - z = 0$. The solution space is two-dimensional. Choosing $(y, z) = (1, 0)$ and $(y, z) = (0, 1)$ gives two linearly independent eigenvectors:

$$v_2 = (-1, 1, 0)^T, \quad v_3 = (1, 0, 1)^T.$$

Diagonalization. The three eigenvectors v_1, v_2, v_3 are linearly independent (eigenvectors with distinct eigenvalues are independent, and v_2, v_3 are independent within $E(1, A)$). Hence A is diagonalizable. Setting

$$C = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

the columns of C are v_1, v_2, v_3 , and

$$C^{-1}AC = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

☺

Question 2: HW1 Q2: Characteristic Polynomial Coefficients

Let $A \in \mathbb{F}^{n \times n}$ and write $p_A(x) = \sum_{i=0}^n c_i x^i$ with $c_n = 1$. Show that

$$c_0 = (-1)^n \det(A) \quad \text{and} \quad c_{n-1} = -\operatorname{tr}(A).$$

Solution: **Item (a).** Substituting $x = 0$,

$$c_0 = p_A(0) = \det(0 \cdot I - A) = \det(-A) = (-1)^n \det(A).$$

Item (b). Apply the Leibniz formula

$$\det(B) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n B_{i, \sigma(i)}$$

to $B := xI - A$. Each term selects one entry from each row, no two from the same column. We track which terms can contribute x^{n-1} .

Claim. For every non-identity permutation $\sigma \neq \operatorname{id}$, the product $\prod_{i=1}^n B_{i, \sigma(i)}$ has degree at most $n - 2$ in x .

Reason. Powers of x come only from diagonal entries $B_{ii} = x - a_{ii}$. So the degree of the product equals the number of fixed points of σ . A permutation cannot fix exactly $n - 1$ points

(the remaining single point would also be fixed), so a non-identity σ fixes at most $n - 2$ points. This proves the claim.

Therefore, writing q for a polynomial of degree $\leq n - 2$,

$$p_A(x) = \text{sgn}(\text{id}) \prod_{i=1}^n (x - a_{ii}) + q(x) = \prod_{i=1}^n (x - a_{ii}) + q(x).$$

Expanding $\prod_{i=1}^n (x - a_{ii})$, the coefficient of x^{n-1} comes from picking x from $n-1$ brackets and $-a_{ii}$ from the remaining one. Summing over the choice of bracket gives $-(a_{11} + \cdots + a_{nn}) = -\text{tr}(A)$. Since q contributes nothing to the x^{n-1} coefficient,

$$c_{n-1} = -\text{tr}(A).$$

☺

Question 3: HW1 Q3: Matrix Square Root

Let $A \in \mathbb{R}^{n \times n}$ be diagonalizable and suppose all eigenvalues of A are nonnegative. Show that there exists $B \in \mathbb{R}^{n \times n}$ such that $B^2 = A$.

Solution: Since A is diagonalizable, there is an invertible C with

$$A = CDC^{-1}, \quad D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

The eigenvalues $\lambda_i \geq 0$ have real square roots, so we may set

$$D_1 := \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) \in \mathbb{R}^{n \times n},$$

and note $D_1^2 = D$. Putting $B := CD_1C^{-1}$, we compute

$$B^2 = (CD_1C^{-1})(CD_1C^{-1}) = CD_1^2C^{-1} = CDC^{-1} = A.$$

☺

Question 4: HW1 Q4: Nonzero Eigenvalues of AB and BA

Let $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times m}$. Show that for every nonzero $\lambda \in \mathbb{F}$, λ is an eigenvalue of AB iff it is an eigenvalue of BA , and the geometric multiplicities agree.

Solution: Same eigenvalues. Suppose $\lambda \neq 0$ is an eigenvalue of AB with eigenvector v , so $ABv = \lambda v$. Then

$$BA(Bv) = B(ABv) = B(\lambda v) = \lambda Bv.$$

We must check $Bv \neq 0$: if $Bv = 0$ then $\lambda v = ABv = A \cdot 0 = 0$, contradicting $\lambda \neq 0$ and $v \neq 0$. So Bv is an eigenvector of BA with eigenvalue λ . Swapping the roles of A, B gives the converse.

Equal geometric multiplicities. Let $v_1, \dots, v_k$ be a basis of $E(\lambda, AB)$. As above, $Bv_1, \dots, Bv_k \in E(\lambda, BA)$. We claim they are linearly independent. Suppose $\sum_{i=1}^k \alpha_i Bv_i = 0$, i.e., $B(\sum_{i=1}^k \alpha_i v_i) = 0$. Then

$$0 = A \cdot 0 = AB \left(\sum_{i=1}^k \alpha_i v_i \right) = \sum_{i=1}^k \alpha_i \lambda v_i.$$

Since $v_1, \dots, v_k$ are independent, $\alpha_i \lambda = 0$ for all i , and as $\lambda \neq 0$, $\alpha_i = 0$. Hence $\dim E(\lambda, AB) \leq \dim E(\lambda, BA)$. Swapping A, B gives the reverse inequality. $\odot$

10.2 Homework 2

Question 5: HW2 Q1: Eigenvector Sums in Invariant Subspaces

Let $T : V \rightarrow V$, let W be a T -invariant subspace, and let $v_1, \dots, v_k$ be eigenvectors with distinct eigenvalues $\lambda_1, \dots, \lambda_k$. Show that if $v_1 + \dots + v_k \in W$, then $v_i \in W$ for all i .

Solution: Induct on k . The case $k = 1$ is trivial. For $k \geq 2$, set $w := v_1 + \dots + v_k \in W$, and apply $T - \lambda_k I$ to both sides. Since $(T - \lambda_k I)v_k = 0$, this kills the v_k -term:

$$\sum_{i=1}^{k-1} (\lambda_i - \lambda_k) v_i = \sum_{i=1}^{k-1} (T - \lambda_k I) v_i = (T - \lambda_k I) w \in W, \quad (*)$$

where the last membership uses T -invariance of W .

The vectors $(\lambda_i - \lambda_k) v_i$ for $i = 1, \dots, k-1$ are eigenvectors with distinct eigenvalues (since cv is still an eigenvector with eigenvalue λ for any nonzero c , and $\lambda_i \neq \lambda_k$ implies $\lambda_i - \lambda_k \neq 0$). Their sum lies in W by $(*)$, so by the induction hypothesis applied to these $k-1$ eigenvectors, each $(\lambda_i - \lambda_k) v_i \in W$, and dividing by the nonzero scalar $\lambda_i - \lambda_k$ gives $v_i \in W$ for $i = 1, \dots, k-1$. Plugging back into $w = v_1 + \dots + v_k$, we conclude $v_k = w - \sum_{i < k} v_i \in W$. $\odot$

Question 6: HW2 Q2: Invariant Subspaces of Diagonalizable Maps

Let V be finite-dimensional, $T : V \rightarrow V$ diagonalizable with distinct eigenvalues $\lambda_1, \dots, \lambda_k$, and $W \subseteq V$ a T -invariant subspace. Show

$$W = \bigoplus_{i=1}^k (W \cap E(\lambda_i, T)),$$

and deduce that $T|_W$ is diagonalizable.

Solution: We use the criterion (proved in class) that T is diagonalizable iff $V = \bigoplus_{i=1}^k E(\lambda_i, T)$.

(a) Since T is diagonalizable,

$$V = \bigoplus_{i=1}^k E(\lambda_i, T). \quad (1)$$

Let $w \in W$. By (1), there are $v_i \in E(\lambda_i, T)$ with $w = v_1 + \dots + v_k$. By 5, each $v_i \in W$, so $v_i \in W \cap E(\lambda_i, T)$. Hence

$$W = \sum_{i=1}^k (W \cap E(\lambda_i, T)).$$

The sum is direct because the larger sum (1) is direct: the only way to write $0 = v_1 + \cdots + v_k$ with $v_i \in W \cap E(\lambda_i, T) \subseteq E(\lambda_i, T)$ is $v_1 = \cdots = v_k = 0$.

(b) Note $E(\lambda_i, T|_W) = W \cap E(\lambda_i, T)$ (a vector $v \in W$ satisfies $T|_W v = \lambda_i v$ iff it satisfies $Tv = \lambda_i v$). Hence $W = \bigoplus_{i=1}^k E(\lambda_i, T|_W)$, so by the criterion, $T|_W$ is diagonalizable. $\odot$

Question 7: HW2 Q3: Inverse as a Polynomial

Let A be an $n \times n$ invertible matrix. Show that there is a polynomial f with $\deg(f) = n-1$ such that $A^{-1} = f(A)$.

Solution: Write $p_A(x) = \sum_{i=0}^n a_i x^i$ with $a_n = 1$. Since A is invertible, 0 is not an eigenvalue, so $p_A(0) \neq 0$, i.e., $a_0 \neq 0$.

By Cayley-Hamilton, $\sum_{i=0}^n a_i A^i = 0$. Moving the constant term and dividing by a_0 ,

$$-\frac{1}{a_0} \sum_{i=1}^n a_i A^i = I.$$

Factoring A out of the sum,

$$A \cdot \left(-\sum_{i=0}^{n-1} \frac{a_{i+1}}{a_0} A^i \right) = I,$$

so $A^{-1} = f(A)$ with

$$f(x) := -\sum_{i=0}^{n-1} \frac{a_{i+1}}{a_0} x^i.$$

The coefficient of x^{n-1} in f is $-a_n/a_0 = -1/a_0 \neq 0$, so $\deg(f) = n-1$. $\odot$

Question 8: HW2 Q4: Coprime Factors on a Kernel

Let $T : V \rightarrow V$ with $\dim(V) < \infty$. Let f, g be relatively prime polynomials and $W := \ker(f(T))$. Show that $g(T)|_W$ is a bijection on W .

Solution: First, $g(T)$ maps W into W : if $v \in W$, then $f(T)g(T)v = g(T)f(T)v = g(T) \cdot 0 = 0$, so $g(T)v \in W$. Now W is finite-dimensional, so it suffices to show $g(T)|_W$ is injective.

Suppose $v \in W$ and $g(T)v = 0$. By Bezout, since f, g are coprime, there are polynomials a, b with $af + bg = 1$. Then

$$v = I \cdot v = (af + bg)(T)v = a(T)f(T)v + b(T)g(T)v = a(T) \cdot 0 + b(T) \cdot 0 = 0,$$

using $v \in \ker(f(T))$ and $g(T)v = 0$. Hence $g(T)|_W$ is injective, and therefore bijective. $\odot$

10.3 Homework 3

Question 9: HW3 Q1: Simultaneous Diagonalization

Let V be finite-dimensional and $S, T : V \rightarrow V$ both diagonalizable. Show TFAE: (a) $ST = TS$; (b) there is a basis B such that $[S]_B$ and $[T]_B$ are both diagonal.

Solution: (b) $\Rightarrow$ (a). If $B = (v_1, \dots, v_n)$ is a common eigenbasis with $Tv_i = \lambda_i v_i$ and $Sv_i = \mu_i v_i$, then

$$STv_i = S(\lambda_i v_i) = \mu_i \lambda_i v_i = T(\mu_i v_i) = TSv_i.$$

Two operators agreeing on a basis are equal, so $ST = TS$.

(a) $\Rightarrow$ (b). Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T . Since T is diagonalizable, $V = \bigoplus_{i=1}^k E(\lambda_i, T)$.

We claim each $E(\lambda_i, T)$ is S -invariant. For $v \in E(\lambda_i, T)$,

$$(T - \lambda_i I)Sv = S(T - \lambda_i I)v = S \cdot 0 = 0,$$

using $ST = TS$. So $Sv \in E(\lambda_i, T)$.

By 6, $S|_{E(\lambda_i, T)}$ is diagonalizable. Pick a basis B_i of $E(\lambda_i, T)$ consisting of eigenvectors of S . Each $v \in B_i$ is also an eigenvector of T (since $v \in E(\lambda_i, T)$). Letting $B = \bigcup_i B_i$, the decomposition above ensures B is a basis of V , and every $v \in B$ is a common eigenvector of S and T . Thus $[S]_B$ and $[T]_B$ are both diagonal. $\odot$

Question 10: HW3 Q2: Stabilized Kernels and Images

Let $T : V \rightarrow V$ with $\dim(V) < \infty$.

- (a) Show $\ker(T^i) \subseteq \ker(T^{i+1})$ and $\text{im}(T^i) \supseteq \text{im}(T^{i+1})$ for all $i \geq 0$.
- (b) Let $k \geq 0$ be minimal with $\ker(T^{k+1}) = \ker(T^k)$. Show:
 - i. $\ker(T^\ell) = \ker(T^k)$ and $\text{im}(T^\ell) = \text{im}(T^k)$ for all $\ell \geq k$;
 - ii. $V = \ker(T^k) \oplus \text{im}(T^k)$;
 - iii. $k \leq \dim(V)$.

Solution: (a) If $v \in \ker(T^i)$, then $T^{i+1}v = T(T^i v) = T \cdot 0 = 0$, so $v \in \ker(T^{i+1})$. If $v \in \text{im}(T^{i+1})$, write $v = T^{i+1}w = T^i(Tw)$, so $v \in \text{im}(T^i)$.

(b).i. We show $\ker(T^\ell) = \ker(T^k)$ for all $\ell \geq k+1$ by induction on ℓ . The case $\ell = k+1$ is the assumption. For $\ell \geq k+2$: by (a), $\ker(T^k) \subseteq \ker(T^\ell)$. For the reverse, let $v \in \ker(T^\ell)$ and put $u := Tv$. Then $T^{\ell-1}u = T^\ell v = 0$, so $u \in \ker(T^{\ell-1}) = \ker(T^k)$ by induction. Hence $T^k u = 0$, i.e., $T^{k+1}v = 0$, so $v \in \ker(T^{k+1}) = \ker(T^k)$.

For images, use rank-nullity: $\dim \text{im}(T^\ell) = \dim V - \dim \ker(T^\ell) = \dim V - \dim \ker(T^k) = \dim \text{im}(T^k)$, combined with $\text{im}(T^\ell) \subseteq \text{im}(T^k)$ from (a), gives equality.

(b).ii. By rank-nullity, $\dim \ker(T^k) + \dim \text{im}(T^k) = \dim V$. So it suffices to show $\ker(T^k) \cap \text{im}(T^k) = \{0\}$. Let $v \in \ker(T^k) \cap \text{im}(T^k)$; write $v = T^k w$. Then $0 = T^k v = T^{2k} w$, so $w \in \ker(T^{2k}) = \ker(T^k)$ by (b).i. Hence $v = T^k w = 0$.

(b).iii. By minimality of k , the chain

$$\ker(T^0) \subsetneq \ker(T^1) \subsetneq \cdots \subsetneq \ker(T^k)$$

is strictly increasing, giving $k + 1$ distinct values $\dim \ker(T^i)$ in $\{0, 1, \dots, \dim V\}$. So $k + 1 \leq \dim V + 1$, i.e., $k \leq \dim V$. $\odot$

Question 11: HW3 Q3: Jordan Canonical Form

Find the Jordan canonical form of

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Solution: **Characteristic polynomial.** Expanding along the first column twice,

$$p_A(x) = \det(xI - A) = (x - 1)^5,$$

so $\lambda = 1$ is the only eigenvalue, with algebraic multiplicity 5.

Number of blocks. We compute

$$A - I = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

This has two zero rows, and rows 1 and 4 are scalar multiples of each other; one checks $\text{rank}(A - I) = 2$, so $\dim \ker(A - I) = 3$. Hence there are exactly 3 Jordan blocks. The block sizes (summing to 5) must be either $(3, 1, 1)$ or $(2, 2, 1)$.

Largest block size. Direct computation gives $(A - I)^2 = 0$. Hence the largest block has size at most 2, so the block sizes are $(2, 2, 1)$. The Jordan canonical form is

$$J = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

$\odot$

10.4 Homework 4

Question 12: HW4 Q1: Cyclic Vector from Coprime Pieces

Let $\dim(V) = n$, $T : V \rightarrow V$, and $V = V_1 \oplus \cdots \oplus V_r$ with each V_i being T -invariant. Suppose the minimal polynomials $q_{T|_{V_i}}$ are pairwise relatively prime, and each $T|_{V_i}$ has a cyclic vector v_i . Show that T has a cyclic vector.

Solution: **Solution via Bezout (cleaner).** Set $v := v_1 + \cdots + v_r$ and put $q_i := q_{T|_{V_i}}$, $Q_i := \prod_{j \neq i} q_j$. Since the q_j are pairwise coprime, $\gcd(q_i, Q_i) = 1$, so by Bezout there are polynomials a, b with $aq_i + bQ_i = 1$. Set $f_i := bQ_i$.

For $j \neq i$, the factor q_j of Q_i kills v_j (since $q_j(T)|_{V_j} = q_{T|_{V_j}}(T|_{V_j}) = 0$), so $f_i(T)v_j = b(T)Q_i(T)v_j = 0$. For $j = i$,

$$f_i(T)v_i = b(T)Q_i(T)v_i = v_i - a(T)q_i(T)v_i = v_i,$$

using $aq_i + bQ_i = 1$ and $q_i(T)v_i = 0$. Therefore

$$f_i(T)v = \sum_{j=1}^r f_i(T)v_j = v_i.$$

Now let $W := \text{span}\{T^k v : k \geq 0\}$ be the cyclic subspace of v . The identity $f_i(T)v = v_i$ shows $v_i \in W$. Since v_i is cyclic for $T|_{V_i}$, the subspace W contains $\text{span}\{T^k v_i : k \geq 0\} = V_i$. As $V = V_1 + \cdots + V_r \subseteq W$, v is cyclic for T .

Alternative solution by contradiction. We give the second proof from the official solution. Set $v = \sum v_i$. Suppose for contradiction $\{v, Tv, \dots, T^{n-1}v\}$ is linearly dependent, so there is a non-zero polynomial f of degree at most $n-1$ with $f(T)v = 0$. Since the V_i are T -invariant, $f(T)v_i \in V_i$, and the direct-sum decomposition gives $f(T)v_i = 0$ for each i . As $T^k v_i \in V_i$ and v_i is cyclic for $T|_{V_i}$, the span $\text{span}\{T^k v_i\} = V_i$, and $f(T) \cdot T^k v_i = T^k \cdot f(T)v_i = 0$, so $f(T)|_{V_i} = 0$. Thus $q_i \mid f$. By pairwise coprimality, $\prod_i q_i \mid f$. But for a map with a cyclic vector the minimal and characteristic polynomials coincide, so $q_i = p_{T|_{V_i}}$ has degree $\dim V_i$, and $\deg \prod_i q_i = \sum_i \dim V_i = n$, contradicting $\deg f \leq n-1$. ☺

Question 13: HW4 Q2: Possible Jordan Forms

Let $A \in \mathbb{R}^{6 \times 6}$ have minimal polynomial $q_A(x) = (x - \lambda)^3(x - \mu)$. List all possible Jordan canonical forms.

Solution: The eigenvalues of A are exactly the roots of the minimal polynomial, so they are λ and μ . The largest Jordan block of type λ has size 3 (the multiplicity of $(x - \lambda)$ in q_A), and every block of type μ has size 1. The blocks must sum to 6, with at least one λ -block of size 3 and at least one μ -block. The remaining $6 - 3 - 1 = 2$ units of size are distributed among additional λ -blocks (each of size ≤ 3) and μ -blocks (each of size 1). The possibilities are:

- $J(\lambda, 3), J(\lambda, 2), J(\mu, 1);$
- $J(\lambda, 3), J(\lambda, 1), J(\lambda, 1), J(\mu, 1);$
- $J(\lambda, 3), J(\lambda, 1), J(\mu, 1), J(\mu, 1);$

- $J(\lambda, 3), J(\mu, 1), J(\mu, 1), J(\mu, 1)$.



Question 14: HW4 Q3: Jordan Basis

Let

$$A = \begin{pmatrix} 3 & 0 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 \\ -1 & 0 & 3 & 0 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 1 & 1 & 1 & 1 & 3 \end{pmatrix}.$$

Find a Jordan basis, the characteristic and minimal polynomials, and the Jordan canonical form.

Solution: A is lower triangular with 3's on the diagonal, so $p_A(x) = (x - 3)^5$. Set

$$N := A - 3I = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix},$$

which is nilpotent. A Jordan basis for N is also a Jordan basis for A .

Powers of N . Direct computation gives

$$N^2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}, \quad N^3 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad N^4 = 0.$$

So the largest block has size 4. The block sizes summing to 5 with maximum 4 are $(4, 1)$.

Jordan basis. We need a basis $\{v_1, v_2\}$ of $\ker(N)$ with $v_1 \in \text{im}(N^3) \cap \ker(N)$ being the start of a length-4 chain. From N^3 , $\text{im}(N^3) = \text{span}\{e_5\}$. From $Nx = 0$,

$$\ker(N) = \{x \in \mathbb{R}^5 : x_1 = x_2 = 0, x_3 + x_4 = 0\} = \text{span}\{e_5, e_3 - e_4\}.$$

So $\text{im}(N^3) \cap \ker(N) = \text{span}\{e_5\}$. Take $v_1 := e_5$ and complete with $v_2 := (0, 0, 1, -1, 0)^T$.

To find the chain: solve $N^3w = v_1 = e_5$. The choice $w = e_1$ works (last row of N^3). Then

$$Ne_1 = (0, 1, -1, 0, 1)^T = e_2 - e_3 + e_5, \quad N^2e_1 = N(e_2 - e_3 + e_5) = e_4 + e_5 - e_5 + 0 = e_4,$$

and $N^3e_1 = Ne_4 = e_5$. Thus a Jordan basis for A is

$$(N^3w, N^2w, Nw, w, v_2) = (e_5, e_4, e_2 - e_3 + e_5, e_1, (0, 0, 1, -1, 0)^T).$$

Canonical form and minimal polynomial. With block sizes $(4, 1)$ both of type $\lambda = 3$,

$$J = \begin{pmatrix} 3 & 1 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}, \quad q_A(x) = (x - 3)^4$$

(the multiplicity of λ in q_A equals the largest size of a Jordan block of type λ). $\ominus$

Question 15: HW4 Q4: Powers of a Jordan Block

Let $J = J(\lambda, n)$. Show that for $m \geq 1$ and $1 \leq i, j \leq n$,

$$(J^m)_{ij} = \binom{m}{j-i} \lambda^{m-j+i},$$

where $\binom{m}{k} = 0$ if $k < 0$ or $k > m$.

Solution: Induct on m . **Base** $m = 1$. The matrix J has λ on the diagonal ($j - i = 0$) and 1 on the superdiagonal ($j - i = 1$), and 0 elsewhere. The formula gives $\binom{1}{0} \lambda^1 = \lambda$ when $j = i$, $\binom{1}{1} \lambda^0 = 1$ when $j = i + 1$, and 0 otherwise (for $j - i < 0$ or $j - i > 1$). This matches.

Inductive step. Assume the formula holds for $m - 1$. From $J^m = J^{m-1}J$,

$$(J^m)_{ij} = \sum_{k=1}^n (J^{m-1})_{ik} J_{kj}. \quad (*)$$

Column j of J has nonzero entries only at $J_{jj} = \lambda$ and (if $j \geq 2$) at $J_{j-1,j} = 1$.

Case $j = 1$: only $k = 1$ contributes, $J_{11} = \lambda$, so $(J^m)_{i1} = (J^{m-1})_{i1} \cdot \lambda$. By IH, $(J^{m-1})_{i1} = \binom{m-1}{1-i} \lambda^{m-i}$, which is 0 for $i > 1$. For $i = 1$, $(J^{m-1})_{11} = \lambda^{m-1}$, so $(J^m)_{11} = \lambda^m = \binom{m}{0} \lambda^m$. Both cases agree with the claimed formula.

Case $2 \leq j \leq n$: only $k = j$ and $k = j - 1$ contribute, giving

$$(J^m)_{ij} = (J^{m-1})_{i,j-1} \cdot J_{j-1,j} + (J^{m-1})_{ij} \cdot J_{jj} = (J^{m-1})_{i,j-1} + \lambda (J^{m-1})_{ij}.$$

Substituting the IH,

$$(J^m)_{ij} = \binom{m-1}{j-1-i} \lambda^{m-j+i} + \lambda \cdot \binom{m-1}{j-i} \lambda^{m-1-j+i} = \lambda^{m-j+i} \left[\binom{m-1}{j-1-i} + \binom{m-1}{j-i} \right].$$

By Pascal's identity, the bracket equals $\binom{m}{j-i}$, completing the induction. $\ominus$

Question 16: HW4 Q5: Parallelogram and Polarization Identities

Let V be an inner product space and $u, v \in V$. Show

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2, \quad \|u + v\|^2 - \|u - v\|^2 = 4\Re\langle u, v \rangle.$$

Solution: Expand using sesquilinearity:

$$\|u + v\|^2 = \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle,$$

$$\|u - v\|^2 = \langle u - v, u - v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle.$$

(In $\langle u, -v \rangle$ and $\langle -v, u \rangle$ we pull out the scalar -1 ; over $\mathbb{C}$, conjugating -1 from the second slot still gives -1 .)

Sum: the cross terms $\pm\langle u, v \rangle$ and $\pm\langle v, u \rangle$ cancel:

$$\|u + v\|^2 + \|u - v\|^2 = 2\langle u, u \rangle + 2\langle v, v \rangle = 2\|u\|^2 + 2\|v\|^2.$$

Difference: the diagonal terms cancel and the cross terms double:

$$\|u + v\|^2 - \|u - v\|^2 = 2\langle u, v \rangle + 2\langle v, u \rangle = 2\langle u, v \rangle + 2\overline{\langle u, v \rangle} = 4\Re\langle u, v \rangle,$$

using $\langle v, u \rangle = \overline{\langle u, v \rangle}$ and $z + \bar{z} = 2\Re z$. ☺

10.5 Homework 5

Question 17: HW5 Q1: Cauchy-Schwarz Inequalities

For positive real numbers $x_1, \dots, x_n, y_1, \dots, y_n > 0$, prove

$$\left(\frac{x_1 + \dots + x_n}{n} \right)^2 \leq \frac{x_1^2 + \dots + x_n^2}{n}, \quad \frac{(x_1 + \dots + x_n)^2}{y_1 + \dots + y_n} \leq \sum_{i=1}^n \frac{x_i^2}{y_i}.$$

Solution: Both follow from Cauchy-Schwarz $\langle u, v \rangle^2 \leq \|u\|^2 \|v\|^2$ for the standard inner product on $\mathbb{R}^n$.

(a) Take $u = (x_1, \dots, x_n)$ and $v = (1/n, \dots, 1/n)$. Then

$$\langle u, v \rangle = \frac{x_1 + \dots + x_n}{n}, \quad \|u\|^2 = \sum_i x_i^2, \quad \|v\|^2 = \sum_i \frac{1}{n^2} = \frac{1}{n}.$$

The Cauchy-Schwarz inequality $\langle u, v \rangle^2 \leq \|u\|^2 \|v\|^2$ gives

$$\left(\frac{\sum x_i}{n} \right)^2 \leq \left(\sum_i x_i^2 \right) \cdot \frac{1}{n} = \frac{\sum_i x_i^2}{n}.$$

(b) Take $u_i = x_i/\sqrt{y_i}$ and $v_i = \sqrt{y_i}$ (well-defined as $y_i > 0$). Then

$$\langle u, v \rangle = \sum_i \frac{x_i}{\sqrt{y_i}} \cdot \sqrt{y_i} = \sum_i x_i, \quad \|u\|^2 = \sum_i \frac{x_i^2}{y_i}, \quad \|v\|^2 = \sum_i y_i.$$

Cauchy-Schwarz gives $(\sum_i x_i)^2 \leq (\sum_i x_i^2/y_i)(\sum_i y_i)$, and dividing by $\sum_i y_i > 0$ gives the result. ☺

Question 18: HW5 Q2: Orthonormal Basis of $\mathbb{R}_2[x]$

Let $V = \mathbb{R}_2[x]$ with $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$. Find an orthonormal basis.

Solution: Apply Gram-Schmidt to $u_0 = 1, u_1 = x, u_2 = x^2$.

Step 0. Set $v_0 := u_0 = 1$. Compute $\langle v_0, v_0 \rangle = \int_0^1 1 dx = 1$.

Step 1. Compute $\langle u_1, v_0 \rangle = \int_0^1 x dx = 1/2$. Set

$$v_1 := u_1 - \frac{\langle u_1, v_0 \rangle}{\langle v_0, v_0 \rangle} v_0 = x - \frac{1}{2}.$$

Then $\langle v_1, v_1 \rangle = \int_0^1 (x - \frac{1}{2})^2 dx = 1/12$.

Step 2. Compute $\langle u_2, v_0 \rangle = \int_0^1 x^2 dx = 1/3$ and

$$\langle u_2, v_1 \rangle = \int_0^1 x^2(x - \frac{1}{2}) dx = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

Set

$$v_2 := u_2 - \frac{\langle u_2, v_0 \rangle}{\langle v_0, v_0 \rangle} v_0 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = x^2 - \frac{1}{3} - \frac{1/12}{1/12} \left(x - \frac{1}{2}\right) = x^2 - x + \frac{1}{6}.$$

Then $\langle v_2, v_2 \rangle = \int_0^1 (x^2 - x + 1/6)^2 dx = 1/180$.

Normalization. The orthonormal basis is

$$w_0 = 1, \quad w_1 = \sqrt{12} \left(x - \frac{1}{2}\right), \quad w_2 = \sqrt{180} \left(x^2 - x + \frac{1}{6}\right).$$

☺

Question 19: HW5 Q3: Adjoint Fundamental Subspaces

Let V be an inner product space and $T : V \rightarrow V$.

- (a) Show $\ker(T^*) = (\operatorname{im} T)^\perp$.
- (b) If $\dim V < \infty$, show $\operatorname{im}(T^*) = (\ker T)^\perp$.

Solution: (a) ($\subseteq$) Let $u \in \ker(T^*)$, and let $w \in \operatorname{im}(T)$, say $w = Tv$. Then

$$\langle w, u \rangle = \langle Tv, u \rangle = \langle v, T^*u \rangle = \langle v, 0 \rangle = 0,$$

so $u \perp w$. Thus $u \in (\operatorname{im} T)^\perp$.

($\supseteq$) Let $u \in (\operatorname{im} T)^\perp$. For every $v \in V$, $\langle Tv, u \rangle = 0$, hence $\langle v, T^*u \rangle = 0$ for every v . Taking $v := T^*u$ gives $\|T^*u\|^2 = 0$, so $T^*u = 0$, i.e., $u \in \ker(T^*)$.

(b) Apply (a) to T^* in place of T : $\ker(T) = \ker((T^*)^*) = (\operatorname{im}(T^*))^\perp$. In finite dimensions, $(W^\perp)^\perp = W$ for every subspace W , so taking $\perp$ of both sides gives $(\ker T)^\perp = \operatorname{im}(T^*)$. ☺

Question 20: HW5 Q4: Projections

Let V be a vector space. A linear $P : V \rightarrow V$ is a *projection* if $P^2 = P$.

- (a) Show $V = \ker(P) \oplus \operatorname{im}(P)$.
- (b) For $V = U \oplus W$, construct the unique projection with kernel U and image W .
- (c) If V is a finite-dimensional inner product space, show P is orthogonal iff $P^* = P$.

Solution: (a) Let $v \in V$ and $u := v - Pv$. Then $Pu = Pv - P^2v = Pv - Pv = 0$, so $u \in \ker(P)$, and $Pv \in \operatorname{im}(P)$. Thus $v = u + Pv \in \ker(P) + \operatorname{im}(P)$.

For triviality of intersection: if $v \in \ker(P) \cap \operatorname{im}(P)$, write $v = Pw$. Then $0 = Pv = P^2w = Pw = v$.

(b) *Existence.* Define $P : V \rightarrow V$ by $P(u+w) := w$ for the unique decomposition $v = u+w$ with $u \in U, w \in W$. Linearity follows because the decomposition is unique. By construction, $\operatorname{im}(P) \subseteq W$, and for $w \in W$ we have $w = 0 + w$, hence $Pw = w \in \operatorname{im}(P)$, so $\operatorname{im}(P) = W$. Also, every $u \in U$ has decomposition $u = u + 0$, so $Pu = 0$, giving $U \subseteq \ker(P)$. Conversely, if $Pv = 0$ and $v = u + w$, then $w = Pv = 0$, so $v = u \in U$, giving $\ker(P) = U$.

Uniqueness. For any projection P with $\operatorname{im}(P) = W$ and $w \in W$, write $w = Pv$; then $Pw = P^2v = Pv = w$. So any projection with image W acts as the identity on W and as 0 on $U = \ker(P)$, which determines it completely on $V = U \oplus W$.

(c) ($\Leftarrow$) If $P^* = P$, by 19 we have $\ker(P) = \ker(P^*) = (\operatorname{im} P)^\perp$, so P is orthogonal.

($\Rightarrow$) If P is orthogonal, then $\ker(P) = (\operatorname{im} P)^\perp = \ker(P^*)$ by 19, and similarly $\operatorname{im}(P^*) = (\ker P)^\perp = \operatorname{im}(P)$ (using V finite-dimensional). By (b), a projection is determined by its kernel and image, so $P^* = P$.

Alternative. Show directly that $\langle Pv_1, v_2 \rangle = \langle v_1, Pv_2 \rangle$. Write $v_i = u_i + w_i$ with $u_i \in \ker(P), w_i \in \operatorname{im}(P)$. Since P is orthogonal, $\langle u_i, w_j \rangle = 0$. Hence

$$\langle Pv_1, v_2 \rangle = \langle w_1, u_2 + w_2 \rangle = \langle w_1, w_2 \rangle = \langle u_1 + w_1, w_2 \rangle = \langle v_1, Pv_2 \rangle.$$

☺

10.6 Homework 6

Question 21: HW6 Q1: Schur Triangularization

Let V be a finite-dimensional complex inner product space. Show that for every $T : V \rightarrow V$, there is an orthonormal ordered basis B such that $[T]_B$ is upper triangular.

Solution: Induct on $\dim V$. For $\dim V = 0$ (or $\dim V = 1$) the claim is trivial.

For the inductive step, $\dim V \geq 1$. Working over $\mathbb{C}$, the characteristic polynomial of T^* splits, so T^* has at least one eigenvalue λ , with eigenvector $v \neq 0$. Set $W := \operatorname{span}\{v\}$. Then W is T^* -invariant, so $W^\perp$ is T -invariant by a class result (orthogonal complement of an S -invariant subspace is S^* -invariant; here $S = T^*$).

Apply the induction hypothesis to $T|_{W^\perp} : W^\perp \rightarrow W^\perp$ (well-defined since $W^\perp$ is T -invariant, and $\dim W^\perp = \dim V - 1$). This gives an orthonormal ordered basis $B' = (v_1, \dots, v_{n-1})$ of $W^\perp$ with $[T|_{W^\perp}]_{B'}$ upper triangular. Set $v_n := v/\|v\|$ and $B := (v_1, \dots, v_{n-1}, v_n)$.

Orthonormality. B' is orthonormal in $W^\perp$, $v_n \in W$ is a unit vector, and $W \perp W^\perp$, so B is orthonormal.

Upper triangular. For $i \leq n-1$, $Tv_i \in W^\perp = \text{span}\{v_1, \dots, v_{n-1}\}$, so the n th coordinate of $[Tv_i]_B$ is 0. Thus

$$[T]_B = \begin{pmatrix} [T|_{W^\perp}]_{B'} & * \\ 0 & \dots & 0 & * \end{pmatrix},$$

with the upper-left block upper triangular by IH. So $[T]_B$ is upper triangular. $\odot$

Question 22: HW6 Q2: Self-Adjointness Criterion

Let V be finite-dimensional complex inner product and $T : V \rightarrow V$. Show T is self-adjoint iff $\langle Tv, v \rangle \in \mathbb{R}$ for all $v \in V$.

Solution: $(\Rightarrow)$ If $T = T^*$, then for every v ,

$$\langle Tv, v \rangle = \langle v, T^*v \rangle = \langle v, Tv \rangle = \overline{\langle Tv, v \rangle}.$$

A complex number equal to its own conjugate is real, so $\langle Tv, v \rangle \in \mathbb{R}$.

$(\Leftarrow)$ Suppose $\langle Tv, v \rangle \in \mathbb{R}$ for all v . For every v ,

$$\langle v, T^*v \rangle = \langle Tv, v \rangle = \overline{\langle Tv, v \rangle} = \langle v, Tv \rangle,$$

where the second equality uses that $\langle Tv, v \rangle$ is real. Subtracting, $\langle v, (T - T^*)v \rangle = 0$ for all v , i.e., $\langle (T - T^*)v, v \rangle = \overline{\langle v, (T - T^*)v \rangle} = 0$ as well.

By the result from Tutorial 6 (Question 2(a)) on complex inner product spaces—if $S : V \rightarrow V$ satisfies $\langle Sv, v \rangle = 0$ for all v , then $S = 0$ —applied to $S := T - T^*$, we get $T - T^* = 0$, i.e., $T = T^*$. $\odot$

Question 23: HW6 Q3: Invariant Subspaces of Normal Operators

Let V be a finite-dimensional complex inner product space, $T : V \rightarrow V$ normal, and $W \subseteq V$ a T -invariant subspace.

- (a) Show W is T^* -invariant.
- (b) Deduce $W^\perp$ is T -invariant.

Solution: (a) Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of T . By the (complex) spectral theorem, T is diagonalizable, so

$$V = \bigoplus_{i=1}^r E(\lambda_i, T).$$

By 6, $W = \bigoplus_{i=1}^r (W \cap E(\lambda_i, T))$.

Let $v \in W$ and write $v = v_1 + \cdots + v_r$ with $v_i \in W \cap E(\lambda_i, T)$. Each v_i is an eigenvector of T with eigenvalue λ_i . Since T is normal, by a class result v_i is also an eigenvector of T^* with eigenvalue $\overline{\lambda_i}$. Hence $T^*v_i \in \text{span}\{v_i\} \subseteq W$. Summing,

$$T^*v = \sum_{i=1}^r T^*v_i \in W.$$

(b) By a class result (proved without normality), if W is S -invariant then $W^\perp$ is S^* -invariant. Apply this with $S := T^*$: by (a), W is T^* -invariant, so $W^\perp$ is $(T^*)^* = T$ -invariant. $\odot$

Question 24: HW6 Q4: Courant-Fischer Minimax

Let V be a finite-dimensional inner product space, $\dim(V) = n$, and $T : V \rightarrow V$ self-adjoint with orthonormal eigenbasis $v_1, \dots, v_n$ and eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$.

(a) Show that for $1 \leq k \leq n$ and every nonzero $x \in \text{span}\{v_1, \dots, v_k\}$,

$$\frac{\langle Tx, x \rangle}{\langle x, x \rangle} \geq \lambda_k.$$

(b) Show $\lambda_k = \max_{\substack{U \subseteq V \\ \dim U = k}} \left(\min_{\substack{x \in U \\ x \neq 0}} \frac{\langle Tx, x \rangle}{\langle x, x \rangle} \right)$.

Solution: (a) Let $x = \sum_{i=1}^k \alpha_i v_i \neq 0$. Since (v_i) is orthonormal,

$$\langle x, x \rangle = \sum_{i,j=1}^k \alpha_i \overline{\alpha_j} \langle v_i, v_j \rangle = \sum_{i=1}^k |\alpha_i|^2, \quad (1)$$

$$\langle Tx, x \rangle = \left\langle \sum_{i=1}^k \alpha_i \lambda_i v_i, \sum_{j=1}^k \alpha_j v_j \right\rangle = \sum_{i,j=1}^k \lambda_i \alpha_i \overline{\alpha_j} \langle v_i, v_j \rangle = \sum_{i=1}^k |\alpha_i|^2 \lambda_i. \quad (2)$$

Since $\lambda_i \geq \lambda_k$ for $i \leq k$, $\langle Tx, x \rangle \geq \lambda_k \sum_i |\alpha_i|^2 = \lambda_k \langle x, x \rangle$, dividing by $\langle x, x \rangle > 0$ gives the result.

(b) For brevity write $f(U) := \min_{0 \neq x \in U} \langle Tx, x \rangle / \langle x, x \rangle$.

Lower bound. Take $U_0 := \text{span}\{v_1, \dots, v_k\}$. By (a), $f(U_0) \geq \lambda_k$, and equality is attained at $x = v_k$ where $\langle Tv_k, v_k \rangle / \langle v_k, v_k \rangle = \lambda_k$. So $f(U_0) = \lambda_k$.

Upper bound. Let $U \subseteq V$ be any k -dimensional subspace, and put $W := \text{span}\{v_k, v_{k+1}, \dots, v_n\}$, with $\dim W = n - k + 1$. Then $\dim U + \dim W = k + (n - k + 1) = n + 1 > n$, so $U \cap W \neq \{0\}$. Pick a nonzero $y \in U \cap W$ and write $y = \sum_{i=k}^n \alpha_i v_i$. By the same computation as (1)-(2),

$$\langle Ty, y \rangle = \sum_{i=k}^n |\alpha_i|^2 \lambda_i \leq \lambda_k \sum_{i=k}^n |\alpha_i|^2 = \lambda_k \langle y, y \rangle,$$

using $\lambda_i \leq \lambda_k$ for $i \geq k$. So $\langle Ty, y \rangle / \langle y, y \rangle \leq \lambda_k$, hence $f(U) \leq \lambda_k$.

Combining the two bounds: $f(U) \leq \lambda_k$ for every k -dimensional U , with equality at $U = U_0$. Therefore $\lambda_k = \max_{\dim U = k} f(U)$. $\odot$

10.7 Homework 7

Question 25: HW7 Q1: PSD Matrices as Gram Matrices

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $A \in \mathbb{F}^{n \times n}$ PSD, and $r := \text{rank}(A)$.

- (a) Show there is $B \in \mathbb{F}^{n \times n}$ with $BB^* = A$ and the last $n - r$ columns of B zero.
- (b) Deduce there are $v_1, \dots, v_n \in \mathbb{F}^r$ with $A_{ij} = \langle v_i, v_j \rangle$ (Gram matrix).

Solution: (a) Since A is PSD (hence self-adjoint), it has an orthonormal eigenbasis $v_1, \dots, v_n$ with eigenvalues $\lambda_1, \dots, \lambda_n \geq 0$. Since $\text{rank } A = r$, exactly r eigenvalues are nonzero; WLOG $\lambda_1, \dots, \lambda_r > 0$ and $\lambda_{r+1} = \dots = \lambda_n = 0$.

Let U be the matrix with columns $v_1, \dots, v_n$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Since the columns of U are orthonormal, U is unitary, so $U^{-1} = U^*$ and

$$A = UDU^*.$$

Define $\sqrt{D} := \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$ (nonnegative real entries; $(\sqrt{D})^* = \sqrt{D}$). Set $B := U\sqrt{D}$. Then

$$BB^* = (U\sqrt{D})(U\sqrt{D})^* = U\sqrt{D} \cdot \sqrt{D}U^* = UDU^* = A.$$

Last columns vanish. For any $Y \in \mathbb{F}^{n \times n}$ and diagonal $X = \text{diag}(x_1, \dots, x_n)$, we have $(YX)_{ij} = \sum_k Y_{ik}X_{kj} = Y_{ij}x_j$, since $X_{kj} \neq 0$ only when $k = j$. Thus column j of YX is x_j times column j of Y . Apply this with $Y = U$, $X = \sqrt{D}$: column j of $B = U\sqrt{D}$ is $\sqrt{\lambda_j} \cdot (\text{column } j \text{ of } U)$. For $j > r$, $\sqrt{\lambda_j} = 0$, so columns $r + 1, \dots, n$ of B are zero.

(b) Let $r_i \in \mathbb{F}^n$ be the i th row of B (as a vector). Since the last $n - r$ columns of B vanish, r_i has nonzero entries only in its first r coordinates. Set $v_i := (B_{i1}, \dots, B_{ir}) \in \mathbb{F}^r$, so $\langle r_i, r_j \rangle_{\mathbb{F}^n} = \langle v_i, v_j \rangle_{\mathbb{F}^r}$. Now

$$(BB^*)_{ij} = \sum_{k=1}^n B_{ik}\overline{B_{jk}} = \sum_{k=1}^n r_i(k)\overline{r_j(k)} = \langle r_i, r_j \rangle = \langle v_i, v_j \rangle.$$

Since $A = BB^*$, $A_{ij} = \langle v_i, v_j \rangle$. ⊕

Question 26: HW7 Q2: Negative Eigenvalues under PSD Perturbation

Let V be finite-dimensional inner product, $S, T : V \rightarrow V$, S self-adjoint, $S + T$ PSD, and $r := \text{rank}(T)$. Show S has at most r negative eigenvalues (counted with multiplicity).

Solution: Since S is self-adjoint, it admits an orthonormal eigenbasis $v_1, \dots, v_n$ with real eigenvalues $\lambda_1, \dots, \lambda_n$.

Suppose for contradiction S has at least $r + 1$ negative eigenvalues; WLOG $\lambda_1, \dots, \lambda_{r+1} < 0$. Set $W := \text{span}\{v_1, \dots, v_{r+1}\}$, so $\dim W = r + 1$. By rank-nullity, $\dim \ker(T) = n - \text{rank}(T) = n - r$. Hence

$$\dim W + \dim \ker(T) = (r + 1) + (n - r) = n + 1 > n,$$

so there is a nonzero $w \in W \cap \ker(T)$.

Since $S + T$ is PSD and $Tw = 0$,

$$0 \leq \langle (S + T)w, w \rangle = \langle Sw, w \rangle + \langle Tw, w \rangle = \langle Sw, w \rangle. \quad (*)$$

On the other hand, write $w = \sum_{i=1}^{r+1} \alpha_i v_i$ (some $\alpha_i \neq 0$, since $w \neq 0$). Then $Sw = \sum_i \alpha_i \lambda_i v_i$ and

$$\langle Sw, w \rangle = \sum_{i,j=1}^{r+1} \lambda_i \alpha_i \overline{\alpha_j} \langle v_i, v_j \rangle = \sum_{i=1}^{r+1} \lambda_i |\alpha_i|^2 < 0,$$

using $\lambda_i < 0$ for $i \leq r + 1$ and $\sum_i |\alpha_i|^2 > 0$. This contradicts (*). ☺

Question 27: HW7 Q3: SVD Computation

Find a singular value decomposition for

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Solution: Note $\text{rank } A = 2$.

Step 1: orthogonally diagonalize A^*A .

$$A^*A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

The characteristic polynomial (expanding along the first row) is

$$p_{A^*A}(x) = (x - 1)((x - 1)(x - 2) - 1) - (x - 1) = (x - 1)(x^2 - 3x) = x(x - 1)(x - 3).$$

So eigenvalues are $\lambda_1 = 3, \lambda_2 = 1, \lambda_3 = 0$.

$\lambda = 3$: $(A^*A - 3I)x = 0$ has solution space $\text{span}\{(1, 1, 2)^T\}$. Normalizing,

$$v_1 = \frac{1}{\sqrt{6}}(1, 1, 2)^T.$$

$\lambda = 1$: $(A^*A - I)x = 0$ gives $x_3 = 0$ and $x_1 + x_2 = 0$, so the solution space is $\text{span}\{(1, -1, 0)^T\}$. Normalizing,

$$v_2 = \frac{1}{\sqrt{2}}(1, -1, 0)^T.$$

$\lambda = 0$: $\ker(A^*A)$ has solution $\text{span}\{(1, 1, -1)^T\}$. Normalizing,

$$v_3 = \frac{1}{\sqrt{3}}(1, 1, -1)^T.$$

So

$$U_1 = \begin{pmatrix} 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \\ 2/\sqrt{6} & 0 & -1/\sqrt{3} \end{pmatrix}.$$

Step 2: singular values and right U_2 . Set $\sigma_i := \sqrt{\lambda_i}$ and $w_i := \sigma_i^{-1}Av_i$ for $i = 1, 2$. Then $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$, and

$$w_1 = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \frac{1}{3\sqrt{2}} \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

$$w_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Thus

$$U_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

and $A = U_2 \Sigma U_1^*$. ☺

Question 28: HW7 Q4: Largest Singular Value Bound

Let V, W be finite-dimensional inner product spaces and $T : V \rightarrow W$ have singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. Show $\|Tv\| \leq \sigma_1\|v\|$ for all $v \in V$.

Solution: By SVD, there are orthonormal bases $\{v_1, \dots, v_n\}$ of V and $\{w_1, \dots, w_m\}$ of W with

$$Tv_i = \begin{cases} \sigma_i w_i & i \leq r, \\ 0 & i > r. \end{cases}$$

Write $v = \sum_{i=1}^n \alpha_i v_i$. Then $Tv = \sum_{i=1}^r \alpha_i \sigma_i w_i$, and using orthonormality of (v_i) and (w_i) ,

$$\|v\|^2 = \sum_{i=1}^n |\alpha_i|^2, \quad \|Tv\|^2 = \sum_{i,j=1}^r \alpha_i \bar{\alpha}_j \sigma_i \sigma_j \langle w_i, w_j \rangle = \sum_{i=1}^r \sigma_i^2 |\alpha_i|^2.$$

Since $\sigma_i \leq \sigma_1$ for all $i \leq r$,

$$\|Tv\|^2 = \sum_{i=1}^r \sigma_i^2 |\alpha_i|^2 \leq \sigma_1^2 \sum_{i=1}^n |\alpha_i|^2 = \sigma_1^2 \|v\|^2.$$

Taking square roots, $\|Tv\| \leq \sigma_1 \|v\|$. ☺

Chapter 11

Additional Exercises

Note:-

These exercises are taken from the past-paper folder and grouped by topic rather than by exam. Each question records its source, and solutions are included so the chapter can be used as a self-contained revision bank.

11.1 Diagonalization, Minimal Polynomials, and Jordan Form

Question 1: Additional Exercise: Midterm Diagonalization

Source: 2026 Midterm 1, Problem 1.

Prove that

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

is diagonalizable by finding an invertible matrix C such that $C^{-1}AC$ is diagonal.

Solution: The monic characteristic polynomial is

$$\det(xI - A) = \begin{vmatrix} x-1 & -2 \\ -2 & x-1 \end{vmatrix} = (x-1)^2 - 4 = (x-3)(x+1).$$

For $\lambda = 3$ an eigenvector is $(1, 1)^T$, and for $\lambda = -1$ an eigenvector is $(1, -1)^T$. Thus

$$C = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{gives} \quad C^{-1}AC = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}.$$



Question 2: Additional Exercise: Diagonalization Concept Checks

Source: 2026 Midterm 1, Problem 2.

Decide whether each statement is true or false.

1. If $X, Y \in F^{n \times n}$ are similar and X is diagonalizable, then Y is diagonalizable.
2. The matrix $\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ is diagonalizable.
3. If $f(T) = 0$ and λ is an eigenvalue of T , then $f(\lambda) = 0$.
4. If $V = U \oplus W$ for T -invariant subspaces U, W , and $T|_U, T|_W$ are diagonalizable, then T is diagonalizable.

Solution: The answers are true, false, true, true.

For (1), similarity to a diagonal matrix is preserved by similarity. For (2), the characteristic polynomial is x^2 ; if the matrix were diagonalizable it would be similar to the zero matrix, hence would itself be zero. For (3), if v is a λ -eigenvector, then $0 = f(T)v = f(\lambda)v$, so $f(\lambda) = 0$. For (4), combine eigenbases of $T|_U$ and $T|_W$; the direct sum makes their union a basis of V . ☺

Question 3: Additional Exercise: Cayley-Hamilton Reduction

Source: 2026 Midterm 1, Problem 3.

1. State the Cayley-Hamilton theorem.
2. Let A be an $n \times n$ matrix and let f be a polynomial. Show that there is a polynomial g with $\deg g < n$ such that $g(A) = f(A)$.

Solution: Cayley-Hamilton says that a matrix satisfies its characteristic polynomial: $p_A(A) = 0$.

Divide f by the monic characteristic polynomial p_A :

$$f = q p_A + r, \quad \deg r < \deg p_A = n.$$

Evaluating at A gives

$$f(A) = q(A)p_A(A) + r(A) = r(A),$$

so take $g = r$. ☺

Question 4: Additional Exercise: A Five-by-Five Jordan Computation

Source: 2023 Final Exam, Problem 1.

For

$$A = \begin{pmatrix} 4 & 0 & 0 & 0 & 0 \\ 3 & 4 & 1 & 1 & 0 \\ -1 & 0 & 1 & 3 & -2 \\ 2 & 0 & 3 & 1 & 0 \\ -1 & 0 & 0 & 0 & -2 \end{pmatrix},$$

find the characteristic polynomial, eigenvalues and eigenspace dimensions, minimal polynomial, and Jordan normal form.

Solution: A direct determinant computation gives

$$\det(xI - A) = (x - 4)^3(x + 2)^2.$$

Thus the eigenvalues are 4 and -2 . Row reduction gives

$$\dim E_4 = 1, \quad \dim E_{-2} = 1.$$

The nullities of powers are

$$\dim \ker(A - 4I) = 1, \quad \dim \ker(A - 4I)^2 = 2, \quad \dim \ker(A - 4I)^3 = 3,$$

and

$$\dim \ker(A + 2I) = 1, \quad \dim \ker(A + 2I)^2 = 2.$$

So the 4-part has one Jordan block of size 3, and the -2 -part has one Jordan block of size 2. Therefore

$$q_A(x) = (x - 4)^3(x + 2)^2,$$

and, up to block order,

$$J = J(4, 3) \oplus J(-2, 2).$$

☺

Question 5: Additional Exercise: Counting Jordan Forms

Source: 2023 Final Exam, Problem 5.

1. State the formula for the minimal polynomial of a complex matrix in terms of Jordan block sizes.
2. Suppose $A \in M_{12 \times 12}(\mathbb{C})$ has

$$q_A(z) = z^3(z - 2)^2(z - 3), \quad p_A(z) = z^4(z - 2)^4(z - 3)^4.$$

How many possible Jordan normal forms are there, up to rearrangement of blocks?

Solution: If the Jordan blocks for eigenvalue λ have sizes $m_{\lambda,1}, \dots, m_{\lambda,r_\lambda}$, then

$$q_A(z) = \prod_{\lambda} (z - \lambda)^{\max_j m_{\lambda,j}}.$$

For $\lambda = 0$, the algebraic multiplicity is 4 and the largest block size is 3, so the only partition is $4 = 3 + 1$. For $\lambda = 2$, the algebraic multiplicity is 4 and the largest block size is 2, so the possible partitions are

$$4 = 2 + 2, \quad 4 = 2 + 1 + 1.$$

For $\lambda = 3$, all blocks have size 1, so the only partition is $1 + 1 + 1 + 1$. Hence there are

$$1 \cdot 2 \cdot 1 = 2$$

possible Jordan normal forms.

☺

Question 6: Additional Exercise: Jordan Form and Change-of-Basis Matrix

Source: 2024 Final Exam, Problem 1.

For

$$A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 2 & -3 & 0 \\ 0 & 0 & 2 & 0 \\ -2 & 0 & 6 & 2 \end{pmatrix},$$

find a Jordan canonical form J and an invertible matrix C such that $C^{-1}AC = J$.

Solution: Let $N = A - 2I$. Then $\det(xI - A) = (x - 2)^4$, $\dim \ker N = 3$, and $N^2 = 0$. Hence there is one Jordan block of size 2 and two Jordan blocks of size 1:

$$J = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

One compatible Jordan basis is

$$\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Therefore one valid choice is

$$C = \begin{pmatrix} 0 & 1 & 0 & 3 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 0 & 0 & 0 \end{pmatrix}, \quad C^{-1}AC = J.$$



Question 7: Additional Exercise: Possible Jordan Forms from p_T and q_T

Source: 2024 Final Exam, Problem 6.

Suppose $\dim_{\mathbb{C}} V = 10$ and

$$p_T(z) = (z - 3)^6(z + 2)^4, \quad q_T(z) = (z - 3)^4(z + 2).$$

Describe the possible Jordan canonical forms of T .

Solution: For $\lambda = 3$, the algebraic multiplicity is 6 and the largest Jordan block has size 4. Hence the possible partitions are

$$6 = 4 + 2, \quad 6 = 4 + 1 + 1.$$

For $\lambda = -2$, the largest block has size 1 and the algebraic multiplicity is 4, so the only possibility is four 1×1 blocks.

Thus the possible Jordan forms are

$$J(3, 4) \oplus J(3, 2) \oplus (-2)I_4$$

or

$$J(3, 4) \oplus J(3, 1) \oplus J(3, 1) \oplus (-2)I_4.$$



Question 8: Additional Exercise: Practice Midterm Jordan Computation

Source: 2026 Midterm 2 Practice Exam, Problem 1.

For

$$A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & -1 & -1 & 1 \end{pmatrix},$$

find the characteristic and minimal polynomials, the Jordan canonical form, and a Jordan basis.

Solution: The monic characteristic polynomial is

$$\det(xI - A) = (x - 2)^4.$$

Let $N = A - 2I$. Then $\dim \ker N = 3$ and $N^2 = 0$, so the Jordan block sizes are 2, 1, 1. Hence

$$q_A(x) = (x - 2)^2, \quad J = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

One Jordan basis is

$$\left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}; \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}; \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} \right\},$$

where semicolons separate Jordan chains.



Question 9: Additional Exercise: Minimal Polynomial Concept Checks

Source: 2026 Midterm 2 Practice Exam, Problem 3.

Decide whether each statement is true or false, proving true statements and giving counterexamples for false ones.

1. If U is T -invariant, then the minimal polynomial of $T|_U$ divides q_T .
2. If f and q_T are relatively prime, then $f(T)$ is invertible.
3. If $\dim V = 14$, then there exists nilpotent N with $\dim \ker N = 4$ and $N^3 = 0$.
4. If $\langle x, v \rangle = \langle y, v \rangle$ for all v , then $x = y$.

Solution: (1) True. Since $q_T(T) = 0$, restricting to U gives $q_T(T|_U) = 0$, so $q_{T|_U} \mid q_T$.

(2) True. Bezout gives $af + bq_T = 1$. Evaluating at T gives $a(T)f(T) = I$, so $f(T)$ is invertible.

(3) False. A nilpotent operator with $N^3 = 0$ has Jordan blocks of size at most 3, and $\dim \ker N$ equals the number of Jordan blocks. Four blocks of size at most 3 span dimension at most 12, not 14.

(4) True. Put $v = x - y$. Then $\langle x - y, x - y \rangle = 0$, so $x - y = 0$. ☺

Question 10: Additional Exercise: Nilpotence and Eigenvalues

Source: 2026 Midterm 2 Practice Exam, Problem 5.

Let V be finite-dimensional and suppose p_T splits. Show that T is nilpotent if and only if every eigenvalue of T is 0.

Solution: If T is nilpotent, then $T^m = 0$ for some m . If $Tv = \lambda v$ with $v \neq 0$, then $0 = T^m v = \lambda^m v$, so $\lambda = 0$.

Conversely, if every eigenvalue is 0 and p_T splits, then the only root of p_T is 0, so $p_T(x) = x^n$. Cayley-Hamilton gives $T^n = 0$, hence T is nilpotent. ☺

Question 11: Additional Exercise: Kernels of Powers of a Diagonalizable Map

Source: 2026 Midterm 2 Practice Exam, Problem 6.

Let T be diagonalizable. Show that $\ker(T^k) = \ker(T)$ for all $k \geq 1$.

Solution: Choose an eigenbasis $v_1, \dots, v_n$ with $Tv_i = \lambda_i v_i$. For $v = \sum_i a_i v_i$,

$$T^k v = \sum_i a_i \lambda_i^k v_i.$$

Thus $T^k v = 0$ exactly when $a_i = 0$ for every $\lambda_i \neq 0$, which is the same condition as $Tv = 0$. Hence $\ker(T^k) = \ker(T)$. ☺

Question 12: Additional Exercise: Official Midterm Jordan Computation

Source: 2026 Midterm 2, Problem 1.

For

$$A = \begin{pmatrix} -1 & 3 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

find the characteristic and minimal polynomials, Jordan canonical form, and a Jordan basis.

Solution:

$$\det(xI - A) = (x + 1)(x - 1)^3.$$

For $\lambda = -1$ there is one block of size 1, with eigenvector e_1 . For $\lambda = 1$,

$$\dim \ker(A - I) = 2,$$

so the three-dimensional generalized eigenspace has two Jordan blocks, of sizes 2 and 1. A basis for $\ker(A - I)$ is

$$\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

and

$$(A - I) \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

Thus

$$J = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad q_A(x) = (x + 1)(x - 1)^2,$$

with Jordan basis

$$\left\{ e_1; \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}; e_4 \right\}.$$

☺

Question 13: Additional Exercise: Roots, Minimal Polynomials, and Orthogonal Complements

Source: 2026 Midterm 2, Problem 3.

Decide whether each statement is true or false.

1. λ is a root of p_T if and only if λ is a root of q_T .
2. If N is nilpotent of degree t , then $\text{Im}(N^{t-1}) = \ker N$.
3. If $q_A = q_B$, then A and B are similar.
4. For all subspaces U, W of an inner product space, $(U + W)^\perp = U^\perp \cap W^\perp$.

Solution: (1) True: both conditions are equivalent to λ being an eigenvalue of T .

(2) False. For $N = J(0, 2) \oplus (0)$, the nilpotence degree is 2, but $\text{Im } N$ is one-dimensional while $\ker N$ is two-dimensional.

(3) False. Nilpotent Jordan forms with block sizes $(2, 2)$ and $(2, 1, 1)$ both have minimal polynomial x^2 , but they are not similar.

(4) True. A vector is orthogonal to $U + W$ iff it is orthogonal to every vector of U and to every vector of W . ☺

Question 14: Additional Exercise: A Nilpotent Chain is Independent

Source: 2026 Midterm 2, Problem 5.

Let $T : V \rightarrow V$ be linear and suppose $T^k v = 0$ but $T^{k-1} v \neq 0$. Show that

$$\{v, Tv, T^2 v, \dots, T^{k-1} v\}$$

is linearly independent.

Solution: Suppose

$$a_0 v + a_1 Tv + \dots + a_{k-1} T^{k-1} v = 0.$$

Let j be the smallest index with $a_j \neq 0$. Applying T^{k-1-j} gives

$$a_j T^{k-1} v = 0,$$

because all later terms contain $T^k v$ or a higher power. Since $T^{k-1} v \neq 0$, this forces $a_j = 0$, a contradiction. Hence all coefficients are zero. ☹

Question 15: Additional Exercise: Determinant of a Nilpotent Perturbation

Source: 2026 Midterm 2, Problem 6.

Let N be nilpotent on a finite-dimensional vector space. Show that

$$\det(I + N) = 1.$$

Solution: All eigenvalues of N are 0, so the only eigenvalue of $I + N$ is 1. Equivalently, in Jordan form N is strictly upper triangular block-by-block, so $I + N$ is upper triangular with all diagonal entries equal to 1. Thus $\det(I + N) = 1$. ☹

Question 16: Additional Exercise: Practice Final Jordan Computation

Source: 2026 Practice Final Exam, Problem 1.

For

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

find the characteristic and minimal polynomials, Jordan canonical form, and a Jordan basis.

Solution:

$$\det(xI - A) = (x - 1)^4(x - 3).$$

For $\lambda = 3$, e_4 is an eigenvector. For $\lambda = 1$,

$$\ker(A - I) = \text{span}\{e_1, e_3\}, \quad \dim \ker(A - I)^2 = 4.$$

Thus the $\lambda = 1$ block sizes are 2, 2. Since

$$(A - I)(-e_2 + e_5) = e_1, \quad (A - I)e_5 = e_3,$$

a Jordan basis is

$$\{e_1, -e_2 + e_5; e_3, e_5; e_4\}.$$

Hence

$$J = J(1, 2) \oplus J(1, 2) \oplus (3), \quad q_A(x) = (x - 1)^2(x - 3).$$

☺

Question 17: Additional Exercise: Cyclic Vectors and Direct Sums

Source: 2026 Practice Final Exam, Problem 6.

Let $V = V_1 \oplus \cdots \oplus V_r$ for T -invariant subspaces V_i . Suppose T admits a cyclic vector v , and write $v = v_1 + \cdots + v_r$ with $v_i \in V_i$. Show that each v_i is cyclic for $T|_{V_i}$.

Solution: Since v is cyclic, every $w \in V$ has the form $w = f(T)v$ for some polynomial f . Fix i and take $w_i \in V_i$. Viewed as an element of V , we can write

$$w_i = f(T)v = f(T)v_1 + \cdots + f(T)v_r.$$

Because each V_i is T -invariant, $f(T)v_i \in V_i$. By uniqueness of the direct-sum decomposition, the V_i -component of w_i is $f(T)v_i$. Hence $w_i = f(T|_{V_i})v_i$, so v_i is cyclic for $T|_{V_i}$. ☺

Question 18: Additional Exercise: Left Multiplication and Nilpotence

Source: 2022 Final Exam, Problem 3.

Let V be finite-dimensional and $S \in \mathcal{L}(V)$. Define $F_S \in \mathcal{L}(\mathcal{L}(V))$ by

$$F_S(M) = SM.$$

Prove that F_S is nilpotent if and only if S is nilpotent.

Solution: For every $k \geq 1$,

$$F_S^k(M) = S^k M.$$

If $S^k = 0$, then $F_S^k = 0$, so F_S is nilpotent. Conversely, if $F_S^k = 0$, evaluate at the identity map:

$$0 = F_S^k(I) = S^k.$$

Thus S is nilpotent. ☺

Question 19: Additional Exercise: Two Commuting Nilpotents

Source: 2022 Final Exam, Problem 6, Bonus.

Suppose $N_1 N_2 = N_2 N_1$ and $N_1^3 = N_2^3 = 0$. Find, with proof, the smallest possible m such that

$$(N_1 + N_2)^m = 0$$

must hold.

Solution: Since N_1 and N_2 commute, the binomial theorem applies:

$$(N_1 + N_2)^m = \sum_{a=0}^m \binom{m}{a} N_1^a N_2^{m-a}.$$

For $m = 5$, each term has either $a \geq 3$ or $m - a \geq 3$, so every term is zero. Thus $(N_1 + N_2)^5 = 0$ always.

The exponent 5 is sharp. On $\mathbb{F}[x, y]/(x^3, y^3)$, let N_1 be multiplication by x and N_2 multiplication by y . Then $N_1^3 = N_2^3 = 0$ and the maps commute, but

$$(N_1 + N_2)^4(1) = 6x^2y^2 \neq 0$$

over $\mathbb{R}$ or $\mathbb{C}$. Hence $m = 5$ is the smallest guaranteed exponent. ☺

11.2 Inner Product Geometry and Orthogonal Projection

Question 20: Additional Exercise: Projection in $\mathbb{R}^3$

Source: 2022 Final Exam, Problem 2.

In $\mathbb{R}^3$, let

$$U = \text{span}\{(1, 0, 1), (2, 4, 0)\}, \quad v = (-1, 0, 1).$$

Find the vector $u \in U$ such that $\|v - u\|$ is as small as possible.

Solution: Let

$$B = \begin{pmatrix} 1 & 2 \\ 0 & 4 \\ 1 & 0 \end{pmatrix}.$$

The closest vector is the orthogonal projection

$$u = B(B^T B)^{-1} B^T v.$$

Computing,

$$B^T B = \begin{pmatrix} 2 & 2 \\ 2 & 20 \end{pmatrix}, \quad B^T v = \begin{pmatrix} 0 \\ -2 \end{pmatrix},$$

so

$$u = \begin{pmatrix} -1/9 \\ -4/9 \\ 1/9 \end{pmatrix}.$$

☺

Question 21: Additional Exercise: Gram-Schmidt and Closest Point in $\mathbb{R}^4$

Source: 2023 Final Exam, Problem 2.

Let $W \subseteq \mathbb{R}^4$ be spanned by

$$v_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix},$$

and let $x = (1, 0, 0, 0)^T$. Find an orthonormal basis for W , the closest vector $y \in W$ to x , and the distance from x to W .

Solution: Gram-Schmidt gives the orthonormal basis

$$e_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \quad e_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad e_3 = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \\ -3 \end{pmatrix}.$$

Thus

$$y = P_W x = \sum_{i=1}^3 \langle x, e_i \rangle e_i = \begin{pmatrix} 3/4 \\ -1/4 \\ -1/4 \\ -1/4 \end{pmatrix}.$$

Therefore

$$x - y = \begin{pmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{pmatrix}, \quad \text{dist}(x, W) = \|x - y\| = \frac{1}{2}.$$

**Question 22: Additional Exercise: Finite-Dimensional Subspaces Have Orthogonal Complements**

Source: 2023 Final Exam, Problem 6.

Let V be a real or complex inner product space, possibly infinite-dimensional. If $W \subseteq V$ is finite-dimensional, show that

$$V = W \oplus W^\perp.$$

Solution: Choose an orthonormal basis $e_1, \dots, e_m$ of W . For $v \in V$, define

$$w = \sum_{j=1}^m \langle v, e_j \rangle e_j.$$

Then $w \in W$, and for each i ,

$$\langle v - w, e_i \rangle = \langle v, e_i \rangle - \sum_{j=1}^m \langle v, e_j \rangle \langle e_j, e_i \rangle = 0.$$

Hence $v - w \in W^\perp$, so $V = W + W^\perp$. The intersection is $\{0\}$ because if $u \in W \cap W^\perp$, then $\langle u, u \rangle = 0$, so $u = 0$. Thus $V = W \oplus W^\perp$. ☺

Question 23: Additional Exercise: Projection Matrices in $\mathbb{R}^3$

Source: 2024 Final Exam, Problem 3.

Let $V = \mathbb{R}^3$ with the standard inner product,

$$v = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad w = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}.$$

Let $W = \text{span}\{v\}$. Find the projection matrix P onto W , the multiple of v closest to w , and the projection matrix Q onto $W^\perp$.

Solution: Since $v^T v = 6$,

$$P = \frac{vv^T}{v^T v} = \frac{1}{6} \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{pmatrix}.$$

The closest multiple of v to w is Pw , namely

$$Pw = \frac{v^T w}{v^T v} v = \frac{1}{2} v = \begin{pmatrix} 1/2 \\ 1 \\ -1/2 \end{pmatrix}.$$

The projection onto $W^\perp$ is $Q = I - P$:

$$Q = \begin{pmatrix} 5/6 & -1/3 & 1/6 \\ -1/3 & 1/3 & 1/3 \\ 1/6 & 1/3 & 5/6 \end{pmatrix}.$$

☺

Question 24: Additional Exercise: Polynomial Inner Product Projection

Source: 2026 Midterm 2 Practice Exam, Problem 2.

Let $V = \mathbb{R}_2[x]$ with

$$\langle f, g \rangle = f(-1)g(-1) + f(1)g(1) + f(2)g(2),$$

and let $W = \text{span}\{1, x^2\}$. Find an orthonormal basis for W and the orthogonal projection of $f(x) = x$ onto W .

Solution: Start with 1 and x^2 . Since $\|1\|^2 = 3$,

$$e_1 = \frac{1}{\sqrt{3}}.$$

Also

$$x^2 - \frac{\langle x^2, 1 \rangle}{\langle 1, 1 \rangle} \cdot 1 = x^2 - 2.$$

Its norm squared is

$$(1-2)^2 + (1-2)^2 + (4-2)^2 = 6,$$

so

$$e_2 = \frac{x^2 - 2}{\sqrt{6}}.$$

Thus

$$P_W(x) = \langle x, e_1 \rangle e_1 + \langle x, e_2 \rangle e_2 = \frac{2}{3} + \frac{2}{3}(x^2 - 2) = \frac{2}{3}x^2 - \frac{2}{3}.$$

☺

Question 25: Additional Exercise: Integral Cauchy-Schwarz

Source: 2026 Midterm 2 Practice Exam, Problem 4.

Prove that for every continuous $f : [-1, 1] \rightarrow \mathbb{R}$,

$$\left(\int_{-1}^1 x^2 f(x) dx \right)^2 \leq \frac{2}{5} \int_{-1}^1 f(x)^2 dx.$$

Solution: Use the L^2 inner product $\langle g, h \rangle = \int_{-1}^1 g(x)h(x) dx$. Then

$$\left(\int_{-1}^1 x^2 f(x) dx \right)^2 = \langle x^2, f \rangle^2 \leq \|x^2\|^2 \|f\|^2$$

by Cauchy-Schwarz. Finally,

$$\|x^2\|^2 = \int_{-1}^1 x^4 dx = \frac{2}{5},$$

which gives the result.

☺

Question 26: Additional Exercise: Projection in $\mathbb{R}^4$

Source: 2026 Midterm 2, Problem 2.

Let $W \subseteq \mathbb{R}^4$ be spanned by

$$u_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix}.$$

Find an orthonormal basis for W , and for $v = (1, -1, 0, 0)^T$, find the closest vector in W to v .

Solution: Gram-Schmidt gives

$$w_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad w_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}.$$

The closest vector is the projection

$$P_W v = \langle v, w_1 \rangle w_1 + \langle v, w_2 \rangle w_2 = -\frac{1}{\sqrt{2}} w_1 + \frac{1}{\sqrt{2}} w_2 = \begin{pmatrix} 1/2 \\ -1/2 \\ -1/2 \\ -1/2 \end{pmatrix}.$$

☺

Question 27: Additional Exercise: A Unit-Vector Inequality

Source: 2026 Midterm 2, Problem 4.

Let V be an inner product space, $T : V \rightarrow V$ linear, and $\|v\| = 1$. Show that

$$|\langle Tv, v \rangle| \leq \|Tv\|,$$

with equality if and only if v is an eigenvector of T .

Solution: By Cauchy-Schwarz,

$$|\langle Tv, v \rangle| \leq \|Tv\| \|v\| = \|Tv\|.$$

Equality in Cauchy-Schwarz occurs exactly when Tv and v are linearly dependent. Since $v \neq 0$, this means $Tv = \lambda v$ for some scalar λ , which is exactly that v is an eigenvector of T . ☺

11.3 Adjoints, Normality, Self-Adjointness, and Unitary Operators

Question 28: Additional Exercise: Final Exam True/False

Source: 2022 Final Exam, Problem 1.

Decide whether each statement is true or false.

1. If N_1, N_2 are nilpotent, then $N_1 + N_2$ cannot be invertible.
2. Every monic degree n polynomial in $\mathbb{C}[x]$ is the minimal polynomial of some operator on $\mathbb{C}^n$.
3. If S^4 and S^6 are isometries, then S is an isometry.
4. If $S^2 = S^3$, then S is diagonalizable.
5. Every operator on V is the sum of at most three isometries.

Solution: The answers are false, true, false, false, false.

For (1), take

$$N_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix};$$

then $N_1 + N_2$ is invertible. For (2), use the companion matrix of the given monic polynomial. For (3), the matrix

$$S = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$$

satisfies $S^2 = I$, so S^4 and S^6 are isometries, but S is not an isometry. For (4), a nonzero nilpotent Jordan block with square zero satisfies $S^2 = S^3 = 0$ but is not diagonalizable. For (5), $4I$ has operator norm 4, while a sum of three isometries has norm at most 3. ☺

Question 29: Additional Exercise: A Positive Operator from Commuting Normal Operators

Source: 2022 Final Exam, Problem 4.

Let T, S be normal operators on a finite-dimensional inner product space and suppose $ST = TS$. Prove that

$$SS^* + ST^* + TS^* + TT^*$$

is positive.

Solution: The expression factors as

$$SS^* + ST^* + TS^* + TT^* = (S + T)(S + T)^*.$$

For any v ,

$$\langle (S + T)(S + T)^*v, v \rangle = \langle (S + T)^*v, (S + T)^*v \rangle = \|(S + T)^*v\|^2 \geq 0.$$

Thus the operator is positive. ☺

Question 30: Additional Exercise: Complexification and Adjoints

Source: 2022 Final Exam, Problem 5.

Let V be a finite-dimensional real inner product space and $T \in \mathcal{L}(V)$. Prove that the characteristic polynomial of $T_{\mathbb{C}}$ is the same as the characteristic polynomial of $T_{\mathbb{C}}^*$.

Solution: Choose an orthonormal basis of the real space V , and let A be the real matrix of T in that basis. Then $T_{\mathbb{C}}$ has the same matrix A , now viewed over $\mathbb{C}$, and $T_{\mathbb{C}}^*$ has matrix A^T . Therefore

$$\det(xI - A^T) = \det((xI - A)^T) = \det(xI - A).$$

So $T_{\mathbb{C}}$ and $T_{\mathbb{C}}^*$ have the same characteristic polynomial. ☺

Question 31: Additional Exercise: Cube Roots of Real Self-Adjoint Matrices

Source: 2023 Final Exam, Problem 4.

Let $A \in M_{n \times n}(\mathbb{R})$ be real self-adjoint, meaning $A = A^T$. Show that there exists a matrix B such that

$$B^3 = A.$$

Solution: By the real spectral theorem, there is an orthogonal matrix Q such that

$$A = QDQ^T, \quad D = \text{diag}(\lambda_1, \dots, \lambda_n)$$

with $\lambda_i \in \mathbb{R}$. Let

$$E = \text{diag}(\sqrt[3]{\lambda_1}, \dots, \sqrt[3]{\lambda_n}),$$

using the real cube root. Then $E^3 = D$, so

$$B := QEQ^T$$

satisfies $B^3 = QE^3Q^T = QDQ^T = A$. ☺

Question 32: Additional Exercise: Products of Orthogonal Projections

Source: 2023 Final Exam, Problem 8, Bonus.

Let V be a finite-dimensional complex inner product space and let P, Q be orthogonal projections.

1. Show that every eigenvalue of PQ is real and lies in $[0, 1]$.
2. For each $\lambda \in [0, 1]$, give an example of orthogonal projections P, Q for which λ is an eigenvalue of PQ .

Solution: Let $PQx = \lambda x$ with $x \neq 0$. If $\lambda = 0$, there is nothing to prove. If $\lambda \neq 0$, then $x = \lambda^{-1}PQx \in \text{Im } P$, so $Px = x$. Hence

$$\lambda\|x\|^2 = \langle PQx, x \rangle = \langle Qx, Px \rangle = \langle Qx, x \rangle = \|Qx\|^2.$$

Thus $\lambda = \|Qx\|^2/\|x\|^2 \in [0, 1]$.

Conversely, choose θ with $\cos^2 \theta = \lambda$. In $\mathbb{C}^2$, let P be projection onto $\text{span}\{e_1\}$ and let Q be projection onto $\text{span}\{\cos \theta e_1 + \sin \theta e_2\}$. Then

$$PQe_1 = \lambda e_1,$$

so λ is an eigenvalue of PQ . ☺

Question 33: Additional Exercise: Self-Adjoint and Unitary Reflections

Source: 2024 Final Exam, Problem 4.

Let V be a finite-dimensional complex inner product space. Prove that T is both self-adjoint and unitary if and only if there is a subspace $W \subseteq V$ such that

$$Tv = \begin{cases} v, & v \in W, \\ -v, & v \in W^\perp. \end{cases}$$

Solution: If T is self-adjoint and unitary, then T is normal, so it has an orthonormal eigenbasis. Self-adjointness forces eigenvalues real, and unitarity forces eigenvalues to have absolute value 1. Hence the only eigenvalues are ± 1 . Let $W = E_1(T)$. Then $W^\perp = E_{-1}(T)$, and T acts as claimed.

Conversely, if T acts as I on W and $-I$ on $W^\perp$, then with respect to an orthonormal basis adapted to $V = W \oplus W^\perp$, the matrix of T is diagonal with diagonal entries ± 1 . Such a matrix is both self-adjoint and unitary. $\odot$

Question 34: Additional Exercise: Kernels of T and T^*

Source: 2024 Final Exam, Problem 5.

Let V be finite-dimensional and $T \in \mathcal{L}(V)$.

1. Prove that $\dim \ker T = \dim \ker T^*$.
2. Give an example where $\ker T \neq \ker T^*$.
3. If T is normal, prove that $\ker T = \ker T^*$.

Solution: (1) Since the matrices of T and T^* are conjugate transposes, they have the same rank. Rank-nullity gives

$$\dim \ker T = \dim V - \text{rank } T = \dim V - \text{rank } T^* = \dim \ker T^*.$$

(2) On $\mathbb{C}^2$, let

$$T = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $\ker T = \text{span}\{e_1\}$ while $\ker T^* = \text{span}\{e_2\}$.

(3) If T is normal, then

$$\|Tv\|^2 = \langle T^*Tv, v \rangle = \langle TT^*v, v \rangle = \|T^*v\|^2.$$

Thus $Tv = 0$ if and only if $T^*v = 0$, so $\ker T = \ker T^*$. $\odot$

Question 35: Additional Exercise: Positive Operators and Polar Decomposition

Source: 2024 Final Exam, Problem 7.

1. If S is positive and U is unitary, show that USU^* is positive and

$$\sqrt{USU^*} = U\sqrt{S}U^*.$$

2. If $T = UR$ is the polar decomposition of T , prove that T is normal if and only if

$$UR = RU.$$

Solution: (1) For any v ,

$$\langle USU^*v, v \rangle = \langle S(U^*v), U^*v \rangle \geq 0,$$

and USU^* is self-adjoint. Also

$$(U\sqrt{S}U^*)^2 = USU^*$$

and $U\sqrt{S}U^*$ is positive, so uniqueness of the positive square root gives the formula.

(2) In the polar decomposition, $R = \sqrt{T^*T}$ is positive and U is unitary. Thus

$$T^*T = R^2, \quad TT^* = UR^2U^*.$$

So T is normal iff $R^2 = UR^2U^*$. Taking the unique positive square root of both sides gives

$$R = URU^*,$$

which is equivalent to $RU = UR$. ☺

Question 36: Additional Exercise: When $P - Q$ is Positive

Source: 2024 Final Exam, Problem 8, Bonus.

Let P, Q be orthogonal projections on a finite-dimensional real or complex inner product space. Show that

$$P - Q \text{ is positive} \iff PQ = Q.$$

Solution: Assume $P - Q$ is positive. If $x \in \text{Im } Q$, then $Qx = x$, so

$$0 \leq \langle (P - Q)x, x \rangle = \langle Px, x \rangle - \|x\|^2 = \|Px\|^2 - \|x\|^2.$$

But an orthogonal projection cannot increase norm, so $\|Px\| \leq \|x\|$. Hence equality holds, so $Px = x$, and $\text{Im } Q \subseteq \text{Im } P$. Therefore $PQ = Q$.

Conversely, if $PQ = Q$, then taking adjoints gives $QP = Q$, so P and Q commute and $P - Q$ is self-adjoint. Also

$$(P - Q)^2 = P - 2Q + Q = P - Q.$$

Thus $P - Q$ is an orthogonal projection, hence positive. ☺

Question 37: Additional Exercise: Normal-Operator True/False

Source: 2026 Practice Final Exam, Problem 3.

Let V be a finite-dimensional complex inner product space. Decide whether each statement is true or false.

1. If T is normal, then $\langle Tu, Tv \rangle = \langle T^*u, T^*v \rangle$ for all u, v .

2. If $\text{span}\{v\}^\perp$ is T^* -invariant, then v is an eigenvector of T .
3. If $\|Tv\| = \|T^*v\|$ for all v , then T is unitary.
4. If T is normal and every root of p_T has multiplicity one, then every eigenbasis for T is orthogonal.

Solution: (1) True, since

$$\langle Tu, Tv \rangle = \langle u, T^*Tv \rangle = \langle u, TT^*v \rangle = \langle T^*u, T^*v \rangle.$$

(2) True for $v \neq 0$. If $W = \text{span}\{v\}^\perp$ is T^* -invariant, then $W^\perp = \text{span}\{v\}$ is T -invariant, so $Tv \in \text{span}\{v\}$.

(3) False. For $T = 2I$, $\|Tv\| = \|T^*v\|$ for all v , but T is not unitary.

(4) True. A normal operator has orthogonal eigenspaces for distinct eigenvalues. If all roots of p_T are simple, each eigenspace is one-dimensional, so any eigenbasis is orthogonal. ☺

Question 38: Additional Exercise: Skew-Adjoint Eigenvalues

Source: 2026 Practice Final Exam, Problem 4.

Let V be a complex inner product space. If $T^* = -T$, show that every eigenvalue of T is purely imaginary.

Solution: Let $Tv = \lambda v$ with $v \neq 0$. Then, using $T^* = -T$,

$$\lambda \langle v, v \rangle = \langle Tv, v \rangle = \langle v, T^*v \rangle = \langle v, -Tv \rangle = -\bar{\lambda} \langle v, v \rangle.$$

Since $\langle v, v \rangle \neq 0$, we get $\lambda = -\bar{\lambda}$. Thus $\text{Re } \lambda = 0$, so $\lambda = ai$ for some $a \in \mathbb{R}$. ☺

Question 39: Additional Exercise: A Unitarity Criterion

Source: 2026 Practice Final Exam, Problem 7.

Let V be a finite-dimensional complex inner product space.

1. Show that if T is normal and every eigenvalue λ of T satisfies $|\lambda| = 1$, then T is unitary.
2. Show that if T is invertible and $T^* = T^k$ for some integer $k \geq 2$, then T is unitary.

Solution: (1) By the spectral theorem, choose an orthonormal eigenbasis e_i with $Te_i = \lambda_i e_i$. Since $|\lambda_i| = 1$, T preserves norms on this basis and

$$T^*Te_i = |\lambda_i|^2 e_i = e_i.$$

Hence $T^*T = I$. Similarly $TT^* = I$, so T is unitary.

(2) From $T^* = T^k$, T commutes with T^* , so T is normal. If $Te = \lambda e$ for a unit eigenvector, then

$$\bar{\lambda}e = T^*e = T^k e = \lambda^k e.$$

Because T is invertible, $\lambda \neq 0$. Taking absolute values gives $|\lambda| = |\lambda|^k$, hence $|\lambda| = 1$. Part (1) now implies that T is unitary. ☺

11.4 SVD, Polar Decomposition, and T^*T

Question 40: Additional Exercise: A Two-by-Two SVD

Source: 2023 Final Exam, Problem 3.

Let

$$A = \frac{1}{\sqrt{50}} \begin{pmatrix} 4 & -7 \\ 8 & 11 \end{pmatrix}.$$

Find the singular values of A and a singular value decomposition $A = U_2 D U_1^{-1}$.

Solution: Compute

$$A^*A = \begin{pmatrix} 8/5 & 6/5 \\ 6/5 & 17/5 \end{pmatrix}.$$

Its eigenvalues are 4 and 1, so the singular values are 2 and 1. Corresponding orthonormal eigenvectors are

$$v_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

Then

$$w_1 = \frac{1}{2} A v_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} -1 \\ 3 \end{pmatrix}, \quad w_2 = A v_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} -3 \\ -1 \end{pmatrix}.$$

Thus one SVD is

$$U_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad U_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} -1 & -3 \\ 3 & -1 \end{pmatrix},$$

with $A = U_2 D U_1^*$. ☺

Question 41: Additional Exercise: Nonzero Eigenvalues of TT^* and T^*T

Source: 2023 Final Exam, Problem 7.

Let V, W be finite-dimensional complex inner product spaces and $T \in \mathcal{L}(V, W)$.

1. If v is an eigenvector of T^*T with nonzero eigenvalue, prove that $Tv \neq 0$.
2. Prove that TT^* and T^*T have the same nonzero eigenvalues with the same multiplicities.

Solution: (1) If $T^*Tv = \lambda v$ with $\lambda \neq 0$, then

$$\|Tv\|^2 = \langle T^*Tv, v \rangle = \lambda \|v\|^2 \neq 0,$$

so $Tv \neq 0$.

(2) Fix $\lambda \neq 0$. If $T^*Tv = \lambda v$, then

$$TT^*(Tv) = T(T^*Tv) = \lambda Tv,$$

and by part (1), $Tv \neq 0$. Thus T maps the λ -eigenspace of T^*T injectively into the λ -eigenspace of TT^* . Similarly, T^* maps the λ -eigenspace of TT^* injectively into the λ -eigenspace of T^*T . Hence the eigenspaces have the same dimension. Since T^*T and TT^* are self-adjoint, they are diagonalizable, so these dimensions are the algebraic multiplicities. ☺

Question 42: Additional Exercise: SVD and Polar Decomposition

Source: 2024 Final Exam, Problem 2.

Let

$$A = \begin{pmatrix} 2 & 0 \\ 3 & 2 \end{pmatrix}.$$

Find an SVD of A and its polar decomposition $A = UR$.

Solution: First

$$A^T A = \begin{pmatrix} 13 & 6 \\ 6 & 4 \end{pmatrix}.$$

The eigenvalues are 16 and 1, so the singular values are 4 and 1. Orthonormal eigenvectors are

$$v_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

Then

$$w_1 = \frac{1}{4} A v_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad w_2 = A v_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

Thus

$$U_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \quad D = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}, \quad U_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix},$$

and $A = U_2 D U_1^T$.

For the polar decomposition,

$$R = \sqrt{A^T A} = U_1 D U_1^T = \begin{pmatrix} 17/5 & 6/5 \\ 6/5 & 8/5 \end{pmatrix}.$$

Then

$$U = A R^{-1} = \begin{pmatrix} 4/5 & -3/5 \\ 3/5 & 4/5 \end{pmatrix},$$

so

$$A = \begin{pmatrix} 4/5 & -3/5 \\ 3/5 & 4/5 \end{pmatrix} \begin{pmatrix} 17/5 & 6/5 \\ 6/5 & 8/5 \end{pmatrix}.$$



Question 43: Additional Exercise: A Rectangular SVD

Source: 2026 Practice Final Exam, Problem 2.

Find a singular value decomposition of

$$A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Solution:

$$A^*A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

whose eigenvalues are 3 and 1. Thus $\sigma_1 = \sqrt{3}$ and $\sigma_2 = 1$, with

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Set $w_i = \sigma_i^{-1}Av_i$. Then

$$w_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad w_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Complete to an orthonormal basis of $\mathbb{R}^3$ with

$$w_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

Therefore one SVD is $A = U_2 \Sigma U_1^*$, where

$$U_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{pmatrix},$$

and

$$\Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

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Question 44: Additional Exercise: Rank of T^*T

Source: 2026 Practice Final Exam, Problem 5.

Let V be finite-dimensional and $T : V \rightarrow V$ linear. Prove, without citing results from class, that

$$\text{rank}(T^*T) = \text{rank}(T).$$

Solution: We first prove that the kernels are equal. If $Tv = 0$, then $T^*Tv = 0$. Conversely, if $T^*Tv = 0$, then

$$0 = \langle T^*Tv, v \rangle = \langle Tv, Tv \rangle = \|Tv\|^2,$$

so $Tv = 0$. Hence

$$\ker(T^*T) = \ker T.$$

By rank-nullity on the finite-dimensional space V ,

$$\text{rank}(T^*T) = \dim V - \dim \ker(T^*T) = \dim V - \dim \ker T = \text{rank } T.$$

⊖